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IVAN AUGUSTO CECILIO E SILVA

**ESSAYS IN CLIMATE AND HEALTH ECONOMICS:
THE IMPACTS OF FLOODS AND HEATWAVES ON
MATERNAL AND CHILD HEALTH SERVICE UTILIZATION**

Porto Alegre
2026

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Tese de Doutorado submetida ao Programa de Pós-Graduação em Economia da Universidade Federal do Rio Grande do Sul, como requisito para obtenção do título de Doutor em Economia.

Orientador: Prof. Dr. Giácomo Balbinotto Neto
Coorientadora: Prof^a. Dr^a. Cássia Kely Favoretto

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Para todas as vítimas das enchentes no Brasil, em especial às de maio de 2024, no Rio Grande do Sul. Aos que perderam seus lares, seus entes queridos e parte de suas histórias — que esta obra seja também um apelo por justiça climática e por uma reconstrução digna.

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*"O saber de vocês é sintético, o nosso é orgânico.
Vocês pensam de forma linear, por isso chegam ao limite.
Nós pensamos de forma circular, por isso a vida não tem limite.
A roda é começo, meio, começo; ela não tem fim."
Antônio Bispo dos Santos*

ABSTRACT

This thesis investigates the impacts of climate change, specifically extreme events such as floods and heatwaves, on the utilization of maternal and child health (MCH) services. Grounded in an interdisciplinary approach that integrates realist-informed evidence synthesis, systems thinking, and quantitative causal inference, the study is structured into three complementary essays. **Essay 1** consists of a realist-informed systematic review aimed at understanding whether, how, and in which contexts extreme events affect MCH service utilization. A total of 36 quantitative and qualitative studies were included and assessed in terms of methodological quality, type of climatic event, study design, and geographic context. The findings indicate that floods and heatwaves negatively affect access to antenatal, delivery, and postnatal care, particularly in socioeconomically vulnerable settings. **Essay 2** employs Causal Loop Diagrams (CLDs) to map, based on the evidence synthesized in Essay 1, the mechanisms through which floods undermine MCH service utilization. Six core mechanisms were identified: travel capacity, displacement, income loss, health risks, service provision, and broader infrastructure disruptions. The qualitatively constructed CLD comprises 18 interconnected components organized into reinforcing and balancing feedback loops, providing an integrated representation of the dynamic interactions between health systems and climate shocks. **Essay 3** applies causal inference methods to estimate the effects of the 2010 floods on antenatal care utilization in the states of Pernambuco and Alagoas, using SINASC data from 2008 to 2012. The results indicate that the main impact of the event was not an increase in exclusion from antenatal care, but rather a reallocation in the intensity of care. We observe a significant reduction in the proportion of women with seven or more visits and an increase in the intermediate category (4–6 visits), with no evidence of growth in the proportion of women with no or few visits. These findings suggest a partial disruption in continuity of care, rather than a collapse in access to antenatal services. The methodological complementarity across the essays strengthens the internal coherence of the thesis: Essay 1 provides the theoretical and empirical foundation for the causal mapping in Essay 2, while Essay 3 empirically tests the mechanisms identified. Taken together, the three essays demonstrate that extreme events exacerbate pre-existing vulnerabilities and compromise continuity of care in MCH services. The thesis advances the literature by proposing an integrated framework that combines qualitative evidence, systems modeling, and quantitative causal analysis to inform more resilient and equitable public policies in the context of climate change.

Keywords: Climate Change. Floods. Heatwaves. Maternal and Child Health. Interrupted Time Series. Causal Loop Diagrams. Realist Systematic Review.

JEL: I18 (Health: Government Policy; Regulation; Public Health), J13 (Fertility; Family Planning; Child Care; Children; Youth), Q54 (Climate; Natural Disasters and Their Management; Global Warming).

RESUMO

Esta tese investiga os impactos das mudanças climáticas, especificamente de eventos extremos como enchentes e ondas de calor, sobre a utilização dos serviços de saúde materno-infantil (SMI). Com base em uma perspectiva interdisciplinar que integra avaliação realista, pensamento sistêmico e métodos quantitativos, o trabalho está estruturado em três ensaios complementares. O **Ensaio 1** consiste em uma revisão sistemática realista-informada que busca compreender se, como e em que contextos eventos extremos afetam a utilização dos serviços de SMI. Foram incluídos 36 estudos quantitativos e qualitativos, analisados quanto à qualidade metodológica, tipo de evento, desenho de estudo e contexto geográfico. Os resultados indicam que as enchentes e ondas de calor afetam negativamente o acesso ao pré-natal, parto e pós-natal, principalmente em contextos de vulnerabilidade socioeconômica. O **Ensaio 2** utiliza diagramas de loops causais (CLDs) para mapear, com base na evidência sintetizada no Ensaio 1, os mecanismos pelos quais enchentes comprometem a utilização dos serviços de SMI. Foram identificados seis mecanismos principais: capacidade de viajar, desalojamento, perda de renda, riscos à saúde, oferta de serviços e infraestrutura em geral. O CLD construído qualitativamente reúne 18 componentes organizados em loops de retroalimentação, oferecendo uma visão integrada das relações dinâmicas entre sistemas de saúde e choques climáticos. O **Ensaio 3** aplica uma análise de inferência causal para avaliar os efeitos das enchentes de 2010 sobre a realização de consultas de pré-natal em Pernambuco e Alagoas, utilizando dados do SINASC entre 2008 e 2012. Os resultados revelam que o principal impacto do evento não foi o aumento da exclusão do pré-natal, mas sim uma realocação na intensidade do cuidado. Observou-se uma redução significativa na proporção de mulheres com sete ou mais consultas e um aumento na categoria intermediária (4–6 consultas), sem evidência de crescimento na proporção de mulheres com nenhuma ou poucas consultas. Esses achados sugerem uma interrupção parcial da continuidade do cuidado, e não um colapso no acesso ao pré-natal. A complementaridade metodológica entre os ensaios reforça a coesão da tese: o Ensaio 1 fundamenta teoricamente o mapeamento causal do Ensaio 2, enquanto o Ensaio 3 testa empiricamente os mecanismos identificados. Em conjunto, os três ensaios evidenciam que eventos extremos agravam vulnerabilidades existentes e comprometem a continuidade do cuidado em SMI. A tese contribui para o avanço do conhecimento ao propor uma abordagem integrada que combina evidência qualitativa, modelagem sistêmica e análise quantitativa para informar políticas públicas mais resilientes e equitativas frente às mudanças climáticas.

Palavras-chave: Mudanças Climáticas. Enchentes. Ondas de Calor. Saúde Materno-Infantil. Séries Temporais Interrompidas. Diagramas de Loops Causais. Revisão Sistemática Realista.

JEL: I18 (Saúde: Política Governamental; Regulação; Saúde Pública), J13 (Fertilidade; Planejamento Familiar; Cuidado Infantil; Crianças; Juventude), Q54 (Clima; Desastres Naturais e sua Gestão; Aquecimento Global).

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LIST OF ABBREVIATIONS AND ACRONYMS

ANC	Antenatal Care
CAS	Complex Adaptive Systems
CI	Confidence Interval
CINAHL	Cumulative Index to Nursing and Allied Health Literature
CITS	Controlled Interrupted Time Series
CLD	Causal Loop Diagram
CMO	Context-Mechanism-Outcome
COBRADE	Brazilian Classification and Coding of Disasters
COP	Conference of the Parties
CRED	Centre for Research on the Epidemiology of Disasters
DEA	Data Envelopment Analysis
DiD	Difference-in-Differences
ED	Emergency Department
EM-DAT	International Disaster Database
EWE	Extreme Weather Event
FGDs	Focus Group Discussions
FHS	Family Health Strategy
GDP	Gross Domestic Product
GHECO	Global Health Economics Centre
GP	General Practitioner
HICs	High-Income Countries
iCCM	Integrated Community Case Management
ICU	Intensive Care Unit
IDIs	In-Depth Interviews
IDPs	Internally Displaced Persons
IPCC	Intergovernmental Panel on Climate Change
IPT	Initial Programme Theory
IRR	Incidence Rate Ratio
ITSA	Interrupted Time Series Analysis

JEL	Journal of Economic Literature
LMICs	Low- and Middle-Income Countries
LPM	Linear Probability Model
MCH	Maternal and Child Health
MeSH	Medical Subject Headings
MMAT	Mixed Methods Appraisal Tool
M1	Mechanism 1
M2	Mechanism 2
M3	Mechanism 3
M4	Mechanism 4
M5	Mechanism 5
M6	Mechanism 6
NICU	Neonatal Intensive Care Unit
OLS	Ordinary Least Squares
OR	Odds Ratio
PAHO	Pan American Health Organization
PDA	Panel Data Analysis
PNC	Postnatal Care
POPJUS	Population and Just Societies
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PROSPERO	International Prospective Register of Systematic Reviews
REACH	Building Resilience to Floods and Heat in the Maternal and Child Health System in Brazil and Zambia
RHCC	Reproductive Health Kit 8; Capacity Building; Community Awareness
RISR	Realist-Informed Systematic Review
SINASC	Live Birth Information System
SUS	<i>Sistema Único de Saúde</i> (Unified Health System)
SRHR	Sexual and Reproductive Health and Rights
SSP	Shared Socioeconomic Pathway
SYRR	Systemic Risk and Resilience
S2iD	Integrated Disaster Information System
UK	United Kingdom

UNFCC	United Nations Framework Convention on Climate Change
UNICEF	United Nations Children's Fund
WHO	World Health Organization
WMO	World Meteorological Organization

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1 INTRODUCTION

Climate change refers to long-term alterations in the state of the climate, identified by statistically significant changes in average conditions and/or their variability over time (typically assessed over a minimum period of 30 years) (Intergovernmental Panel on Climate Change, 2018). This encompasses shifts in surface temperature, precipitation patterns, wind, humidity, cloud cover, solar radiation (insolation), atmospheric pressure, the frequency and intensity of extreme events (e.g., hurricanes, droughts, floods, heatwaves), ice and snow cover, sea level (due to thermal expansion and glacier melt), and greenhouse gas concentrations (World Meteorological Organization, 2025). While the climate of Earth has always varied naturally, the current changes are largely driven by human activities (especially the combustion of fossil fuels, deforestation, and other land-use changes such as large-scale agriculture, pasture expansion, urbanization, wetland drainage, slash-and-burn practices, infrastructure development, and forest fragmentation) that have increased the concentrations of greenhouse gases in the atmosphere (Roy et al., 2022).

Extreme weather events (EWEs) are meteorological occurrences that deviate markedly from a region's typical weather patterns in terms of intensity, duration, or spatial extent, and usually lie in the far tail of the historical distribution. Statistically, these events are often defined as those that occur in the highest or lowest percentile of observations. Examples include heatwaves, heavy rainfall events, floods, droughts, tropical cyclones, and severe storms. As the climate warms, the atmosphere's capacity to hold moisture increases and altered circulation patterns can lead to more frequent and intense extreme events. This means that events which were once considered "improbable" are becoming more common, contributing to higher risks of damage to ecosystems, infrastructure, and human lives (Seneviratne et al., 2021).

Maternal and child health (MCH) systems are comprehensive networks of services, policies, and resources designed to promote the health and well-being of women throughout pregnancy, childbirth, and the postpartum period, as well as to ensure the healthy growth and development of infants and children. These systems encompass a continuum of care that includes preconception counseling, antenatal care (such as regular health check-ups, nutritional support, and screening for complications), skilled attendance during labor and delivery, and postnatal services for both the mother and child. In addition, MCH systems integrate preventive interventions like immunizations, growth monitoring, nutrition programs, and health education initiatives that target both mothers and children. For instance, an effective MCH system might involve community health workers providing outreach in rural areas, primary care clinics offering routine antenatal services, and referral networks that link to higher-level facilities for emergency obstetric care. This integrated approach is crucial for reducing maternal, newborn, and child morbidity and mortality, a priority highlighted by the World Health Organization (WHO) in its Global Strategy for Women's, Children's and Adolescents' Health (World Health Organization, 2015).

This thesis explores the intersection of climate change, EWEs and MCH systems through three complementary essays. The first essay is a realist-informed systematic review examining how heat waves and floods affect MCH systems globally. It synthesizes current evidence globally and identifies research gaps. Building on these findings, the second essay employs a causal loop diagram (CLD) to map the theoretical framework and illustrate how floods influence MCH utilization. The third essay uses interrupted time series and panel data analyses to assess the impact of floods on prenatal health service use in Pernambuco, Brazil. Each essay is original in nature and contributes uniquely to understanding the relationship between climate extremes and MCH. The primary goal of this thesis is to enhance our understanding of how shifting climate patterns and extreme weather events impact maternal and child health system indicators across diverse contexts, thereby providing critical insights to inform the development of targeted mitigation and adaptation strategies. The three essays outlined here share a solid theoretical and empirical foundation based on frontier research in public health (Romanello et al., 2025; Masters et al., 2025) and health economics (Borghi e Kuhn, 2024; Hensher, 2023).

1.1 STYLIZED FACTS

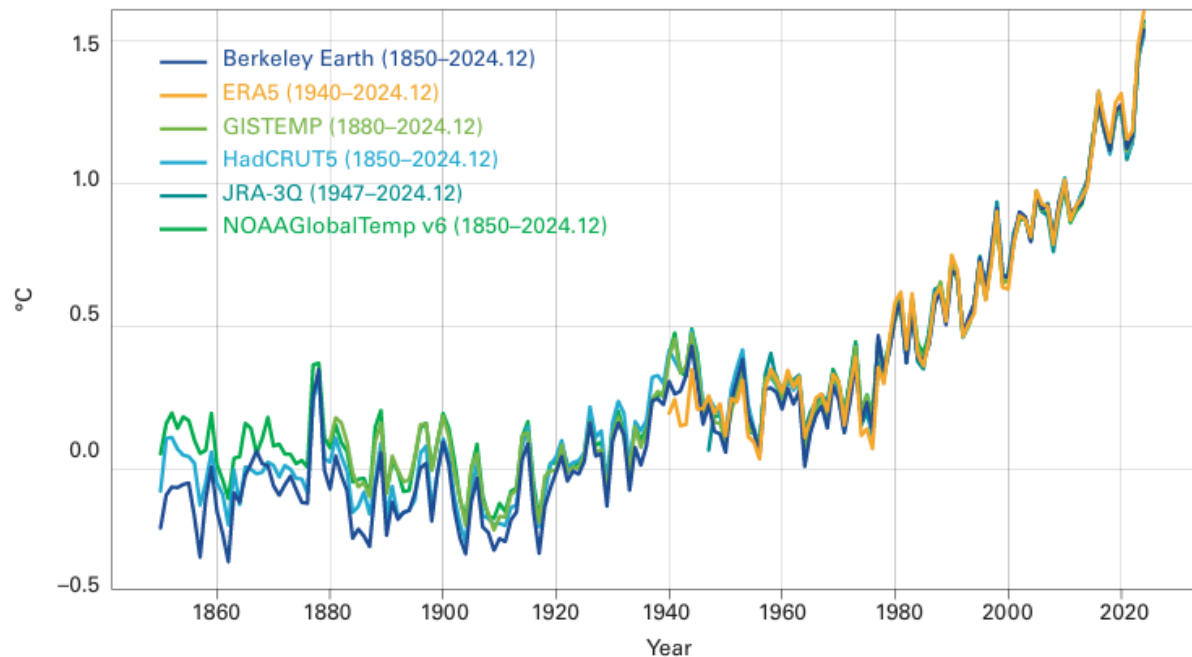
Before delving deeper into any theories and models, it is imperative to shed light on some stylized facts. Stylized facts are simplified, general empirical observations that capture robust, recurring patterns in data across different contexts or time periods. They are not precise in every individual instance but rather describe the broad regularities that hold on average, and these regularities serve as benchmarks for developing, testing, and refining theoretical models. These empirical patterns not only characterize the core of the problem but also provide the solid foundation from which all subsequent development will unfold (Kaldor, 1961; Hirschman, 2016).

1.1.1 In 2024, the global temperature exceeded the 1.5°C threshold above pre-industrial levels (1850-1900) for the first time

The World Meteorological Organization (2025) synthesized six international datasets and reported a global average increase of approximately $1.55^{\circ}\text{C} \pm 0.13^{\circ}\text{C}$, marking the first calendar year above 1.5°C (Figure 1.1). While a single-year exceedance does not equate to crossing the Paris Agreement's¹ long-term target (based on 20-year averages), it signals, with high confidence, an era where 1.5°C global warming is likely to persist unless ambitious mitigation is rapidly implemented (Bevacqua, Schleussner e Zscheischler, 2025).

¹ The Paris Agreement is an international treaty adopted under the United Nations Framework Convention on Climate Change (UNFCCC) at the 21st Conference of the Parties (COP21) in December 2015. It seeks to strengthen the global response to climate change by “holding the increase in the global average temperature to well below 2°C above pre-industrial levels and pursuing efforts to limit the temperature increase to 1.5°C ” (United Nations Framework Convention on Climate Change, 2015). In addition to its mitigation target, the treaty promotes enhanced adaptive capacity, greater resilience to climate impacts, and the alignment of financial flows with low-emission, climate-resilient development pathways.

Figure 1.1 – Annual global temperature anomalies (1850–2024; baseline 1850–1900)

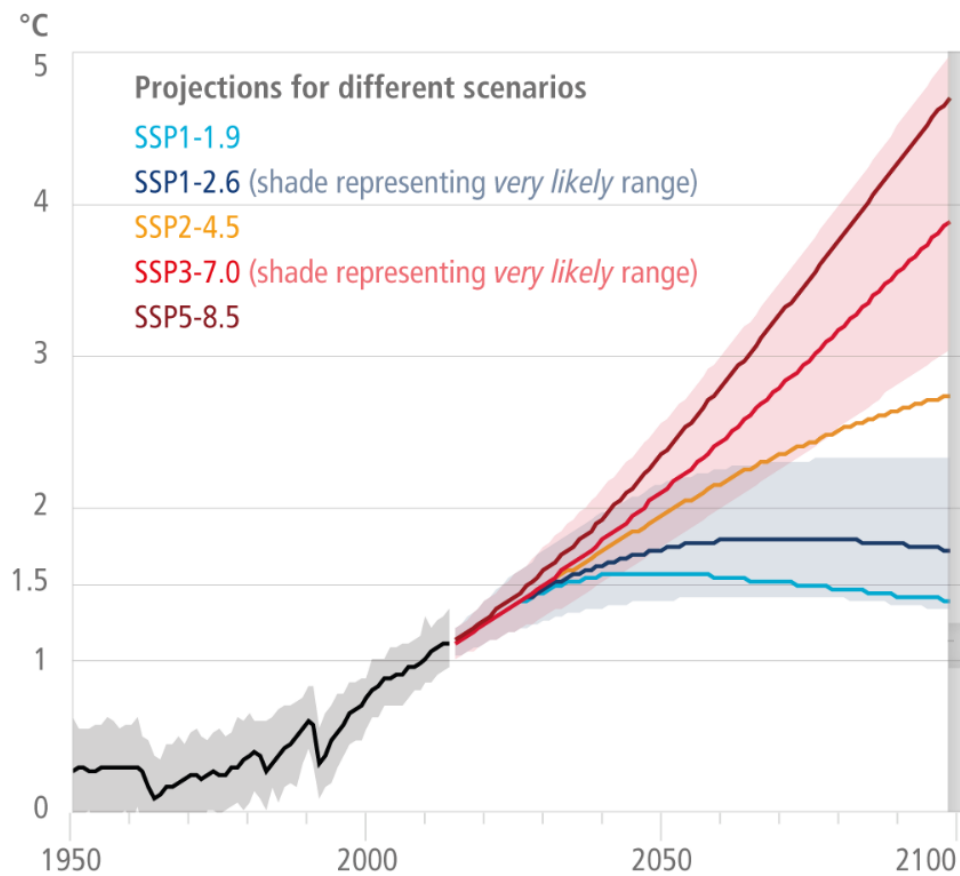


Source: World Meteorological Organization (2025).

Figure 1.1 shows that the planet's average temperature is increasing rapidly. This increase is primarily attributed to human activities, particularly the burning of fossil fuels (such as coal, oil, and gas) for energy generation, in addition to deforestation and other industrial processes (Rom e Pinkerton, 2021). Greenhouse gases like carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O), resulting from these activities, act like a "blanket" in the atmosphere, allowing solar radiation in but hindering heat from escaping, leading to the planet's gradual warming (Kweku et al., 2018). The consequences of this temperature increase are profound and far-reaching, including ecosystem degradation and biodiversity loss (Marselle et al., 2019), the disappearance of water resources, food insecurity, human population displacement, health damage, loss of economic productivity, and social and cultural impacts (Kemp et al., 2022).

The Intergovernmental Panel on Climate Change (IPCC) project a range of possible future temperature increases based on different greenhouse gas emissions trajectories, called Shared Socioeconomic Pathway (SSP) scenarios (Figure 1.2).

Figure 1.2 – Global surface temperature change, increase relative to the period 1850-1900



Source: Pörtner et al. (2022).

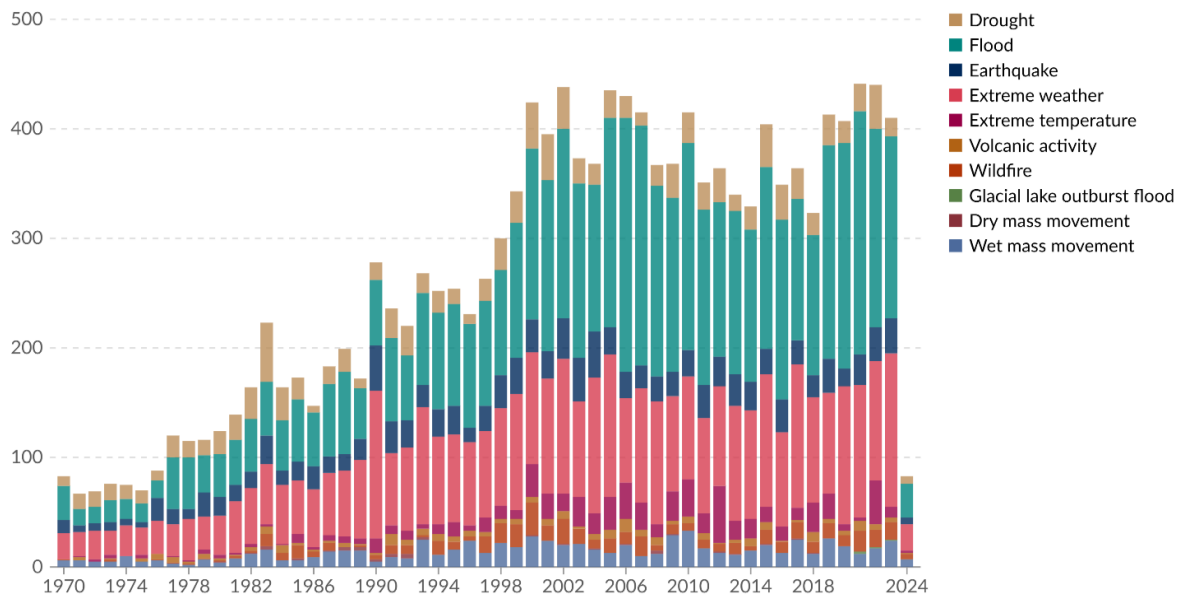
In Figure 1.2, the black line shows the observed global temperature anomaly (relative to a pre-industrial baseline) through recent decades, while each coloured line represents a different SSP scenario, with shaded areas indicating the “very likely” range (often the 5th–95th percentile) of model projections. SSP1-1.9 and SSP1-2.6 assume ambitious mitigation, with emissions peaking soon and then rapidly declining, potentially keeping warming near or below 2°C above pre-industrial levels. SSP2-4.5 represents a “middle-of-the-road” scenario where moderate policies lead to around 2.5–3°C of warming. SSP3-7.0 envisions limited cooperation and higher emissions, driving temperatures toward 3.5–4°C. Finally, SSP5-8.5 is a worst-case, fossil-fuel-intensive pathway where temperatures could exceed 4°C, reflecting minimal climate policy and continued high emissions (Pörtner et al., 2022).

1.1.2 The number of reported natural disasters has been increasing in the last 50 years

Figure 1.3 shows the global number of reported natural disasters by type from 1970 to 2024, indicating an apparent rise from fewer than 100 events per year in the 1970s to over 400 in more recent decades. Floods and extreme weather (storms, hurricanes, tornadoes, etc.) make up a large share of reported events, with other disaster types (earthquakes, droughts, extreme temperatures, etc.) contributing smaller but still notable portions. The increase over time, however,

largely reflects improvements in data collection and reporting practices (particularly the capture of smaller disasters) rather than a definitive surge in actual disaster occurrence (Ritchie e Rosado, 2024).

Figure 1.3 – Global reported natural disasters by type (1970 – 2024)



Source: Ritchie, Rosado e Roser (2022).

Note: Data includes disasters recorded up to April 2024.

Figure 1.3 data is sourced from the International Disaster Database (EM-DAT), which classifies a disaster as any situation or event that overwhelms local capacity and requires external assistance at the national or international level. These disasters typically strike suddenly and cause substantial damage, destruction, and human suffering. EM-DAT focuses on major disasters, which must meet at least one of the following criteria: at least 10 fatalities, at least 100 people affected, a declaration of a state of emergency, or a call for international assistance (Centre for Research on the Epidemiology of Disasters, 2026; Delforge et al., 2025).

1.1.3 Brazil is a country prone and vulnerable to natural disasters

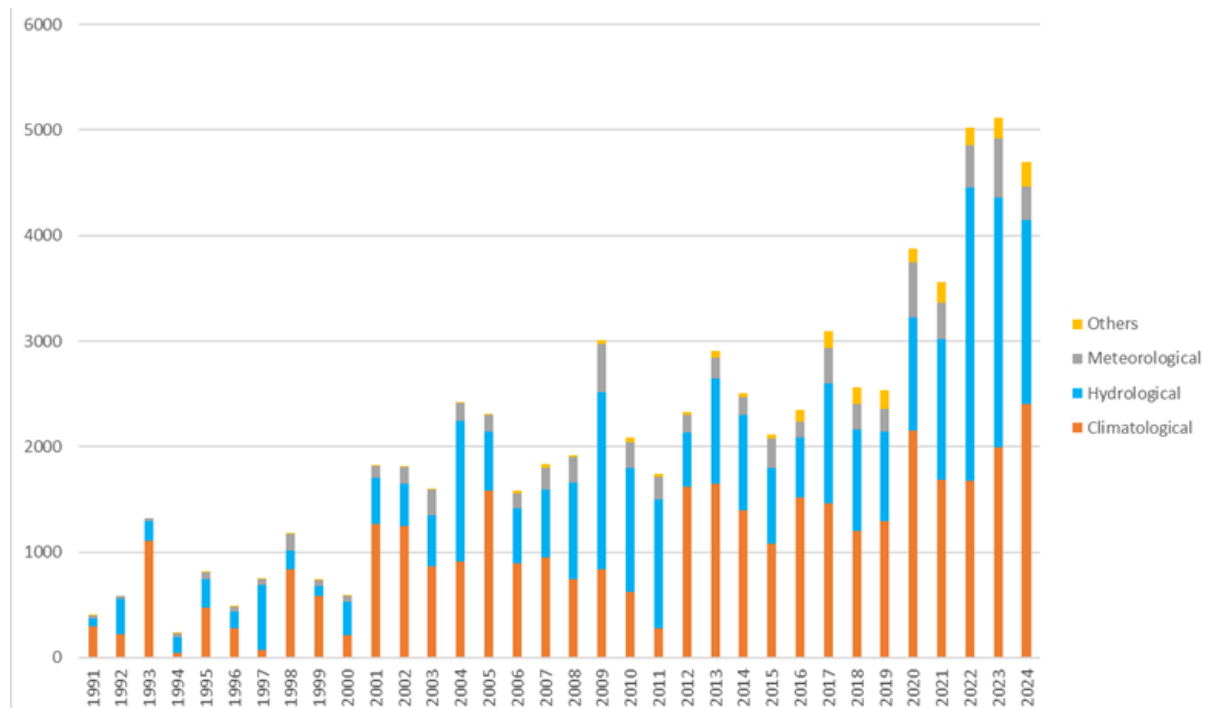
In Brazil, natural disasters are categorized based on the Brazilian Classification and Coding of Disasters (COBRADE) and divided into five groups: geological, hydrological, meteorological, climatological, and biological.

Hydrological disasters include floods, flash floods, and urban flooding, each caused by different water flow dynamics and precipitation patterns. Meteorological disasters encompass large-scale weather systems, storms, and extreme temperatures. These include cyclones, cold fronts, tornadoes, hailstorms, heavy rainfall, and strong winds, as well as heatwaves and cold waves. Climatological disasters mainly refer to droughts, including dry spells, severe droughts,

forest fires, and low air humidity, all of which significantly impact ecosystems and human activities. Geological disasters include earthquakes, volcanic emissions, mass movements (such as landslides, rockfalls, and subsidence), and erosion (coastal, riverbank, and continental). Biological disasters involve epidemics (caused by viral, bacterial, parasitic, or fungal infections) and infestations of animals, algae, or other organisms (BRASIL, 2022).

Figure 1.4 shows the number of emergency requests by municipalities affected by natural disasters in Brazil as recorded in the Integrated Disaster Information System (S2iD) and made available in the Digital Atlas of Disasters in Brazil (BRASIL, 2025).

Figure 1.4 – Municipal Emergency Requests for Natural Disasters in Brazil (1991–2024)



Source: BRASIL (2025).

Figure 1.4 illustrates a significant rise in emergency requests by municipalities affected by natural disasters in Brazil from 1991 to 2024, with record peaks in 2022 and 2023. Over the full period, climatological disasters accounted for 49.33% of requests by municipalities, followed by hydrological disasters at 38.47%, meteorological disasters at 9.43%, and other disasters at 2.77%. To be even more specific, over the full period, droughts and dry spells were the most frequent, accounting for 44.47% of municipalities affected. Flooding-related events (including general flooding, flash floods, and river flooding) comprised 25.03%, while heavy rainfall represented 11.25%. Strong winds and cyclones accounted for 6.26%, followed by forest fires (4.41%) and hailstorms (2.97%). Mass movements contributed 2.34%, and erosion accounted for 1.01%. Less frequent occurrences included cold waves (0.35%), tornadoes (0.15%), heatwaves and low humidity (0.14%), and others (1.62%). These figures show that climatological and hydrological disasters have been the most prevalent in Brazil over the years. However, it's important to note

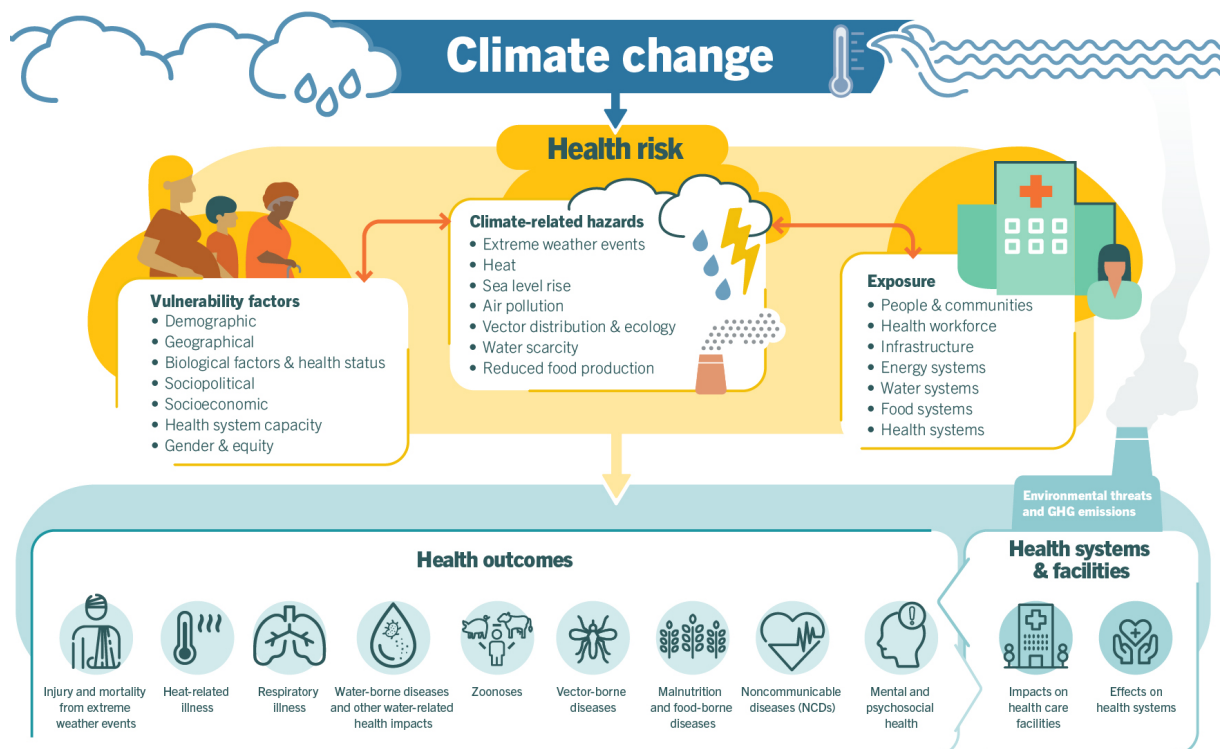
that the S2iD database is based solely on municipal emergency requests, which may lead to underrepresentation of certain events (such as heatwaves) that are harder to detect and report.

1.1.4 Climate change is considered the greatest threat to human health in the 21st century

Long-term changes in temperature, precipitation, wind, and other weather measures have harmful consequences for human health and survival conditions (Rocque et al., 2021). Due to climate change, approximately 250,000 additional deaths per year are projected between 2030 and 2050 from malnutrition, malaria, diarrhea, and heat stress, with direct health costs estimated between 2 and 4 billion dollars per year by 2030 (World Health Organization, 2023). In recent years, there has been growing recognition of the significant impact of climate change on human health, with research focusing on understanding the interaction between climate change and public health, aiming to enhance resilience and reduce risks (Romanello et al., 2025; Myhre et al., 2025; Golden et al., 2025).

Climate change impacts health through multiple exposure pathways, which are the specific mechanisms by which climate-related hazards affect human health (Figure 1.5).

Figure 1.5 – Climate-Sensitive Health Risks: Pathways and Vulnerabilities



Source: World Health Organization (2023).

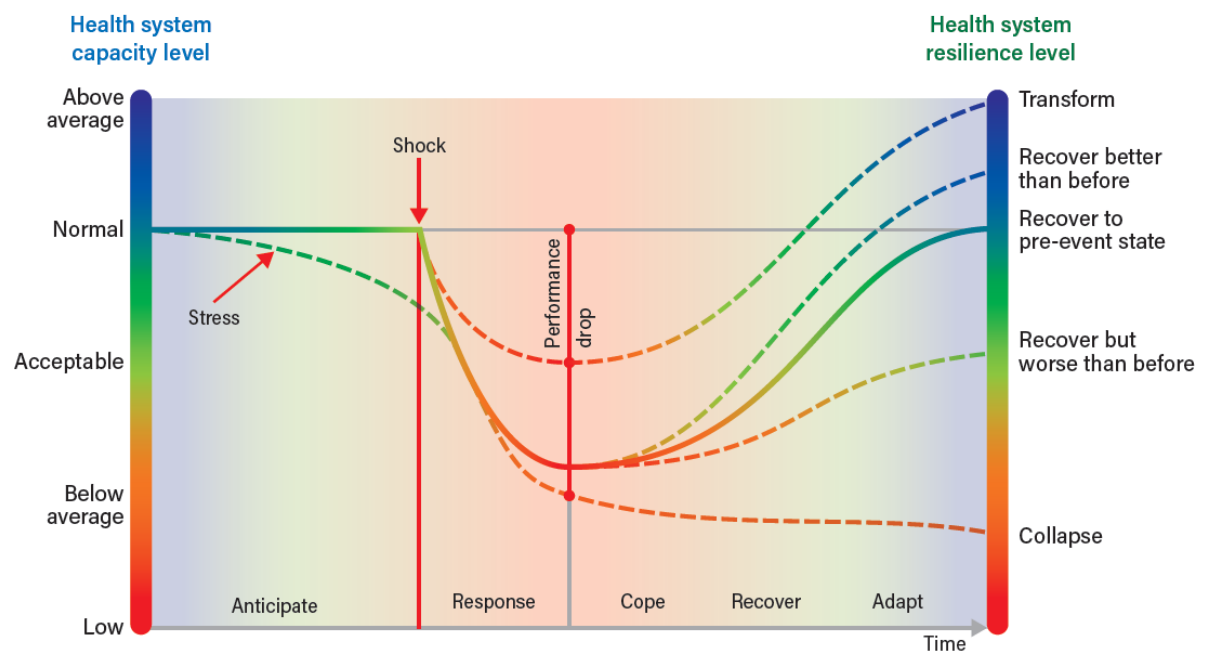
These include direct exposures such as extreme events like storms, floods, and heat stress, as well as indirect exposures like air quality degradation, changes in water quality and availability, food security challenges, and shifts in vector distribution and ecology (World Health Organization, 2023).

These environmental factors interact with vulnerability factors, which are characteristics that increase an individual's or group's susceptibility to the adverse effects of climate change. These include demographic aspects (such as age and gender), geographic location, biological susceptibility, sociopolitical conditions, socioeconomic status, and existing health conditions. These factors influence the capacity to anticipate, cope with, and recover from climate-related stressors, and thus determine how different populations are affected (World Health Organization, 2023).

The interaction between exposure pathways, vulnerability factors, and system resilience ultimately shapes health outcomes, the health-related consequences of climate change. These outcomes can include injuries and mortality, heat-related and respiratory diseases, foodborne and waterborne illnesses, malnutrition, zoonotic infections, non-communicable diseases, and mental and psychosocial health challenges (Ebi et al., 2021; World Health Organization, 2023; Romanello et al., 2025).

Health system resilience plays a crucial role in mitigating these impacts. A climate-resilient health system is one that can prepare for, respond to, and recover from climate shocks while maintaining its core functions and services (Kruk et al., 2015; Witter et al., 2023). Figure 1.6 illustrates how a health system's performance evolves over time when exposed to a shock:

Figure 1.6 – Health System Resilience



Source: World Health Organization (2022).

On the left vertical axis, we see the health system capacity level, ranging from low to above average. Initially, the system operates at a “normal” level but may experience stress even before the shock occurs. When the shock hits, performance drops sharply. The size of this drop reflects the system's immediate vulnerability. The bottom axis represents time and the stages

of response: anticipate, response, cope, recover, and adapt. After the shock, different recovery trajectories are possible: the system may collapse, recover but worse than before, return to its pre-event level, recover better than before, or even transform to a higher level of resilience. The right vertical axis represents the health system resilience level, emphasizing that resilience is not just about bouncing back, but about how well the system absorbs the shock, minimizes performance loss, and adapts or improves over time (World Health Organization, 2022).

Figure 1.7 presents the building blocks for climate-resilient health systems.

Figure 1.7 – Building blocks of climate-resilient health systems



Source: World Health Organization (2022).

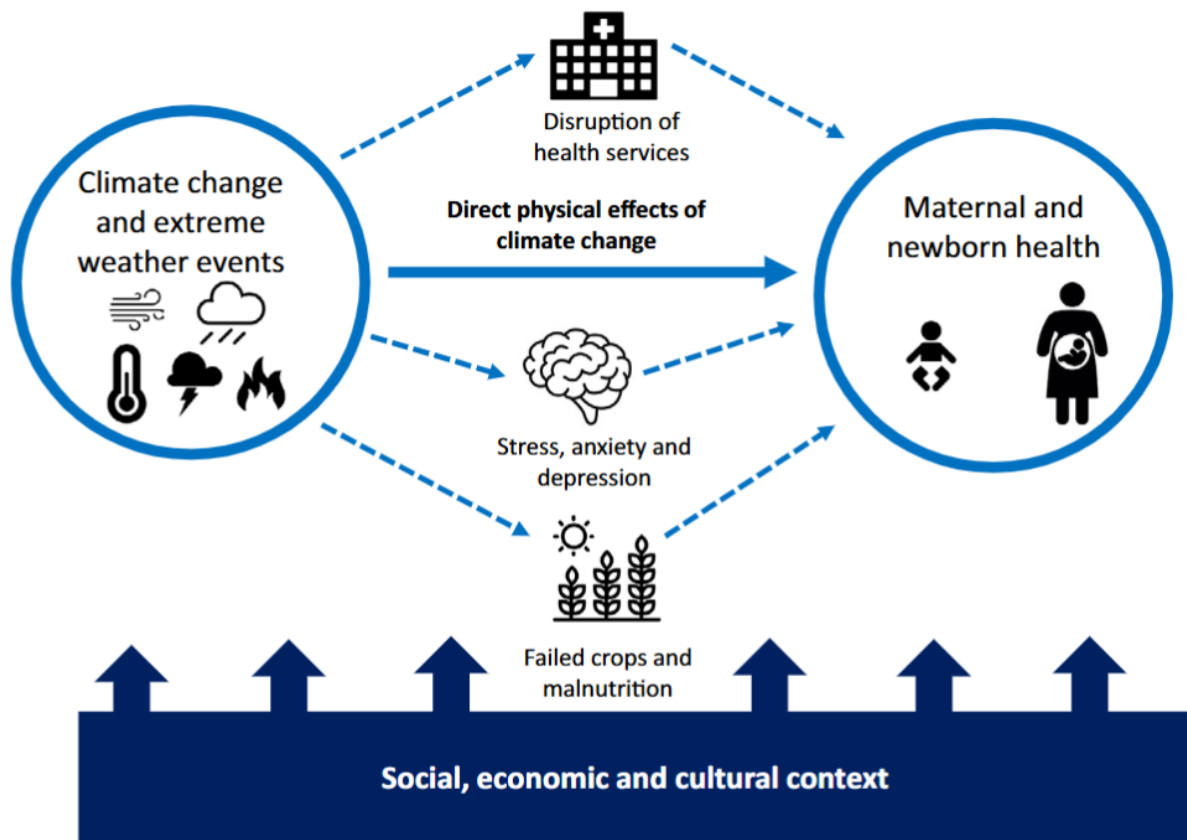
According to the World Health Organization (2022), climate-resilience depends on strengthening the core building blocks of the health system so that it can anticipate, prepare for, respond to, and adapt to climate-related shocks. These building blocks include: leadership and governance, which provide direction, coordination and accountability for climate action; health financing, which ensures sustainable resources for preparedness and adaptation; the health workforce, trained and supported to manage climate-sensitive health risks; health information systems, capable of integrated risk monitoring, early warning and vulnerability assessment; essential medical products and technologies, including climate-resilient and sustainable infrastructure;

and service delivery, which must remain accessible and responsive even during extreme events. Around these core components, climate resilience also requires emergency preparedness and management, climate-informed health programmes, research, risk monitoring, and attention to environmental determinants of health (World Health Organization, 2022).

1.1.5 Pregnant women and children are most vulnerable to climate change

Maternal and child health is directly affected by physical health impacts, mental health stressors, disruptions in healthcare services, and food insecurity caused by crop failures and malnutrition. These layers are all embedded within a broader social, economic, and cultural context, which shapes vulnerabilities and access to resources. Figure 1.8 illustrates the multiple impacts of climate change on maternal and child health, emphasizing how various factors interact.

Figure 1.8 – Impacts of climate change on maternal-child health



Source: Roos et al. (2021).

Systematic literature reviews suggest that women may be at greater risk of complications during pregnancy (preterm birth, low birth weight, or stillbirth) when exposed to high temperatures, due to the compromised thermoregulation ability of their bodies (Kuehn e McCormick, 2017; Chersich et al., 2020; Rocque et al., 2021). In addition, its indirect effects such as infrastructure damage, resource scarcity, and disruptions to healthcare systems can severely impact maternal–child well-being, especially in developing nations and marginalized communities (Arpin

et al., 2021; Wheeler et al., 2022). Women in resource-poor nations, already disproportionately affected by poverty, malnutrition, and limited education, comprise most displaced persons and face heightened risks during disasters, as strained healthcare systems with inadequate facilities and emergency resources further compromise prenatal care, safe delivery, and maternal-infant health outcomes (Nour, 2011; Pappas, Kovats e Ranganathan, 2024).

Despite these concerns, pregnant women and children are frequently overlooked as a vulnerable group in climate change research and policy discussions. Their specific needs receive limited attention, particularly in low- and middle-income countries, where constrained funding and other resource limitations hinder rigorous investigation into climate-related maternal–child health outcomes (Helldén et al., 2021; Amekpor et al., 2025; Zangerl et al., 2024; Persson et al., 2025). Many pregnant women in low- and middle-income countries live in large urban slums or remote rural areas, the environments in which they will raise their children, and will be increasingly subject to EWEs in the coming years (Food and Agriculture Organization of the United Nations, 2024; Bhanbhro e Ali, 2025).

1.2 A HEALTH ECONOMICS PERSPECTIVE ON CLIMATE IMPACTS: EVIDENCE FROM BRAZIL

Despite increasing global attention to climate impacts on health, empirical research on climate extremes and maternal–child health outcomes in Brazil remains limited. Early work in health economics examined climatic determinants of infant mortality: Rosenberg et al. (2000) used ordinary least squares (OLS) regression to assess the effects of temperature, precipitation, altitude, and coastal proximity on infant mortality across all Brazilian municipalities in 1996. Pereda, Menezes e Alves (2014) found, through panel data econometrics, that both excessive and insufficient rainfall increase neonatal mortality rates and the incidence of low birth weight in Brazilian states. Using the same method, Rocha e Soares (2015) found that water scarcity (lack of rainfall) in municipalities in Brazil’s semi-arid region is strongly correlated with higher infant mortality, lower birth weight, and shorter gestation periods. Mata et al. (2023) examined a large-scale adaptation intervention in Brazil’s semi-arid region, showing that access to cisterns improved birth outcomes, underscoring the role of resilience interventions in mitigating climatic stressors.

More recent observational evidence strengthens the case for climate impacts on pregnancy outcomes. In a nationally representative study of nearly 10 million live births, Requia e Papatheodorou (2024) found that ambient temperature exposure during pregnancy was associated with increased risk of preterm birth across multiple regions of Brazil, particularly in the North and Midwest, indicating that heat exposure may provoke earlier labor and regionally heterogeneous vulnerability patterns. Complementing this, Feldens e Oliveira (2026) uses nearly 45 million births across Brazilian municipalities (2000–2020) to demonstrate that temperature extremes and deviations from historical weather patterns significantly reduce birth weight, with the strongest effects in isolated and underserved areas and during the third trimester. Parfitt e Goulart (2026),

using administrative data on 25 million births in Brazilian public hospitals (2008–2019), shows that extreme heat also affects the organization of care within maternity wards, reducing overall cesarean rates, shifting deliveries from physicians to nurses, and increasing pre-labor cesareans among physician-led births. Together, these findings point to intertwined physiological stress responses and operational health system adjustments as distinct yet complementary mechanisms of maternal–child climate vulnerability in Brazil.

An important feature of this emerging literature is the application of econometric methods to identify causal effects of climate variability on health outcomes. Economists frequently conceptualize extreme weather events as exogenous shocks (quasi-natural experiments) that generate plausibly random variation in exposure across time and space. By exploiting high-frequency climate data combined with administrative health records, researchers are able to implement fixed-effects models, difference-in-differences strategies, and other causal inference techniques to isolate the impact of temperature spikes, rainfall anomalies, or floods on health service utilization and birth outcomes. This framework has been successfully applied in other contexts to link climate shocks to disruptions in maternal and child health service use (Haque, Parr e Muhidin, 2020a; Haque, Parr e Muhidin, 2020b; Chandir et al., 2023), demonstrating how environmental stressors can alter both demand-side behavior and supply-side constraints within health systems.

From a health economics standpoint, the proposed topic is both timely and relevant. Economists have recently begun to realize that children’s health starting in the prenatal (gestational) period is a valuable form of human capital, since children who are healthy from birth live longer, attain higher levels of education, become more productive adults, and earn higher wages in the future (Currie, 2020; Carrets, 2021). Maternal health interventions have proven effectiveness and cost-effectiveness, as they not only save mothers’ lives but also reduce neonatal mortality and morbidity, help combat inequality and improve the overall performance of health systems (Borghi et al., 2006). Furthermore, it is suggested that improvements in child and maternal health may generate positive economic benefits for gross domestic product (GDP) per capita, especially in developing countries (Trondillo, 2016; Klobodu et al., 2019).

Climate change and extreme weather events pose a multifaceted challenge to maternal and child health service utilization, and economic theory and methods provide a coherent framework to understand and address this challenge. By viewing healthcare as a system where providers supply services and families demand care, we can identify how climate shocks alter incentives, constraints, and outcomes. Supply represents the healthcare system’s capacity to provide services such as hospitals, clinics, medical personnel, equipment, and medications. Demand represents the need and willingness of individuals to seek and use those services, influenced by factors like perceived benefits, costs (monetary and non-monetary), and accessibility. Even though healthcare often isn’t a classical market (many services are subsidized or free at point of use), the core principles still apply. Utilization will be high when services are readily available (ample supply) and people are able and willing to access care (strong demand). Conversely, if supply is

constrained or demand is suppressed, utilization falls. Importantly, climate change and extreme weather events can act as shocks to either side of this equation, upsetting the balance. By examining how supply-side constraints (like damaged infrastructure and strained resources) and demand-side barriers (like reduced access and altered priorities) coincide during climate events, we uncover the mechanisms, analogous to shifts in supply and demand, behind observed drops in service use.

The economic perspective is not just an academic exercise; it has practical and political implications. It directs us to ask: How can we relax the supply constraints (perhaps through resilient design and surge staffing)? How can we lower the barriers on the demand side (perhaps through mobile clinics, transport, or emergency cash support)? The global nature of climate change means these questions are relevant from Bangladesh to Brazil, even as local context modifies specifics. In Brazil's case, we see that a strong public health system alone is not enough. Resilience against climate shocks must be built through smart investments and policies (Golden et al., 2025), many of which boil down to allocating resources efficiently under uncertainty. In the language of economics, it means moving the supply and demand curves in the right direction through investment, incentives, and innovation so that the equilibrium level of MCH service utilization remains high, safeguarding maternal and child health in a changing climate. This is where evidence-based public policies become essential, as they must be designed and implemented on the basis of rigorous data, scientific analysis, and empirical evaluation rather than intuition or political convenience. By grounding decisions in robust evidence, policymakers can more effectively target resources, anticipate unintended consequences, and design interventions that are not only cost-effective but also equitable and contextually appropriate, which is especially critical in the face of complex and evolving climate-health challenges (Hess et al., 2014).

1.3 RESEARCH AIMS AND THESIS STRUCTURE

The multidisciplinary approach adopted in this thesis, integrating perspectives from health sciences, economics, and climate science, reflects the inherent complexity of climate-related health challenges and the need for analytically integrated frameworks (Arthur, 2021; Nason, 2023; Martins et al., 2024; Dagneau, Ehgartner e Gulis, 2025). A growing body of research has documented associations between extreme weather events and adverse maternal and child health outcomes (Lakhoo et al., 2025; Persson et al., 2025; Masters et al., 2025; Pappas, Kovats e Ranganathan, 2024). Nevertheless, important gaps persist in the literature (Meherali et al., 2024).

First, the existing evidence remains fragmented across disciplines and geographic contexts, with limited synthesis specifically addressing how heatwaves and floods affect MCH service utilization and health system functioning (Valente et al., 2026; Martins et al., 2024; Dewi et al., 2024; Safajou, Nahidi e Ahmadi, 2023). Second, although disruptions in care have been observed, the mechanisms through which climate shocks translate into changes in health care use remain insufficiently theorized and conceptually integrated (Dagneau, Ehgartner e Gulis, 2025; Myhre et al., 2025; Ansah et al., 2024). Third, rigorous empirical evidence from low- and middle-income

countries, where exposure and vulnerability are often greatest, remains limited, both in terms of methodological robustness and geographic coverage (Feldens e Oliveira, 2026; Parfitt e Goulart, 2026; Chandir et al., 2023; Orderud et al., 2022; Fernandes et al., 2022; Baten et al., 2020).

This thesis addresses these gaps through an interdisciplinary health economics lens, examining how extreme weather events affect the utilization of maternal and child health services. Rather than treating climate shocks as isolated events, the thesis conceptualizes them as systemic disruptions that simultaneously alter both the demand for and the supply of health services. By combining evidence synthesis, systems modeling, and econometric analysis, the study advances a more comprehensive understanding of how climate-related shocks propagate through health systems.

The three essays are unified by a common research question: **how do extreme climate events disrupt maternal and child health service utilization, through which mechanisms, and in which contexts?** Each essay contributes a distinct but complementary component to answering this overarching question.

- **Essay 1** synthesizes the global literature on the effects of heatwaves and floods on MCH service utilization. Using a realist-informed systematic review, it identifies patterns of impact, contextual moderators, and recurring mechanisms across diverse settings.
- **Essay 2** builds on this evidence to develop a dynamic causal framework. Through the construction of a Causal Loop Diagram (CLD), it maps the reinforcing feedback processes through which floods and heatwaves affect health service access, demand, infrastructure, and system resilience.
- **Essay 3** provides original empirical evidence on the effects of the 2010 floods in Pernambuco and Alagoas, Brazil. Using administrative data and causal inference methods, it estimates the impact of floods on antenatal care utilization and explores heterogeneous effects across levels of baseline adherence.

The essays demonstrate that climate shocks exacerbate pre-existing vulnerabilities, disproportionately affect women with lower access to care, and generate systemic disruptions that extend beyond immediate infrastructure damage. The thesis therefore contributes not only to the empirical literature on climate and health, but also to the economic analysis of health system resilience. By integrating qualitative synthesis, systems thinking, and quantitative causal analysis, it provides actionable evidence to inform adaptive, equity-oriented public policies aimed at safeguarding maternal and child health in the context of climate change.

Accordingly, the general objectives of the thesis are:

- **Essay 1:** To synthesize and critically assess the global evidence on the impacts of heatwaves and floods on maternal and child health service utilization.

- **Essay 2:** To identify and model the causal mechanisms through which extreme weather events affect maternal and child health service use.
- **Essay 3:** To estimate the causal effect of the 2010 floods on antenatal care utilization in Pernambuco and Alagoas, Brazil.

To achieve these objectives, the thesis is organized into three complementary essays, following this introduction (Chapter 1). Chapter 2 presents the realist-informed systematic review. Building on this foundation, Chapter 3 develops the theoretical framework through a causal loop diagram that formalizes the identified mechanisms. Chapter 4 presents the empirical analysis, providing causal evidence from Brazil using interrupted time series, and difference-in-differences methods. The thesis concludes in Chapter 5, which synthesizes the findings, discusses policy implications, and outlines directions for future research. A summary of each essay is provided in Table 1.1.

Table 1.1 – Summary of Thesis Essays

Essay 1	Realist-Informed Systematic Review
Research Question	What are the impacts of heatwaves and floods on maternal and child health (MCH) service utilization?
Hypothesis	Heatwaves and floods disrupt MCH service utilization by impairing both access to care and service delivery, leading to altered utilization patterns.
Method	Realist-informed systematic review.
Data Sources	MEDLINE, Embase, Web of Science, EconLit, Global Health, Global Index Medicus, and CINAHL.
Essay 2	Causal Loop Diagram
Research Question	Through which mechanisms do floods and heatwaves affect MCH service utilization, and how do these effects vary across socioeconomic and geographic contexts?
Hypothesis	Floods and heatwaves influence MCH service utilization through multiple interrelated mechanisms that vary across local contexts.
Method	Causal Loop Diagram (CLD).
Data Sources	Evidence synthesized in Essay 1.
Essay 3	Empirical Analysis
Research Question	What is the causal effect of the 2010 floods on antenatal care utilization in Pernambuco and Alagoas, Brazil?
Hypothesis	Flood exposure leads to a measurable decline in antenatal care utilization, reflecting disruptions in service provision and changes in health-seeking behavior across affected municipalities.
Method	Interrupted time series, and difference-in-differences analysis.
Data Sources	SINASC (Live Birth Information System) and S2iD (Integrated Disaster Information System), 2008–2012.

2 REALIST-INFORMED SYSTEMATIC REVIEW

ABSTRACT

Background: Heatwaves and floods are increasing globally, posing risks to healthcare systems and disrupting maternal and child health (MCH) service utilisation. Understanding whether, how, and in which contexts floods and heatwaves affect MCH is essential for effective adaptation planning.

Methods: We conducted a realist-informed systematic review across seven databases, identifying 5,280 records. After dual screening, 36 studies met inclusion criteria, reporting the effects of floods and/or heatwaves on routine and emergency MCH service utilisation (23 floods, 12 heatwaves, 1 both). No restrictions were applied by language, setting, or study design. Findings were synthesised narratively.

Results: Evidence for heatwaves is stronger for emergency and inpatient care, including maternal hospitalisations, obstetric emergencies, neonatal intensive care admissions, and under-five emergency department visits, while evidence on routine care is very limited. In contrast, floods are associated with reduced routine services, including antenatal care, institutional delivery, and postnatal care; floods also impede access to emergency obstetric care, leading to delayed referrals and preventable deaths.

Conclusion: Floods and heatwaves affect MCH service utilisation through shared and hazard-specific pathways shaped by contextual vulnerability. Key gaps remain, including limited qualitative evidence on heatwaves, scarce research on child health services beyond acute care, and uneven geographic coverage. These findings can inform targeted strategies to strengthen climate-resilient MCH systems.

Keywords: Heatwaves. Floods. Maternal health. Child health. Healthcare utilisation. Health systems resilience. Emergency planning.

JEL: I18 (Health: Government Policy; Regulation; Public Health), J13 (Fertility; Family Planning; Child Care; Children; Youth), Q54 (Climate; Natural Disasters and Their Management; Global Warming).

HIGHLIGHTS

- First realist-informed review linking heatwaves and floods to MCH service use.
- Heatwaves are linked to emergency hospital use; floods disrupt routine MCH services.
- Evidence is stronger for maternal than child use; few causal or dual-hazard studies.
- Mechanisms are shaped by context, including health systems, socioeconomy and climate.

2.1 INTRODUCTION

The objective of this essay is to explore whether, how, and in which contexts heatwaves and floods affect the utilization of maternal and child healthcare services. By mapping the current evidence, identifying key pathways, and emphasizing the contextual variability of these effects, this review seeks to inform future policies and interventions to mitigate the adverse economic and human capital impacts associated with climate-induced disruptions in maternal and child healthcare.

A heatwave refers to a period during which an area experiences excessive and prolonged heat, marked by several unusually hot days in a row. These events involve persistently high maximum and minimum temperatures, which may extend from a few days to even months, depending on the region. Both daytime and nighttime temperatures are critical because when nights remain hot, the body has less chance to recover, and extreme temperatures tend to occur earlier in the day and persist for longer periods (World Meteorological Organization, 2026a).

Floods are natural disasters that occur when an overflow of water submerges land that is usually dry. The three most common types of floods are flash floods (caused by rapid, heavy rainfall), river floods (caused by prolonged rain or snowmelt overflowing rivers), and coastal floods (caused by storm surges or tsunamis) (World Meteorological Organization, 2026b).

Examples of service utilization for mothers and women of reproductive age include antenatal check-ups, institutional delivery care (for normal and complicated births), postnatal follow-up visits, and family planning services. Care for newborns and children under five years comprises routine health screening, essential nutrition services, and regular vaccinations. Thus, MCH utilization is the extent to which healthcare services designed to improve the health outcomes of mothers and children are accessed and used by the target population (Sarikhani, Najibi e Razavi, 2024).

The intersection of climate factors and human health has gained prominence in recent years, prompting a meticulous examination of how these changes affect vulnerable groups (World Health Organization, 2023; Romanello et al., 2025). Pregnant women and children are especially at risk because of their heightened physiological and developmental vulnerabilities (Rylander, Odland e Sandanger, 2013; Kuehn e McCormick, 2017; Roos et al., 2021), which amplify both direct and indirect health consequences during extreme weather events.

The impacts of adverse shocks in early childhood include an increased likelihood of developing chronic illnesses (such as diabetes, hypertension, and cardiovascular diseases), greater incidence of mental health disorders (like depression and anxiety), reduced cognitive performance, physical limitations, and a higher risk of premature mortality in adulthood (Carrets, 2021). Exposure to environmental shocks during early childhood (from the prenatal period through age five) can lead to long-lasting and potentially irreversible impacts across the life cycle by disrupting fetal development, impairing cognitive and physical health, reducing educational attainment, and diminishing labor market outcomes in adulthood, with effects compounded over time through biological damage, reduced parental investment, and socioeconomic vulnerability

(Currie et al., 2014; Carrets, 2021).

Floods can increase the likelihood of adverse birth outcomes, pregnancy complications, and mortality (Mallett e Etzel, 2018), while heatwaves can overwhelm thermoregulatory mechanisms, leading to higher maternal morbidity and mortality (Chersich et al., 2023). Moreover, heat exposure during pregnancy has been linked to increased risk of miscarriage, stillbirth and preterm birth (Kuehn e McCormick, 2017; Chersich et al., 2020; Rocque et al., 2021).

Amid these clinical risks, emerging evidence also highlights disruptions in accessing routine maternal and child healthcare services. For instance, Baharav et al. (2023) report a marked increase in neonatal intensive care unit admissions during extreme heat events, and Conway et al. (2024) show that elevated temperatures in sub-Saharan Africa can deter pregnant women from seeking care, especially where travel distances to health facilities are long.

Flood-related displacement similarly jeopardizes healthcare utilization, as uprooted populations may lack the infrastructural support, financial resources, and local networks necessary for receiving antenatal or postnatal care (Daalen et al., 2021; Safajou, Nahidi e Ahmadi, 2023). Further exacerbating this situation, women of reproductive age lose access to essential reproductive health services when floods undermine health system infrastructure, particularly in contexts with limited policy frameworks and inadequate data collection to guide interventions (Safajou, Nahidi e Ahmadi, 2023). Despite these indications of service disruption, few studies have systematically examined how climate-related events lead to reduced access to and utilization of maternal and child healthcare and a comprehensive synthesis is still lacking (Meherali et al., 2024).

This gap in the literature is problematic for several reasons. First, understanding the “how” and “why” behind service disruptions is crucial for designing effective adaptation strategies and public policies, as MCH systems are critical to future human capital and carry significant economic implications (Jamaluddin, 2024; Ekinici, 2025). Second, most research so far has concentrated on specific regions (primarily North America and Asia) while underexploring variation across diverse socioeconomic and infrastructural contexts (Meherali et al., 2024). Third, many studies have focused on clinical outcomes rather than service utilization, leaving a critical gap in our comprehension of the mechanisms by which extreme weather events disrupt MCH services (Segal e Giudice, 2022; Dalugoda et al., 2022; Veenema, Hoepner e Geer, 2023).

To address this gap, this essay provides a systematic literature review with a rigorous and transparent method to identify, select, evaluate, and synthesize existing evidence (Page et al., 2021). However, a traditional systematic review might overlook the deeper contextual and mechanistic drivers of healthcare disruptions. Consequently, this article employs a realist-informed systematic review, which integrates the strengths of a systematic approach with a focus on understanding the mechanisms and contexts underlying observed outcomes (Pawson et al., 2005). Unlike a full realist review, this method strikes a balance between empirically synthesizing existing studies and exploring the “how” and “why” behind the impacts of heatwaves and floods on maternal and child healthcare.

Accordingly, beyond this Introduction (Section 2.1), the essay is structured as follows.

Section 2.2 presents the methodological approach, including the initial theoretical framework, search strategy, inclusion and exclusion criteria, data extraction procedures, and methods of analysis and synthesis. Section 2.3 describes the characteristics and design of the included studies, assesses their quality, and summarizes the main findings. Section 2.4 interprets these findings, and Section 2.5 offers the concluding remarks.

2.2 METHODS

This section outlines the methods used to examine how floods and heatwaves affect the utilisation of maternal and child healthcare services. We conducted a realist-informed systematic review that combined rigorous procedures for identifying, selecting, appraising, and extracting evidence with a theory-driven analysis of the contexts and mechanisms shaping the observed outcomes. The subsections describe the initial theoretical framework, search strategy, study selection and appraisal, data extraction, and evidence synthesis.

2.2.1 Realist-informed systematic review

Conventional systematic reviews and meta-analyses excel at estimating average effects, yet their conclusions can be unstable when effects vary across settings, populations, and health systems. Complexity-oriented frameworks contend that policy-relevant evidence must model interdependence, non-linearity, and context–mechanism interactions rather than merely quantify mean effects (Rutter et al., 2017). A Realist-Informed Systematic Review (RISR) addresses this challenge by combining the procedural rigour of systematic reviewing (transparent searching, explicit eligibility criteria, critical appraisal) with realist logic, which asks what works, for whom, in what contexts, through which mechanisms, and why (Pawson et al., 2005; Wong et al., 2013; Wong et al., 2016). In practice, RISR retains systematic processes for searching, selection, appraisal, and extraction, while structuring questions, data coding, synthesis, and reporting around realist constructs and theory refinement. A RISR draws on scientific/critical realism¹, which assumes that social programs work (or not) through underlying causal mechanisms whose activation depends on contexts. A Context–Mechanism–Outcome (CMO) configuration is the basic “unit” of realist explanation: in context C, mechanism M is triggered, producing outcome O (Pawson et al., 2005; Wong et al., 2013).

Mechanisms (M) are the generative causal processes that produce change when triggered in a given context. A widely used operationalisation is resources + reasoning: programmes introduce resources (e.g., tools, opportunities, constraints) that interact with actors’ reasoning (e.g., beliefs, norms, capabilities), yielding change. Distinguishing mechanisms from programme

¹ Scientific/critical realism is a philosophy of science which holds that a mind-independent world exists (ontological realism). Reality is stratified into the real (underlying structures and generative mechanisms), the actual (events those mechanisms generate), and the empirical (what we observe). Our knowledge is theory-laden and fallible (epistemic relativism), yet we can still judge between explanations by how well they account for the evidence across settings (judgmental rationality). See Bhaskar (1975) - *A Realist Theory of Science*, and Pawson & Tilley (1997) - *Realistic Evaluation*, for foundational overviews.

activities (typically treated as part of the context) is essential for valid causal explanation (Dalkin et al., 2015; Shaw et al., 2018).

Context (C) comprises the conditions that enable or constrain mechanisms: organisational structures, resources, governance, incentives, professional norms, sociodemographics, place, and timing. Context is not mere “setting”; it denotes the features that shape mechanism activation, which may be conceived as relatively static triggers/constraints or as dynamic, relational, multilevel processes evolving over time, two narratives with distinct implications for review design, aims, and inference (Greenhalgh e Manzano, 2022).

Outcomes (O) are the observable consequences of mechanism activation in each context - intended or unintended, beneficial or harmful, and occurring at proximal, intermediate, or distal points in time. In realist synthesis, outcomes are often conceptualized as chains: early shifts (e.g., in actors’ reasoning or behaviour) can cumulate into later organisational or system-level effects or be dampened when contextual features inhibit the mechanism (Pawson, 2006).

RISR seeks to build an interlocking set of CMO configurations that together constitute a programme theory of how an intervention or phenomenon generates its effects. Reviews typically articulate initial programme theories (IPTs) during scoping and iteratively refine them via testing, confirmation, refutation, and elaboration into middle-range theories that “travel” across settings while specifying boundary conditions (Pawson, 2010; Wong et al., 2013). Because mechanisms are contingent, realist reviewers expect demi-regularities (recurrent but non-universal patterns) rather than uniform effects, and they use these patterned regularities to strengthen explanatory propositions (Pawson, 2006).

Realist synthesis is deliberately method-plural: instead of ranking designs, it values any study that helps test or refine C–M–O explanations, judged on relevance and rigour. Quantitative studies map patterns and boundaries, qualitative/process studies reveal how people reason and which contextual features matter, and mixed-methods link the two; reviewers then weave this evidence to compare and strengthen CMO explanations (Wong et al., 2013).

Although often presented linearly, realist-informed reviews are iterative and theory-driven, usually following a six steps process (Wong, 2018; Pawson, 2006):

- a) Locate existing theories to develop the IPTs (drawing on conceptual literature, prior evaluations, and stakeholder expertise).
- b) Search for data using comprehensive and purposive, theory-refining strategies (often with an information specialist).
- c) Select documents based on relevance (does the source inform C, M, O, or the links between them for theory building/testing?) and rigour (are the inferences warranted given the methods and data?).
- d) Extract and organise data with templates that code for C, M (resources/ reasoning), O, and CMO linkages; log rival mechanisms and contextual contingencies.

- e) Analyse and synthesise using retroductive reasoning, i.e., moving between data and theory to infer the most plausible underlying mechanisms and enabling/ constraining contexts; adjudicate between rival CMOs on explanatory coherence and reach.
- f) Test and refine programme theory into transferable middle-range theory, making explicit the contexts under which mechanisms are likely (or unlikely) to produce desired outcomes.

This study follows the six steps for conducting an iterative realist-informed review as described by Pawson (2006) and Wong (2018) and implemented by Borghi et al. (2018) and Singh et al. (2021). The systematic approach follows the guideline of the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA 2020), as proposed by Page et al. (2021). The review protocol was published in PROSPERO (Silva et al., 2024).

2.2.2 Initial theoretical framework

We began by developing an initial programme theory. Based on the literature gap, we identified two climate hazards (heat waves and floods) and one set of outcomes (utilization of MCH services). Our initial hypothesis is that heat waves and floods disrupt the utilization of maternal and child healthcare (MCH) services by creating both supply- and demand-side barriers. On the supply side, these events can undermine service availability (for example, infrastructure destruction or staff shortages), while on the demand side, they can affect affordability (income loss or increased out-of-pocket costs) and acceptability (safety concerns or norms around seeking care). Building on this understanding, we hypothesize the pathways through which exposure to floods and heatwaves may shape MCH utilization. These pathways were identified through discussion with experts on the effects of flood/heat effects on maternal and child health and health systems, and constitute an initial theoretical framework.

We hypothesize that these pathways rarely operate in isolation. For instance, displacement and migration can reduce household income (thereby affecting affordability) while also overburdening healthcare services in receiving areas and exacerbating supply-side constraints. Similarly, damage to transportation networks limits the ability to travel for both providers and patients, fueling a cycle in which decreased availability of care further dampens demand. Contextual factors - such as urban or rural location, degree of remoteness, health system decentralization, staffing and funding levels, and overall income level - further determine the severity and nature of these disruptions.

2.2.3 Search strategy

The ideal search strategy combines the database's controlled vocabulary (e.g., Medline's MeSH terms) with text words from the articles. It's crucial to use both medical and lay terms, synonyms, varied word forms, plurals, and abbreviations. The truncation symbol (*) captures different endings of a root word (e.g., mone* = money, monetary, monetization). Based on the

question's components, relevant terms and synonyms that increase search sensitivity are identified and combined using Boolean operators ("AND," "OR," "NOT"). Here, "AND" narrows results to those containing all terms, while "OR" broadens results to articles containing any of the terms.

We systematically searched the following electronic bibliographic databases: Medline (via Ovid), Embase, Web of Science, EconLit, Global Health, Global Index Medicus and CINAHL. We combined search terms and controlled vocabulary related to (Heat Waves OR Floods) AND Maternal and Child Health AND (Healthcare Utilization OR Healthcare Affordability) in our search strategy. This strategy was initially developed for Medline and then adapted for the other databases. The complete search strategy was reviewed by an information specialist and is detailed in Appendix 2.A. We included studies from the earliest available year up to the date of the search (Feb 13th, 2024), considering publications from any country in any language and using electronic translation tools as needed.

2.2.4 Selecting and appraising documents

We applied predefined eligibility criteria to select studies for this review (Table 2.1).

We included primary research (qualitative, quantitative, or mixed methods) examining how floods and/or heatwaves affect maternal and child healthcare utilisation. Eligible populations were pregnant and postpartum women (up to six weeks post-delivery, according to recommended guidelines from World Health Organization (2010)) and children aged 0–5 years (or studies in which outcomes for this age group could be clearly isolated). Outcomes had to relate to the use of MCH services, including antenatal, delivery and postnatal care, routine child health services (e.g. vaccination, growth monitoring, nutrition visits), and emergency care.

We excluded studies that: (i) did not assess floods or heatwaves as the primary exposure; (ii) did not examine MCH service utilisation; (iii) focused on populations outside the defined maternal or early-childhood period; (iv) not based on primary research (e.g., commentaries, reviews, opinion pieces); or (v) lacked discussion of mechanisms/pathways or contextual factors linking exposure to outcomes. Study quality was considered during interpretation of the evidence, rather than used as a threshold for exclusion.

To minimize selection bias, twelve independent reviewers conducted title/abstract and full-text screening in pairs. Before screening, all reviewers were trained on the inclusion and exclusion criteria, and any disagreements within pairs were resolved by a third reviewer. The lead author coordinated pair assignments, training, compiled decisions, and cross-checked for overall consistency.

For title and abstract screening, we used ASReview, an open-source tool that uses machine learning to rank study relevance, to streamline the screening process due to the high number of references (Schoot et al., 2021; ASReview Team, 2024). Each review pair provided the same "prior knowledge" for training the algorithm, including two known relevant references (Baten et al., 2020; Xu et al., 2017) and one known irrelevant reference (Sungkyu e Jinyoung, 2020). All references were screened, with no stopping criteria, to ensure comprehensive coverage of

Table 2.1 – Eligibility Criteria for Study Selection

Category	Inclusion Criteria	Exclusion Criteria
Population	Pregnant women; postpartum women (up to 6 weeks); children aged 0–5 years; studies including broader age ranges when data for ages 0–5 are extractable.	Non-pregnant women; postpartum period beyond 6 weeks; children over 5 years; studies without age-disaggregated data.
Exposure	Heatwaves or extreme heat events; floods.	Other natural hazards or unrelated weather events.
Comparison	Not required (observational and qualitative designs included).	Not applicable.
Outcome	Utilization of maternal and child health (MCH) services, including antenatal care, delivery care, postnatal follow-up, hospital admissions, child health screenings, vaccinations, and cost-related or affordability dimensions of service use.	Studies not examining MCH service utilization (e.g., disease prevalence, clinical outcomes only, general health indicators).
Timing	Studies published up to the predefined search date.	Studies published after the search date.
Study Type	Qualitative, quantitative, and mixed-methods empirical studies.	Commentaries, opinion pieces, editorials, policy briefs, narrative reviews, or studies lacking empirical or contextual analysis.
Source	Peer-reviewed journal articles; theses and dissertations.	Inaccessible full texts or conference abstracts without sufficient methodological detail.
Geography	All geographic regions.	No geographic exclusions.
Language	Any language, provided translation was feasible.	Languages for which reliable translation was not feasible.

potentially relevant studies. EndNote 21 (Clarivate, 2013) was used for reference management and duplicate removal.

2.2.5 Extracting data

For each included study, we extracted key details into an electronic spreadsheet, including the study design (qualitative, quantitative, or mixed-methods), method, event type, study period, country, geographical scale (national or subnational), area of focus (urban or rural), study population, outcomes, mechanisms and context. For quantitative studies, we also extracted the measured effect sizes. The complete extraction table is available in Appendix 2.B.

2.2.6 Analysing and synthesising data

We assessed the methodological quality and risk of bias of the included studies using the Mixed Methods Appraisal Tool (MMAT), which is specifically designed for systematic reviews that incorporate qualitative, quantitative, and mixed-methods research (Hong, Gonzalez-Reyes e Pluye, 2018). Given the substantial heterogeneity in study designs, exposures, outcomes, contexts, and analytical strategies, a quantitative meta-analysis was not appropriate. Instead, we employed a narrative (descriptive) synthesis. Narrative synthesis is a well-established and recommended approach for integrating diverse forms of evidence, as it enables the identification of recurring patterns, underlying mechanisms, and contextual influences across studies (Popay et al., 2006).

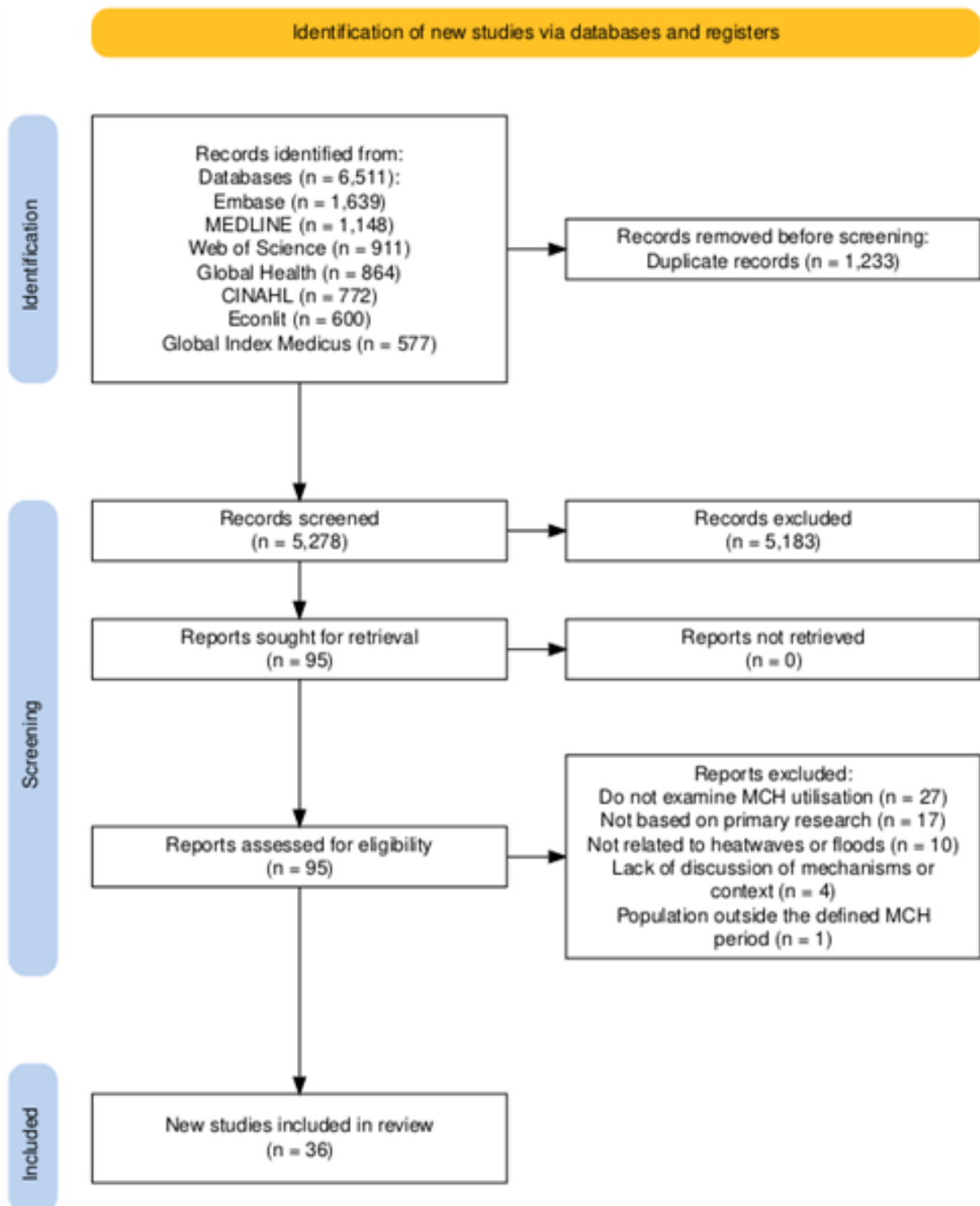
We first categorized the studies by methodological approach (quantitative, qualitative, or mixed-methods) and then grouped them according to the specific MCH outcomes examined (e.g., antenatal, delivery, and postnatal care), as well as by geographical and socioeconomic context (urban versus rural settings and high-income versus low- and middle-income countries). We extracted detailed methodological information, including statistical models used in quantitative analyses (e.g., logistic regression, difference-in-differences, Poisson and segmented regression, and distributed lag non-linear models), qualitative analysis techniques (e.g., thematic coding and framework analysis), and integration strategies applied in mixed-methods studies. The findings were then synthesized narratively to highlight common themes, methodological trends, and contextual variations, thereby providing a coherent and comprehensive account of the evidence.

2.3 RESULTS

This section presents the findings of the review, beginning with the study selection process and the methodological, geographical, and temporal characteristics of the included evidence. It then examines the effects of heatwaves and floods on routine and emergency maternal and child healthcare utilisation, before synthesising the main patterns, mechanisms, and contextual factors identified across studies.

Figure 2.1 shows the results of the systematic review process.

Figure 2.1 – PRISMA Flow Diagram



Source: This figure was generated by the author using the Shiny app developed by Haddaway et al. (2022)

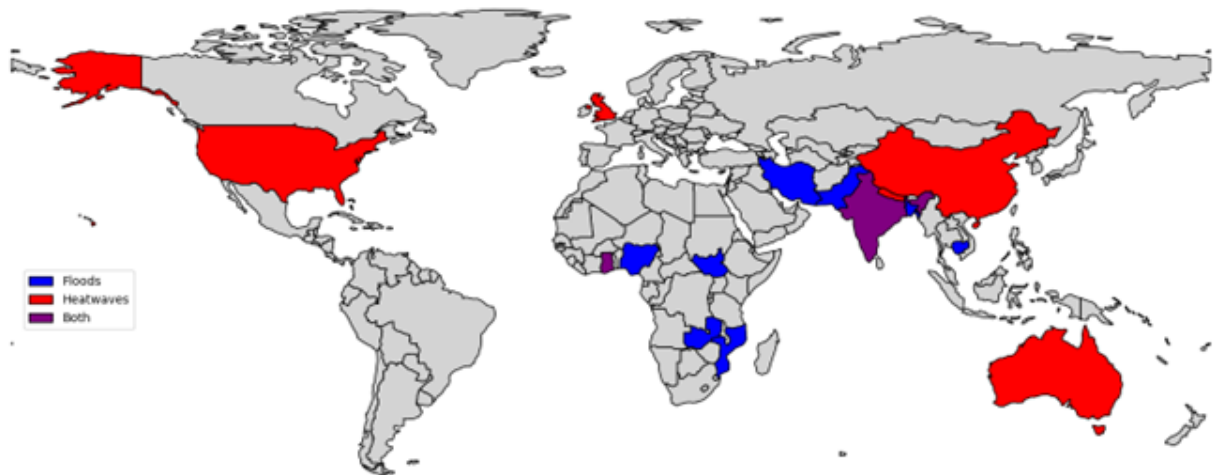
On February 13, 2024, our search strategy retrieved 6,511 records (1,148 from Medline [17.6%], 1,639 from Embase [25.2%], 864 from Global Health [13.3%], 600 from EconLit [9.2%], 911 from Web of Science [14.0%], 772 from CINAHL [11.9%], and 577 from Global Index Medicus [8.9%]). After removing 1,233 duplicates (18.9%), 5,278 references remained.

Of these, 5,183 (98.2%) were excluded during title and abstract screening, leaving 95 for full-text review. Among the full texts, 59 (62.1%) were excluded for the following reasons: no discussion of MCH service utilization or affordability (30), not based on primary research (16), unrelated to heat waves or floods (9), lacking discussion of mechanisms/pathways or contextual factors (3), or focusing on a population/age group outside our scope (1). The list of papers excluded by full-text eligibility is available in Appendix 2.C.

This process ultimately yielded 36 records for inclusion in the review: 23 on floods, 12 on heatwaves and 1 examining both hazards. The methodological quality of the 36 studies was strong across qualitative, quantitative, and mixed-methods designs, as assessed using the MMAT (Appendix 2.D). Flood-related studies were methodologically diverse, including quantitative (50%), qualitative (37.5%), and mixed-methods designs (12.5%). Heatwave studies, by contrast, were predominantly quantitative (92.3%). Only one study addressed both hazards using mixed-methods. The limited number of studies reflects the inherent challenges of disaster-related research, where randomisation is both unethical and infeasible. In the absence of randomisation, causal inference for climate-related hazards must rely on well-established quasi-experimental designs that exploit within-unit comparisons and exogenous temporal variation. However, in this body of literature, most evidence remains associational rather than causal.

Figure 2.2 shows the geographic distribution of the countries where the included studies were conducted.

Figure 2.2 – Flood and Heatwave Studies by Country



Source: Prepared by the author.

Note: Map lines only delineate study areas and do not necessarily depict accepted national boundaries.

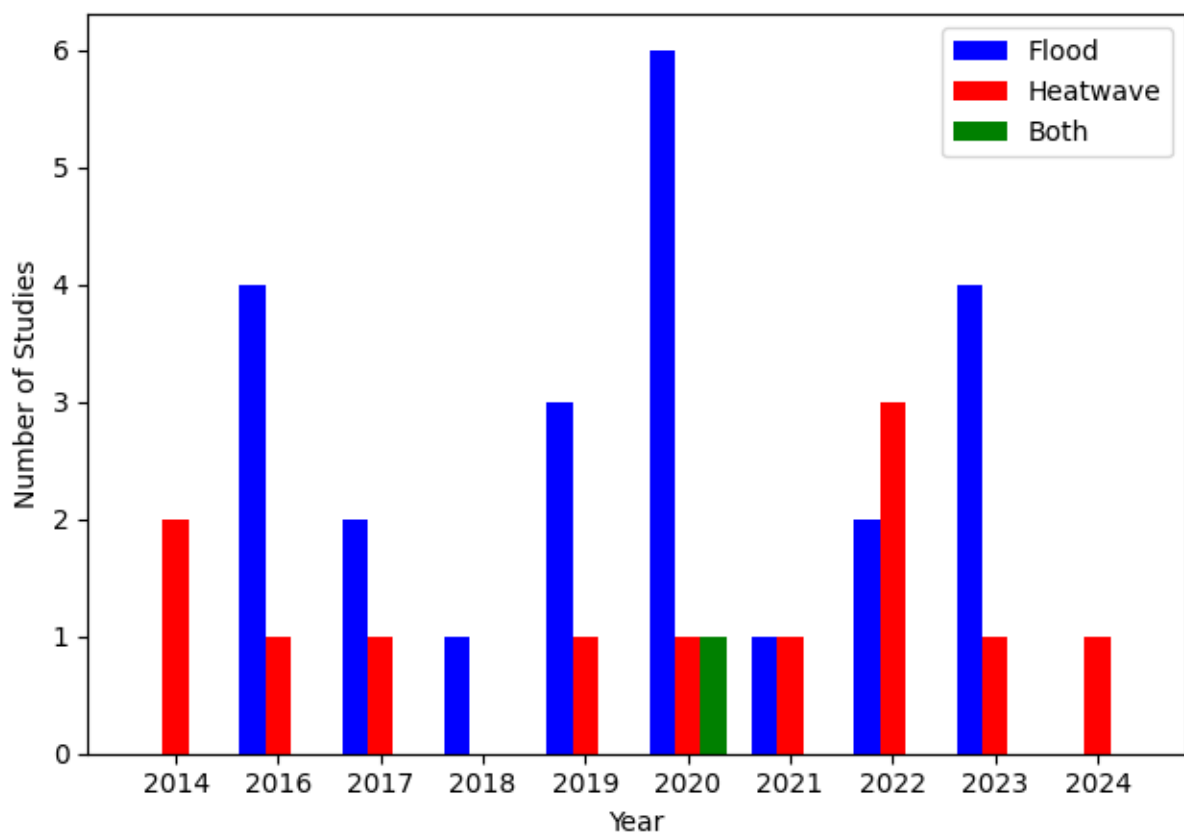
Flood studies were predominantly conducted in Asia (17 studies, 70.8%) and Africa (7 studies, 29.2%). Within Asia, Bangladesh accounted for the largest share with 9 studies (37.5%), followed by Pakistan with 3 studies (12.5%), and Cambodia and India with 2 studies each (8.3%). Iran contributed 1 study (4.2%). In Africa, Mozambique and South Sudan each featured in 2 studies (8.3%), while Nigeria, Ghana, and Zambia each contributed 1 study (4.2%).

Geographically, most studies were conducted at the subnational level (21 studies, 87.5%), with only 3 at the national level (12.5%). The majority focused on rural areas (14 studies, 58.3%), while 1 study (4.2%) focused on urban settings and 9 studies (37.5%) examined both rural and urban settings.

Heat studies were conducted across various regions, with 30.8% each in North America (United States, 4 studies) and Oceania (Australia, 4 studies). Asia accounted for 23% of the studies (China, Nepal, and India, 1 study each), while Europe (England, 1 study) and Africa (Ghana, 1 study) contributed 7.7% each. Geographically, 4 studies (30.8%) were conducted at the national level, and 9 studies (69.2%) focused on subnational contexts. Urban areas were the focus in 6 studies (46.15%), rural areas in 1 study (7.7%), and both rural and urban settings in 6 studies (46.15%). Most studies were conducted in high-income countries (HICs, 69.2%) - United States, Australia, and England - while 30.8% were from low- and middle-income countries (LMICs), including China, Nepal, India, and Ghana.

Figure 2.3 presents a bar chart illustrating the distribution of included studies by year of publication.

Figure 2.3 – Flood and Heatwave Studies by Year



Source: Prepared by the author.

Notably, 94% of the studies analyzed were published within the last 10 years (since 2016) and 61% within the last 5 years (since 2020), indicating that this is a rapidly evolving field.

2.3.1 Heatwaves

Heatwaves are commonly defined as extended periods of abnormally high temperatures lasting from several consecutive days to, in some contexts, even months, during which both maximum and minimum temperatures substantially exceed the typical range for a given location (World Meteorological Organization, 2026a). Importantly, however, the studies included in this review operationalised heat exposure using markedly different definitions, many of which do not strictly correspond to the meteorological definition of a heatwave. By definition, a “wave” requires at least two consecutive days of elevated temperatures; under this criterion, only five studies applied a duration-based heatwave definition (Campbell et al., 2019; Davis et al., 2020; Smith et al., 2016; Xu et al., 2014; Xu et al., 2017). The remaining studies instead focused on high-, hot-, or extreme-temperature days, rather than sustained heat events. We did not include studies focused on isolated high temperature (non-daily measures), as these would deviate too far from the conceptual definition of heatwaves.

Although this heterogeneity in exposure definitions is not ideal, we deliberately adopted broader inclusion criteria to capture the full scope of evidence linking daily heat exposure to MCH service use. Restricting inclusion to studies using strict heatwave definitions would have resulted in an evidence base too limited to support meaningful synthesis or inference. Temperature metrics also varied substantially across studies: some employed absolute thresholds, while others relied on relative, percentile-based measures; most used daily maximum or mean temperatures, and very few incorporated minimum temperatures. This diversity shows the lack of consensus in the literature regarding the most appropriate operationalisation of heatwave exposure, despite existing guidance from the World Meteorological Organization (2026a).

2.3.1.1 Impacts on routine care

Only one study examined the association between environmental heat and routine care use. Bishwakarma (2022) showed that in Nepal, each additional high-temperature day (daily mean temperature $\geq 32^\circ\text{C}$) during pregnancy is associated with a decline of 0.011–0.016 antenatal care visits per woman, on average, depending on model specification. While a one-day increase in heat exposure represents a very marginal change and thus yields a small per-day effect in a linear model, exposure is cumulative and baseline antenatal care utilisation is already low in this setting, where many women attend only 2–4 visits. As a result, even modest per-day effects translate into substantively meaningful declines (for example, 0.11–0.16 fewer visits per additional hot days), particularly given that missed visits tend to cluster among vulnerable groups and that heat exposure is persistent rather than isolated; thus, the effect is small on a daily basis but non-trivial at the population level.

Bishwakarma (2022) did not directly test mechanisms but interprets the decline in antenatal care use in light of the well-documented access constraints in rural Nepal, where most households are located and reaching a health post often requires long travel up to four hours,

frequently on foot. The author argued that high temperatures likely intensify these mobility barriers by increasing physical discomfort and heat-related stress, thereby discouraging routine antenatal visits. This interpretation is context-specific, applying primarily to remote rural settings where walking or prolonged travel is necessary, and is consistent with qualitative evidence showing that pregnant women are reluctant to walk long distances for care in hot conditions (Scorgie et al., 2023), suggesting that thermal stress amplifies existing access barriers rather than reducing care-seeking preferences.

2.3.1.2 Impacts on emergency care

More evidence supports a link between extreme heat and negative impacts on health, which are indicated by acute emergency hospital admissions. Kim, Lee e Rossin-Slater (2021) found that in the United States, each additional extremely hot day during pregnancy (≥ 3 standard deviations above the county-month mean temperature) is associated with approximately 1 additional prenatal hospitalization per 1,000 pregnancies (0.101 percentage points; 95% CI 0.036-0.166) and 0.7 additional emergency or urgent hospitalizations per 1,000 pregnancies (0.071 percentage points; 95% CI 0.024-0.118). These effects correspond to relative increases of 2.2% for any prenatal hospitalization and 2.3% increase for emergency or urgent hospitalizations, compared with baseline risks (sample means) of 45.3 and 30.8 hospitalizations per 1,000 pregnancies, respectively. Importantly, because the rise in hospitalizations was driven by emergency and urgent admissions rather than scheduled care, Kim, Lee e Rossin-Slater (2021) indicate that extreme heat triggers acute pregnancy-related complications (e.g., hypertension and excessive vomiting) reflecting deteriorating maternal health rather than increased healthcare access.

Quilty et al. (2024) reported that in the humid tropical zone of northern Australia (Northern Territory), the incidence of aeromedical obstetric retrievals was 49% higher in the hottest temperature quartile compared with the coldest quartile (IRR = 1.49, CI 1.13-1.96), while there was no association with temperature for obstetric retrievals in the arid zone. Quilty et al. (2024) suggests this result reflects heightened heat-related vulnerability of pregnant women combined with limited local obstetric capacity in remote communities, with persistent heat and high humidity worsening acute complications and increasing reliance on emergency evacuation.

Chen et al. (2022) further showed that in China, maternal hospitalization costs rise with cumulative extreme-temperature exposure before delivery, with larger estimated effects for longer pre-delivery windows: the coefficient increases from 0.027 for the prior 4 weeks ($\approx 2.7\%$, 95% CI 0.2–5.2%) to 0.070 for the prior 8 weeks ($\approx 7.0\%$, 95% CI 3.7–10.3%) when moving from no extreme-temperature days to extreme temperature every day in the window. The 4-week estimate implies roughly 0.096% higher costs per additional extreme-heat day, although these results reflect overall maternal inpatient expenditures rather than cause-specific obstetric admissions.

Evidence also links extreme heatwave exposure to hospitalisation associated with acute neonatal and infant morbidity. Cil e Kim (2022) estimated that, in the United States, each additional gestational day with an average temperature above 90°F increases neonatal intensive

care unit (NICU) admissions by 0.153 per 1,000 live births (95% CI: -0.023 to 0.329), implying approximately 1.5 additional NICU admissions per 1,000 live births for every 10 additional hot days during pregnancy; trimester-specific analyses suggest that second-trimester exposure is most strongly associated with this increase. Kakkad et al. (2014) found that, during the 2010 heatwave in Ahmedabad, India, NICU admissions increased sharply (24 vs. 8 in 2009 and 4 in 2011), with each 1 °C rise above 42 °C associated with a 43% increase (95% CI: 9.2-88%). In Brisbane, Xu et al. (2017) found that only intense heatwaves ($\geq$ 97th percentile) increased infant hospitalizations, with relative risk rising from 1.05 for 2 days (95% CI: 1.01-1.10) to 1.18 for 4 days (95% CI: 1.05-1.32), while no significant effects were observed for less intense heat events (defined by the 90th or 95th percentiles).

Multiple studies document higher rates of emergency department (ED) visits for children during extreme heat, particularly among those under 5 years old. Campbell et al. (2019) compared emergency department presentations occurring on severe/extreme heatwave days with presentations on non-heatwave days matched by day of week, month, and year, finding 19% higher odds (OR: 1.19; 95% CI: 1.04-1.36) of ED visits among children aged 0–5 during heatwaves in Tasmania. Similarly, Xu et al. (2014) found that ED admissions among under-5 children rose sharply at temperatures above 27°C in Australia, with more than a threefold increase in the risk of EDAs during heatwaves compared to moderate temperatures. In the United States, Davis et al. (2020) observed a 20% increase in ED visits among children aged 1–4 on heatwave days (OR = 1.20; 95% CI: 1.05-1.37), while Stowell et al. (2023) reported a small but non-significant increase in heat-related ED admissions among children aged 0-5 years (OR = 1.06; 95% CI: 0.93-1.20). Similarly, one study that looked at the impact of heat on primary care visits. Smith et al. (2016) reported that primary care (general practitioner; GP) consultations for heat illness in children under-5 were three times higher during the 2013 UK heatwave compared to non-heatwave years (IRR = 3.0; 95% CI: -7.5 TO 13.5), although the overall burden of heat was low.

2.3.2 Floods

Floods are hydro-meteorological event that occurs when water overflows and submerges land that is normally dry. The most common flood types include flash floods, caused by intense and short-duration rainfall; riverine (fluvial) floods, resulting from rivers exceeding their channel capacity; and coastal floods, typically driven by storm surges or tsunamis (World Meteorological Organization, 2026b).

Across the studies reviewed, the empirical evidence is overwhelmingly concentrated on riverine floodplain flooding, most often seasonal and monsoon-driven, reflecting the dominant hydrological reality of many low- and middle-income settings in South Asia and sub-Saharan Africa. Recurring riverine flooding is the primary focus for studies in Bangladesh (Abdullah et al., 2019; Baten et al., 2020; Ray-Bennett et al., 2021; Orderud et al., 2022; Purno et al., 2023), India (Barman, Sarif e Saha, 2023; Surekha, Anbazhagan e Kiran, 2020), Pakistan (Jacquet et al., 2016; Sadia et al., 2016), Nigeria (Amoo et al., 2018), South Sudan (Wilunda et al., 2016;

Wilunda et al., 2017), Zambia (Mroz et al., 2023), Cambodia (Saulnier et al., 2020; Saulnier et al., 2022), and Mozambique (Firoz et al., 2016; Makanga et al., 2017).

Several studies emphasise riverbank erosion as a core element of riverine flooding dynamics in Bangladesh, highlighting its role in land loss, displacement, and prolonged vulnerability (Haque, Parr e Muhidin, 2019; Haque, Parr e Muhidin, 2020a; Haque, Parr e Muhidin, 2020b). In contrast, only a limited number of studies explicitly address compound flooding, in which riverine floods are intensified by pluvial rainfall, flash flooding, or coastal storm surges, as documented in Pakistan (Chandir et al., 2023), northern Iran (Peyravi, Rahnavard e Marzaleh, 2020), and coastal Bangladesh (Shah, Miller e Mothabbir, 2019). Urban pluvial flooding is particularly underrepresented, appearing in only a single study focused on low-income urban communities in Ghana (Codjoe et al., 2020). Moreover, this body of literature largely overlooks coastal flooding as a standalone hazard, as well as chronic waterlogging and groundwater-driven inundation, despite the increasing importance of these flood types under climate change and sea-level rise.

2.3.2.1 Impacts on routine care

Antenatal care (ANC) and delivery care are the most frequently reported outcomes in the literature on floods. Floods were associated with a reduction in ANC utilisation, particularly among displaced populations (Baten et al., 2020; Haque, Parr e Muhidin, 2020a). Repeated displacement exposure over recent years, including before or between pregnancies, was associated with substantially reduced ANC use, as women displaced three or more times had lower odds of receiving any ANC (OR = 0.32; 95% CI: 0.17-0.64) and of completing four or more ANC visits (OR = 0.38; 95% CI: 0.18-0.78), while one or two displacements showed no significant effect (Haque, Parr e Muhidin, 2020a). However, Orderud et al. (2022) found no significant average effects of flood exposure on antenatal or delivery care, except for a small decline in at least one ANC visit 25–36 months post-flood (-2.1%; 95% CI: -4.2% to -0.1%). In Nigeria, ANC coverage among displaced women fell from 82.4% pre-flood to 53.3% post-flood due to transport and income loss (Amoo et al., 2018). While most studies highlight negative effects, Baten et al. (2020) found that women in flood-affected areas were more likely to complete four or more ANC visits (adjusted OR = 1.20; 95% CI: 1.04–1.38), likely reflecting mitigation through humanitarian support, but repeated flooding reduced ANC use, with women in twice-flooded areas having substantially lower odds of adequate ANC compared with once-flooded areas (OR = 0.57; 95% CI: 0.46–0.70). In Rumbek North (South Sudan), floods during the wet season make travel to ANC unsafe or impossible and can disrupt supplies, so attendance drops most in remote villages and cattle camps where distance, lack of transport, insecurity, domestic burdens, and husbands' restrictions compound the access shock (Wilunda et al., 2017).

Across studies, floods are linked to lower rates of institutional delivery and skilled birth attendance (Sadia et al., 2016; Baten et al., 2020; Haque, Parr e Muhidin, 2020b; Orderud et al., 2022). In Pakistan, only 41.9% of births were attended by a skilled provider six months after the major 2010 floods, with access especially poor for rural and low-income households (Jacquet

et al., 2016). In two union councils in Nowshera that were among the areas most affected by Pakistan's 2010 floods, a cross-sectional household survey conducted in September 2013–January 2014 found very low maternal health service use (ANC 45%, skilled birth attendance 26%, facility delivery 19.7%); however, these figures are descriptive and cannot be attributed causally to flooding, as they may also reflect pre-existing under-resourcing and socioeconomic disadvantage (Sadia et al., 2016). In Rumbek North (South Sudan), floods during the rainy season render roads impassable and paralyse referral systems, forcing planned facility births to revert to home or cattle-camp deliveries and leading to reported maternal and newborn deaths when women must be carried “on shoulders” and complications cannot be referred (Wilunda et al., 2016).

Flood exposure also reduces postnatal care (PNC). In Bangladesh, women in flood-affected areas were 24% less likely to receive maternal PNC (OR 0.76, 95% CI: 0.67–0.87) and their babies 18% less likely to receive newborn PNC (OR 0.82, 95% CI: 0.72–0.93); repeated floods further worsened these odds (once- vs four-times flooded: maternal PNC OR 0.43 [0.32–0.59]; newborn PNC OR 0.41 [0.31–0.55]) (Baten et al., 2020). Among households displaced by flooding three or more times, neonates had 63% lower odds of receiving PNC from a trained provider (OR 0.37, 95% CI: 0.15–0.92) (Haque, Parr e Muhidin, 2020b). A national analysis for Bangladesh found no average association between district-level flood exposure on maternal care overall once socioeconomic/demographic differences were controlled for (and did not report PNC specifically), but subgroup analyses restricted to women who had lived 8 or more years in the same area (to investigate whether migration could affect results) showed some negative associations: flood exposure 25–36 months before birth was linked to lower likelihood of skilled birth attendance (OR –0.026; 95% CI: –0.051, –0.001) and facility delivery (OR –0.032; 95% CI: –0.059, –0.006), consistent with potential harms from specific floods in particular contexts despite a null national-average effect (Orderud et al., 2022).

Floods can disrupt routine child health services. Childhood immunization coverage dropped by 19% following a flood disaster in Pakistan, leading to approximately 11,500 missed doses (Chandir et al., 2023). Treatments for childhood pneumonia and diarrhoea fell by 30% during flood months compared to pre-flood months in rural Bangladesh due to clinic closures and supply chain breakdowns (Shah, Miller e Mothabbir, 2019). Healthcare-seeking behaviours is low among flood-displaced families: 15.5% of ill children received no care or only home treatment, and among those taken for care outside the home, treatment by a trained provider was less common for displaced than non-displaced children (10.3% vs 35.0%), consistent with a lower likelihood of using a trained provider (OR 0.44; 95% CI 0.26–0.73) (Haque, Parr e Muhidin, 2019).

Sexual and reproductive health and rights (SRHR), including family planning and post-abortion care, are often neglected during crises, even though disruptions can rapidly translate into preventable morbidity and mortality (Purno et al., 2023; Ray-Bennett et al., 2021). Floods disrupt already fragile SRHR infrastructure, impeding short- and long-term access to essential services. Evidence from Bangladesh suggests these gaps can be mitigated: the Reproductive Health in

Crises Consortium (RHCC) package (emergency medical kit + staff training + community awareness) improved provider capacity and post-abortion care availability/quality across three flood-prone primary health care facilities during the 2017 floods (Ray-Bennett et al., 2021), while a post-2022 flood programme deploying trained midwives to remote union centres (with supportive supervision, logistics, and community volunteers) delivered high volumes of SRHR care and increased service delivery versus the same period in the prior year (Purno et al., 2023).

2.3.2.2 Impacts on emergency care

Floods were found to impede access to hospital-based maternal care for emergency obstetric needs (Makanga et al., 2017; Mroz et al., 2023). Abdullah et al. (2019) documented maternal deaths during flood periods that were directly linked to flood-induced transport disruptions and community isolation, with women dying on route to facilities due to prolonged delays caused by reliance on boat transport and impassable roads. Flooded roads, high transportation costs, and a lack of ambulances hindered timely referrals (Shah, Miller e Mothabbir, 2019; Peyravi, Rahnavard e Marzaleh, 2020). Midwives played a crucial role in stabilizing emergencies and initiating referrals (Purno et al., 2023).

2.3.3 Narrative synthesis

Across 36 included studies, floods and heatwaves disrupted MCH service utilisation through hazard-specific pathways and context-dependent mechanisms. Evidence on extreme heat was predominantly centred on acute care, with multiple studies reporting increases in emergency department visits and inpatient utilisation for pregnant women and infants during high-temperature days and heatwaves (Campbell et al., 2019; Davis et al., 2020; Xu et al., 2014; Smith et al., 2016; Stowell et al., 2023). These patterns were most frequently documented in urban, high-income settings, where heat island dynamics may intensify exposure and where hospital-based administrative datasets allow timely detection of acute events (Chen et al., 2022; Codjoe et al., 2020).

Evidence on heat impacts on *routine* maternal care was limited: only one study explicitly examined missed ANC visits and delayed care-seeking during heat events (Bishwakarma, 2022). Across the studies that addressed care delivery constraints, facility-level conditions appeared to shape heat-related risks and service safety, including high indoor temperatures, unreliable cooling or power supply, and overcrowding affecting obstetric and neonatal care (Kakkad et al., 2014; Codjoe et al., 2020). Household resources and social stratification also conditioned vulnerability, including unequal access to cooling and patterns of racial and socioeconomic disadvantage in exposure and care-seeking (Cil e Kim, 2022; Kim, Lee e Rossin-Slater, 2021; Chen et al., 2022). A small subset of studies described adaptations such as heat-adapted service design, community outreach, and aeromedical retrieval, which supported continuity of safe care in specific contexts (Kakkad et al., 2014; Quilty et al., 2024).

In contrast, flood-related evidence more consistently reported reductions in routine MCH services (particularly ANC, institutional delivery, postnatal care, and childhood immunisations) often linked to disrupted transport networks, impaired referral pathways, and displacement (Makanga et al., 2017; Mroz et al., 2023; Baten et al., 2020). These barriers were most acute in rural floodplains, where sparse facility distribution and constrained mobility already limited access (Makanga et al., 2017; Barman, Sarif e Saha, 2023). Several studies described delayed recovery of service coverage after floods due to staff shortages, supply disruptions, and temporary relocations (Shah, Miller e Mothabbir, 2019; Peyravi, Rahnavard e Marzaleh, 2020; Chandir et al., 2023). Reported adaptations included mobile clinics, redistribution of midwifery capacity, cash transfers, service provision on higher ground, and transport solutions (boats, 4×4 vehicles, aircraft) to mitigate delays in emergency obstetric care (Ray-Bennett et al., 2021; Purno et al., 2023; Saulnier et al., 2022; Wilunda et al., 2016; Quilty et al., 2024).

Across both hazards, studies suggested that disruption patterns were conditioned by baseline system capacity and social vulnerability, including differential access by socioeconomic status, gender norms, and household resources, as well as variation in the governance and functioning of referral pathways (Haque, Parr e Muhidin, 2019; Amoo et al., 2018; Firoz et al., 2016; Cil e Kim, 2022; Kim, Lee e Rossin-Slater, 2021).

2.4 DISCUSSION

This realist-informed systematic review advances the literature by moving beyond documenting associations between extreme weather events and MCH outcomes to explicitly theorising how floods and heatwaves disrupt service utilisation through context–mechanism–outcome (CMO) configurations. The urgency of strengthening such causal understanding is underscored by major scientific assessments: climate change is already increasing hazardous heat and heavy precipitation extremes, disproportionately affecting vulnerable populations and placing growing pressure on health systems (Pörtner et al., 2022; Romanello et al., 2025). Within this risk landscape, MCH services are particularly sensitive to disruptions in mobility, facility readiness, and household constraints, because continuity of preventive and time-sensitive care (ANC, delivery, PNC, and immunisation) depends on functioning referral pathways, reliable access, and trusted service delivery.

Prior reviews have demonstrated adverse effects of extreme weather on maternal health in LMICs (Pappas, Kovats e Ranganathan, 2024) and on healthcare utilisation in rural and remote settings (Dewi et al., 2024), but they largely synthesise outcomes or broad themes without systematically mapping the causal pathways that connect hazards to health system disruption. In parallel, the heat-in-pregnancy evidence base has been synthesised mainly around biomedical outcomes (e.g., preterm birth, stillbirth, neonatal complications), including recent large-scale meta-analytic work (Lakhoo et al., 2025) and studies focusing on early-life growth (Bonell et al., 2024). These syntheses strengthen confidence in heat-related risks, yet they provide limited insight into service continuity and system-level disruption. Reviews of MCH services during emergencies

and disasters often aggregate heterogeneous shocks (pandemics, conflicts, and natural hazards) and therefore tend not to distinguish hazard-specific mechanisms or the conditions under which disruptions persist (Sahoo et al., 2021). Finally, broader syntheses of extreme weather impacts on health services highlight that service disruption remains under-synthesised relative to individual health outcomes (Valente et al., 2026).

Flood impacts predominantly operate through structural barriers to access and continuity: damaged roads, disrupted transport networks, impaired referral pathways, and displacement undermine routine care-seeking and service delivery. This mechanism profile is consistent with the stronger and more consistent evidence of declines in ANC, institutional delivery, PNC, and immunisation coverage observed in flood-affected settings (Makanga et al., 2017; Baten et al., 2020; Chandir et al., 2023). Importantly, the flood pathway is not necessarily short-lived: several studies suggest recovery lags well beyond physical repairs due to staff shortages, supply-chain disruptions, and temporary relocation of services (Shah, Miller e Mothabbir, 2019; Peyravi, Rahnavard e Marzaleh, 2020; Chandir et al., 2023). From a health economics perspective, floods produce both direct constraints (physical access) and indirect constraints (income shocks, displacement, heightened opportunity costs), which can amplify delays and increase preventable morbidity and mortality (Haque, Parr e Muhidin, 2019; Amoo et al., 2018; Firoz et al., 2016).

In contrast, extreme heat more frequently manifests as increased acute service utilisation via heightened maternal and neonatal risk, particularly in urban and high-income contexts where hospital administrative data are readily available and heat island dynamics intensify exposure (Campbell et al., 2019; Davis et al., 2020; Chen et al., 2022; Codjoe et al., 2020). However, the limited evidence on routine maternal care during heat events represents a critical blind spot. This asymmetry likely reflects both methodological convenience (hospital datasets capturing acute episodes) and a research emphasis on biomedical outcomes rather than continuity of primary care. The single identified study on missed ANC during heat events suggests that, in settings with weak transport and limited household adaptation capacity, heat may also act as a barrier to routine care, but the current evidence base is insufficient to characterise this pathway robustly (Bishwakarma, 2022).

Across both hazards, facility conditions emerge as an intermediate layer shaping utilisation and outcomes. High indoor temperatures, unreliable cooling and power, and overcrowding compromise the delivery and safety of obstetric and neonatal care (Kakkad et al., 2014; Codjoe et al., 2020). This distinction matters for interpretation: increased acute utilisation under heat can reflect increased need (risk exposure) rather than improved access, while declines in routine utilisation under floods can reflect barriers to both demand and supply.

Health system resilience theory provides a coherent lens for integrating these findings and clarifying policy-relevant levers. Following Kruk et al. (2015), resilience is the capacity of health actors, institutions, and populations to prepare for, respond to, and recover from shocks while maintaining core functions. Blanchet et al. further conceptualise resilience as absorptive, adaptive, and transformative capacity under stress (Blanchet et al., 2017). Within our evidence

base, many responses (mobile outreach, redeployment of staff, alternative transport, temporary service relocation) primarily reflect absorptive and adaptive capacities (Ray-Bennett et al., 2021; Purno et al., 2023; Saulnier et al., 2022; Wilunda et al., 2016; Quilty et al., 2024). Where floods are repeated or severe and baseline infrastructure and governance are weak, these capacities appear insufficient, resulting in sustained declines in service coverage and quality (Barman, Sarif e Saha, 2023; Baten et al., 2020; Chandir et al., 2023).

Critically, resilience is not value-neutral. The resilience literature cautions that “coping” can transfer costs to households and frontline workers and can obscure equity trade-offs if not explicitly addressed (Witter et al., 2023). This critique aligns with our findings: disruptions and adaptations are consistently conditioned by social vulnerability (unequal access to cooling, differential mobility constraints, gendered decision-making), shaping who bears the burden of shocks and who retains access to safe care (Cil e Kim, 2022; Kim, Lee e Rossin-Slater, 2021; Haque, Parr e Muhidin, 2019; Firoz et al., 2016). A realist, equity-centred resilience framing therefore supports a dual focus: maintaining service availability and readiness (supply-side resilience) while reducing household-level constraints and opportunity costs that shape care-seeking under shock (demand-side resilience).

The heterogeneity in exposure measurement and study design reflects the fragmented state of climate–health evidence. Flood exposure is quantified using diverse approaches (remote sensing, administrative databases, self-reported exposure), while heatwave definitions vary widely (absolute thresholds, percentiles, composite indices), limiting comparability across settings and constraining meta-analytic synthesis. A coordinated methodological framework aligned with established hazard definitions would strengthen cumulative knowledge production (Pörtner et al., 2022). Moreover, the dominance of cross-sectional designs in flood studies and retrospective acute-care analyses in heat studies limits inference on long-term, cumulative, and compounding impacts, including repeated exposure and multi-hazard contexts.

These methodological patterns plausibly contribute to the observed evidence asymmetry: heat research skews toward acute outcomes in hospital datasets from high-income contexts, while flood research concentrates in rural LMIC settings and focuses on routine service coverage. This distribution narrows understanding of (i) heat impacts on routine maternal care and (ii) flood impacts in urban contexts and underrepresented regions such as Latin America and the Caribbean.

Our findings support a dual-track strategy to strengthen MCH resilience to climate shocks. On the supply side, priorities include climate-resilient infrastructure (including thermal safety and reliable power for cooling), strengthened logistics and supply chains, and workforce protection and surge capacity to maintain service continuity during disruptions (Kakkad et al., 2014; Codjoe et al., 2020). On the demand side, social protection mechanisms buffering income loss and displacement, community-tailored preparedness plans, and trusted risk communication are central to reducing delays in care-seeking and preventing harmful care avoidance or unsafe care substitution (Ray-Bennett et al., 2021; Purno et al., 2023). Because utilisation under heat

may increase through heightened morbidity, planning should anticipate demand surges for acute obstetric and neonatal care while explicitly protecting routine preventive services from being crowded out.

Macro assessments highlight that climate hazards increasingly impose systemic burdens and economic costs, reinforcing the value of proactive adaptation rather than reactive response (Romanello et al., 2025; Pörtner et al., 2022). For MCH specifically, delays in routine care generate downstream costs through avoidable complications, while acute surges strain budgets and capacity, both of which are central concerns in health economics and health system performance.

Future research should prioritise: (1) standardised hazard metrics (consistent thresholds for heatwaves and flood severity) and transparent exposure operationalisation; (2) longitudinal and mixed-methods designs that connect lived experiences to system-level outcomes and identify causal mechanisms more credibly; (3) broader geographic coverage, including underrepresented regions (Latin America and the Caribbean, for instance) and low-resource urban settings; (4) explicit modelling of social vulnerability (income, gender, displacement, housing quality, access to cooling) and care substitution dynamics (informal care, delayed care, service switching); and (5) multi-hazard and compounding-risk analyses that reflect real-world climate risk profiles rather than single-event retrospection.

2.5 CONCLUSION

Overall, this realist-informed systematic review shows that floods and heatwaves disrupt MCH service utilisation through distinct but overlapping pathways that are strongly conditioned by baseline system capacity and social vulnerability. The evidence is most consistent for floods suppressing routine services (ANC, facility delivery, PNC, immunisation) via mobility constraints, displacement, income loss, and service disruption, whereas heat is more often linked to increases in acute and emergency utilisation that reflect heightened maternal and neonatal risk and operational stress. These findings imply that climate-resilient MCH strategies must protect both access and facility functionality while reducing household-level barriers through preparedness and social protection. Major gaps remain, including limited evidence on heat-related disruptions to routine care, sparse research on child services beyond acute use, and uneven geographic coverage, underscoring the need for standardised exposure metrics and more longitudinal, mixed-methods, and causal studies.

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APPENDIX 2.A – SEARCH STRATEGY

Ovid MEDLINE(R) ALL <1946 to February 12, 2024>

1 (flood* or inundation* or overflow* or rain* or precipitation or hurricane* or typhoon* or cyclone* or monsoon or "storm surge*" or "urban runoff*").mp. 199480

2 ("heat wave*" or heatwave* or heat-wave* or "extreme heat" or "heat stress" or heatstroke or "heat stroke" or "heat exposure" or "heat strain" or "heat index" or "heat episode*" or "heat event*" or "heat day*" or "hot day*" or "high temperature*" or "hot temperature*" or "overheat*").mp. 201228

3 exp climate change/ or climate/ or weather/ or seasons/ or extreme weather/ or extreme hot weather/ or extreme heat/ or hot temperature/ or exp heat stress disorders/ or cyclonic storms/ or floods/ or rain/ 316629

4 1 or 2 or 3 544698

5 ((health* or care or service*) adj3 (utili#ation or usage or access* or visit* or outpatient or inpatient or admission* or demand or barrier* or coverage or reach* or availab* or disrupt* or damage* or afford* or econ* or mone* or cost* or charge* or expen* or financ* or pric* or "out-of-pocket"))).mp. 571146

6 exp "health care economics and organizations"/ or delivery of health care/ or health services accessibility/ or healthcare disparities/ 1816116

7 5 or 6 2076191

8 ((matern* or mother* or child* or infant* or baby or babies or pregnan* or perinatal or post-natal or postnatal or postpartum or delivery) adj3 (health* or care or service*)).mp. 489428

9 child health/ or family health/ or infant health/ or maternal health/ or exp Maternal Health Services/ or exp Child Health Services/ 111625

10 8 or 9 535872

11 4 and 7 and 10 1148

Embase Classic+Embase <1947 to 2024 February 12>

1 (flood* or inundation* or overflow* or rain* or precipitation or hurricane* or typhoon* or cyclone* or monsoon or "storm surge*" or "urban runoff*").mp. 264435

2 ("heat wave*" or heatwave* or heat-wave* or "extreme heat" or "heat stress" or heatstroke or "heat stroke" or "heat exposure" or "heat strain" or "heat index" or "heat episode*" or "heat event*" or "heat day*" or "hot day*" or "high temperature*" or "hot temperature*" or "overheat*").mp. 112652

3 exp climate change/ or climate/ or weather/ or severe weather/ or extreme weather/ or extreme hot weather/ or heat wave/ or high temperature/ or exp heat stress/ or hurricane/ or flooding/ or rain/ or rainy season/ 222358

4 1 or 2 or 3 477143

5 ((health* or care or service*) adj3 (utili#ation or usage or access* or visit* or outpatient or inpatient or admission* or demand or barrier* or coverage or reach* or availab* or disrupt* or damage* or afford* or econ* or mone* or cost* or charge* or expen* or financ* or pric* or "out-of-pocket"))).mp. 917891

6 exp economics/ or exp health economics/ or exp health care need/ or health care availability/ or exp health care access/ or health care delivery/ or health care disparity/ 1453230
7 5 or 6 1862793

8 ((matern* or mother* or child* or infant* or baby or babies or pregnan* or perinatal or post-natal or postnatal or postpartum or delivery) adj3 (health* or care or service*)).mp. 699202

9 child health/ or family health/ or maternal health service/ or exp maternal care/ or exp child health care/ 208047

10 8 or 9 780358

11 4 and 7 and 10 1639

Global Health <1910 to 2024 Week 05>

1 (flood* or inundation* or overflow* or rain* or precipitation or hurricane* or typhoon* or cyclone* or monsoon or "storm surge*" or "urban runoff*").mp. 66203

2 ("heat wave*" or heatwave* or heat-wave* or "extreme heat" or "heat stress" or heatstroke or "heat stroke" or "heat exposure" or "heat strain" or "heat index" or "heat episode*" or "heat event*" or "heat day*" or "hot day*" or "hot temperature*" or "high temperature*" or "overheat*").mp. 17603

3 exp climate change/ or climate/ or weather/ or exp environmental temperature/ or climatic factors/ or meteorological factors/ or heat/ or heat stress/ or heat exhaustion/ or heat shock/ or hurricanes/ or floods/ or rain/ or wet season/ or monsoons/ 60593 4 1 or 2 or 3 114571

5 ((health* or care or service*) adj3 (utili#ation or usage or access* or visit* or outpatient or inpatient or admission* or demand or barrier* or coverage or reach* or availab* or disrupt* or damage* or afford* or econ* or mone* or cost* or charge* or expen* or financ* or pric* or "out-of-pocket"))).mp. 180213

6 exp economics/ or exp economic analysis/ or health care utilization/ or health care costs/ or health inequalities/ or exp health services/ 219806

7 5 or 6 299151

8 ((matern* or mother* or child* or infant* or baby or babies or pregnan* or perinatal or post-natal or postnatal or postpartum or delivery) adj3 (health* or care or service*)).mp. 114141

9 child health/ or exp infants/ or children/ or maternity services/ or pregnant women/ or pregnancy/ or gestation period/ 657136

10 8 or 9 679549

11 4 and 7 and 10 864

Econlit <1886 to January 25, 2024>

1 (flood* or inundation* or overflow* or rain* or precipitation or hurricane* or typhoon* or cyclone* or monsoon or "urban runoff*" or "heat wave*" or heatwave* or heat-wave* or "extreme heat" or "heat stress" or heatstroke or "heat stroke" or "heat exposure" or "heat strain" or "heat index" or "heat episode*" or "heat event*" or "heat day*" or "hot day*" or "overheat*" or weather* or climat* or temperature* or storm*).mp. 52880

2 ((matern* or mother* or child* or infant* or baby or babies or pregnan* or perinatal or post-natal or postnatal or postpartum or delivery) adj3 (health* or care or service*)).mp. 35558

3 1 and 2 600

Web of Science (Date Run: Feb 13 2024)

#1 TS=(flood* or inundation* or overflow* or rain* or precipitation or hurricane* or typhoon* or cyclone* or monsoon or "urban runoff*" or "heat wave*" or heatwave* or heat-wave* or "extreme heat" or "heat stress" or heatstroke or "heat stroke" or "heat exposure" or "heat strain" or "heat index" or "heat episode*" or "heat event*" or "heat day*" or "hot day*" or "overheat*" or weather* or climat* or temperature* or storm*)

#2 TS=((health* or care or service*) NEAR/3 (utili?ation or usage or access* or visit* or outpatient or inpatient or admission* or demand or barrier* or coverage or reach* or availab* or disrupt* or damage* or afford* or econ* or mone* or cost* or charge* or expen* or financ* or pric* or "out-of-pocket"))

#3 TS=((matern* or mother* or child* or infant* or baby or babies or pregnan* or perinatal or post-natal or postnatal or postpartum or delivery) NEAR/3 (health* or care or service*))

#1 AND #2 AND #3 911 results

CINAHL (February 13, 2024)

S1 flood* or inundation* or overflow* or rain* or precipitation or hurricane* or typhoon* or cyclone* or monsoon or "storm surge*" or "urban runoff*" 32,451

S2 "heat wave*" or heatwave* or heat-wave* or "extreme heat" or "heat stress" or heatstroke or "heat stroke" or "heat exposure" or "heat strain" or "heat index" or "heat episode*" or "heat event*" or "heat day*" or "hot day*" or "high temperature*" or "hot temperature*" or "overheat*" 10,706

S3 (MH "Climate+") OR (MH "Weather") OR (MH "Extreme Weather") OR (MH "Heat") OR (MH "Heat Stress Disorders+") OR (MH "Heat Stroke") OR (MH "Heat Exhaustion") OR (MH "Rain") OR (MH "Water Rescue") OR (MH "Natural Disasters") OR (MH "Seasons") 43,543

S4 S1 OR S2 OR S3 72,225

S5 (health* or care or service*) N3 (utili?ation or usage or access* or visit* or outpatient or inpatient or admission* or demand or barrier* or coverage or reach* or availab* or disrupt*

or damage* or afford* or econ* or mone* or cost* or charge* or expen* or financ* or pric* or "out-of-pocket") 466,129

S6 (MH "Costs and Cost Analysis+") or (MH "Health Services Needs and Demand+") or (MH "Health Resource Utilization+") 183,937

S7 S5 OR S6 512,787

S8 (matern* or mother* or child* or infant* or baby or babies or pregnan* or perinatal or post-natal or postnatal or postpartum or delivery) N3 (health* or care or service*) 313,495

S9 (MH "Child Health") or (MH "Family Health") or (MH "Maternal-Child Health") or (MH "Maternal Health Services+") or (MH "Child Health Services") 71,785

S10 S8 OR S9 331,419

S11 S4 AND S7 AND S10 772

GLOBAL INDEX MEDICUS (13/02/2024)

((tw:(("heat wave*" or heatwave* or heat-wave* or "extreme heat" or "heat stress" or heatstroke or "heat stroke" or "heat exposure" or "heat strain" or "heat index" or "heat episode*" or "heat event*" or "heat day*" or "hot day*" or "high temperature*" or "hot temperature*" or "overheat*" or flood* or inundation* or overflow* or rain* or precipitation or hurricane* or typhoon* or cyclone* or monsoon or "storm surge*" or "urban runoff*")) OR (mh:(("Weather" OR "Climate" OR "Temperature" OR "Heat Wave (Meteorology)" OR "Floods")))) AND (((tw:(health* or care or service*)) AND (tw:(utilization or utilisation or usage or access* or visit* or outpatient or inpatient or admission* or demand or barrier* or coverage or reach* or availab* or disrupt* or damage* or afford* or econ* or mone* or cost* or charge* or expen* or financ* or pric* or "out-of-pocket")))) OR (mh:(("Economics" OR "Health Care Economics and Organizations" OR "Costs and Cost Analysis" OR "Health Services Needs and Demand" OR "Health Services Accessibility" OR "Delivery of Health Care" OR "Healthcare Disparities")))) AND (((tw:(health* or care or service*)) AND (tw:(matern* or mother* or child* or infant* or baby or babies or pregnan* or perinatal or post-natal or postnatal or postpartum or delivery)))) OR (mh:(("Child Health" OR "Family Health" OR "Infant Health" OR "Maternal Health" OR "Maternal Health Services" OR "Child Health Services"))))

Results: 577

APPENDIX 2.B – EXTRACTION TABLE

Table 2.B.1 – Summary of included studies

Reference	Event	Methods	Country / Period	Population	Outcome	Effect size	Mechanisms	Context
Abdullah et al. (2019)	Flood	Qualitative: FGDs and IDIs.	Bangladesh (Jul–Sep 2015)	Community members (pregnant women, guardians, teachers, local leaders, village doctors, TBAs) in two rural communities (Khaliaghuri and Netrakona).	Maternal morbidity/-mortality risk linked to inadequate ANC and delayed referrals.	Not reported.	Boat-only transport; delays, high costs, and reluctance of boatmen impede timely facility access; deaths often occur en route.	Rural perception of disaster-related maternal deaths as “natural”; limited preparedness for maternal care during floods.
Amoo et al. (2018)	Flood	Quantitative: Cross-sectional survey; descriptive statistics.	Nigeria (Oct–Nov 2012)	432 household heads, nursing mothers, and pregnant women in 17 IDP camps.	Antenatal care (ANC).	ANC declined from 82% pre-flood to 53% post-flood.	Displacement, income loss, and transport barriers reduced care access.	58% were farmers; 69.2% lost livelihoods; average income fell to \$35.20.
Barman, Sarife Saha (2023)	Flood	Quantitative: Spatial analyses; logistic and geographically weighted regressions.	India (2019–2021)	176,843 neonates (NFHS-5); 707 districts, focus on hazard-prone areas.	Postnatal care (PNC).	PNC coverage 75.9% in flood-affected areas vs. 81.1% nationally; women 15% less likely to receive PNC.	Facility damage, transport disruption, displacement, and diversion of resources to emergency response.	Frequent multi-hazard exposure; remote terrain, weak infrastructure, poverty, and low education constrain access.

Continued on next page

Reference	Event	Methods	Country / Period	Population	Outcome	Effect size	Mechanisms	Context
Baten et al. (2020)	Flood	Quantitative: Multivariate logistic regression.	Bangladesh (2011–2014)	4,620 women with recent live birth; varying flood exposure frequency.	ANC, institutional delivery, maternal and newborn PNC.	Flood exposure reduced likelihood of skilled ANC (-19%), facility delivery (-12%), C-section (-17%), maternal PNC (-24%), and newborn PNC (-18%); effects intensified with repeated floods.	Infrastructure damage, transport disruption, displacement, safety concerns, and delayed system recovery under repeated shocks.	Repeated exposure fostered adaptation and NGO support; greater use of outreach/mobile services in heavily affected areas.
Bishwakarma (2022)	Heatwave	Quantitative: Linear fixed-effects model.	Nepal (2009–2014)	4,704 children from 3,725 households (506 clusters).	Antenatal care (ANC).	Each additional high-temperature day reduced ANC visits by 0.0163 (0.0110 adjusted); cumulative exposure substantially lowers utilization.	Heat discourages travel due to discomfort and health risks; resource strain may deprioritize care-seeking.	Rural context with long travel times (82% of households), weak infrastructure, low maternal education, and social inequality heighten vulnerability.
Campbell et al. (2019)	Heatwave	Quantitative: Case-crossover; conditional logistic regression.	Australia (2008–2016)	841,965 ED visits at two Tasmanian hospitals; focus on children <5 years.	Emergency Department (ED) visits (children <5).	Heatwaves increased ED visits for children <5 by 19% (OR 1.19; 95% CI: 1.04–1.36), the largest rise across age groups.	Immature thermoregulation, higher dehydration risk, and exacerbation of respiratory conditions increase vulnerability; caregivers may seek urgent care more readily.	Tasmania's cooler climate and low historical heat exposure may reduce population adaptation, increasing susceptibility during extreme heat events.

Continued on next page

Reference	Event	Methods	Country / Period	Population	Outcome	Effect size	Mechanisms	Context
Chandir et al. (2023)	Flood	Quantitative: Difference-in-differences (DiD).	Pakistan (Jul 2019–Nov 2022)	2.38 million children aged 0–23 months in 12 low-coverage districts (Sindh).	Childhood immunization.	Vaccination rates declined 19% more in flooded towns vs. non-flooded; ~11,500 missed immunizations; persistent deficits despite catch-up.	Infrastructure damage, blocked access, and service disruption reduce vaccine delivery; recovery incomplete even after three months.	Flood-prone districts with historically low immunization coverage; compounded by COVID-19 waves and health worker strikes; systemic fragility slows recovery.
Chen et al. (2022)	Heatwave	Quantitative: OLS with time fixed effects.	China (2017–2019)	64,270 neonatal observations from a major hospital in Guangzhou.	Newborn hospitalization costs.	Each additional extreme-heat day (4–8 weeks pre-delivery) increased costs by 0.096% (CNY 147.87 per newborn; 1.79% of total expenses).	Maternal heat stress and dehydration impair placental oxygen/nutrient flow, increasing neonatal complications and lowering Apgar scores.	Urban heat island effects, rapid urbanization, and socioeconomic disparities; stronger effects among mothers with low education.
Cil e Kim (2022)	Heatwave	Quantitative: Regression with difference-in-differences.	United States (2009–2018)	34.7 million births nationwide.	NICU admissions and extremely preterm birth.	Each additional day >90°F increased NICU admissions by 0.153 per 1,000 births (0.2% of mean); ≥2 SD summer heat raised admissions by 0.4%; stronger effects among Hispanic (up to 0.7%) and Black mothers (0.5% rise in extremely preterm births).	Heat exposure disrupts placental blood flow, elevates maternal stress, and triggers complications (e.g., preeclampsia, early contractions), increasing prematurity and NICU need.	Racial and socioeconomic disparities in heat exposure and adaptive capacity persist despite AC access; structural inequalities heighten vulnerability.

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Reference	Event	Methods	Country / Period	Population	Outcome	Effect size	Mechanisms	Context
Codjoe et al. (2020)	Heatwave	Mixed-methods: Field observations, temperature measurements, semi-structured interviews.	Ghana (May–Jul 2018)	Providers from public and private facilities in four low-income, high-density urban communities (Accra and Tamale).	MCH service provision.	Not reported.	High indoor temperatures reduce service quality, increase infection risk and staff fatigue; power outages disrupt drug/vaccine refrigeration; overcrowding delays essential services.	Under-resourced facilities, poor ventilation, unreliable electricity/water, urban heat islands, and high patient density amplify heat-related constraints.
Codjoe et al. (2020)	Flood	Mixed-methods: Field observations, rainfall records, semi-structured interviews.	Ghana (May–Jul 2018)	Providers from public/private facilities in four low-income, flood-prone urban communities (Accra and Tamale).	MCH service provision.	Not reported.	Floods damage infrastructure, supplies, and generators, causing power outages and service disruption; flooded roads increase transport costs, break referral chains, and shift demand toward informal providers.	Rapid urbanization, inadequate infrastructure, and recurrent climate shocks in dense informal settlements weaken system resilience.
Davis et al. (2020)	Heatwave	Quantitative: Time-stratified case-crossover.	United States (2010–2017)	Residents of Roanoke, Virginia, visiting Carilion Clinic hospitals; focus on children aged 1–4.	Emergency Department (ED) visits (ages 1–4).	Heatwaves increased ED visits by 20% (OR 1.20; 95% CI: 1.05–1.37); strongest effect at lag 0.	Young children’s limited thermoregulation increases dehydration and respiratory risks; immediate physiological stress drives acute care use.	Exposure varies by housing quality, cooling access, and socioeconomic status despite widespread AC availability.

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Reference	Event	Methods	Country / Period	Population	Outcome	Effect size	Mechanisms	Context
Firoz et al. (2016)	Flood	Qualitative: FGDs and in-depth interviews.	Mozambique (Dec 2013–Apr 2014)	Pregnant women, women of reproductive age, TBAs, male partners, health workers, community leaders.	Maternal health (community perspectives).	Not reported.	Flood-related livelihood loss reduces affordability and increases dependence on male partners; damaged facilities and staff shortages weaken service delivery; malaria and poor sanitation heighten risk.	Rural poverty, post-conflict fragility, gender inequality, and restrictive norms constrain women's autonomy and care-seeking.
Haque, Parre Muhidin (2019)	Flood	Quantitative: Multivariate logistic regression.	Bangladesh (Jan–May 2017)	1,003 children (<15 years) from displacement-prone and non-prone rural households.	Child healthcare-seeking.	Children of displaced parents were 83% more likely to receive no/only home care (OR 1.83; 95% CI: 1.08–3.12) and 56% less likely to see a trained provider (OR 0.44; 95% CI: 0.26–0.73).	Income loss, higher travel costs, reduced provider availability, and disrupted trust in providers reduce formal care use.	Displacement increases isolation and economic hardship; displaced families face weaker provider networks and greater reliance on informal care.
Haque, Parre Muhidin (2020a)	Flood	Quantitative: Multivariate logistic regression.	Bangladesh (Jan–May 2017)	611 women with recent live birth; 289 displaced (high-risk areas), 322 non-displaced.	Antenatal care (ANC).	Women displaced ≥ 3 times had 68% lower odds of any ANC and 62% lower odds of ≥ 4 skilled visits; 1–2 displacements non-significant.	Repeated displacement increases geographic isolation, financial barriers, and disrupts provider relationships; joint decision-making improves ANC use.	Displacement-prone areas are poorer, remote, and face limited female provider availability; entrenched gender norms influence utilization.

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Reference	Event	Methods	Country / Period	Population	Outcome	Effect size	Mechanisms	Context
Haque, Parre Muhidin (2020b)	Flood	Quantitative: Multivariate logistic regression.	Bangladesh (Jan–May 2017)	599 rural women (displaced and non-displaced).	Facility delivery and postnatal care (PNC).	Women displaced ≥ 3 times were 75% less likely to deliver in facilities and 63% less likely to receive trained PNC (OR 0.37; $p < 0.05$); facility delivery increased trained PNC odds 11.7-fold.	Relocation to remote areas, damaged infrastructure, low income, and broken provider ties reduce skilled delivery; reduced ANC mediates lower PNC use.	Displaced households are poorer, less educated, and have limited media and provider access, compounding maternal health inequities.
Jacquet et al. (2016)	Flood	Quantitative: Descriptive statistics and multivariate analysis.	Pakistan (Jan–Feb 2011)	1,569 flood-affected households (predominantly rural, low-income, low education; widespread home damage and income loss).	Skilled birth attendance (delivery).	Among births within six months post-flood (15.4% severe income loss and rural residence constrain access; small birth subsample limited statistical power to detect predictors.	Low-income households were >13 times more likely to lack care; rural families faced >4 times the risk; larger families and low education compounded barriers.	
Kakkad et al. (2014)	Heatwave	Quantitative: Segmented regression analysis.	India (2009–2011)	2,025 neonates in a low-income hospital (554 NICU admissions; 36 heat-related).	NICU admissions (heat-related).	During the 2010 heatwave, heat-related NICU admissions rose to 24 (vs. 8 in 2009; 4 in 2011); each 1°C above 42°C increased admissions by 43% (95% CI: 9.2–88%); relocating the ward reduced admissions by 64%.	Neonatal thermoregulatory immaturity and non-air-conditioned environments increase hyperthermia risk; prenatal heat exposure may induce fetal stress or mild prematurity.	Low-income urban hospital without cooling infrastructure; top-floor ward intensified heat exposure; malnutrition and vulnerability heightened risk.

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Reference	Event	Methods	Country / Period	Population	Outcome	Effect size	Mechanisms	Context
Kim, Lee e Rossin-Slater (2021)	Heatwave	Quantitative: Fixed-effects regression with difference-in-differences.	United States (2003–2013)	2.24 million mothers in Arizona, New York, and Washington.	Maternal hospitalizations.	Each additional extreme-heat day increased hospitalization risk by 0.1 percentage points (2.2%); stronger effect among Black women (0.4 pp; 5%); effects observed in cooler but not hotter counties.	Heat stress during pregnancy increases hypertension, dehydration, early labor, and infection risk, leading to emergency hospitalizations.	Racial disparities in heat exposure and adaptive capacity; absence of effects in hotter counties suggests behavioral and infrastructural adaptation.
Makanga et al. (2017)	Flood	Quantitative: Geospatial analysis.	Mozambique (Mar 2013–Oct 2014)	Women of reproductive age in CLIP trial regions (rural southern Mozambique).	Geographical access to maternal care.	Wet season reduced access within 1 hour to primary care from 49% to 40% (walking) and 87% to 82% (transport); access to secondary care within 2 hours declined from 64% to 53% (walking).	Flooding blocks roads and slows travel, disproportionately affecting women reliant on walking or public transport during obstetric emergencies.	Rural terrain, poor roads, and seasonal rains limit access; geography identified as primary barrier after accounting for cultural and financial constraints.
Mroz et al. (2023)	Flood	Quantitative: Geographical health access modelling and agent-based modelling (ABM).	Zambia (Oct 2017–Oct 2018)	Women of reproductive age on the Barotse Floodplain.	Access to delivery sites and EmOC.	Flooding reduced timely walking access to delivery sites from 55% to 29%; emergency referral access within 1 hour fell from 65% to 23%; by peak flooding, 76% lacked EmOC access.	Floodwaters render walking paths and roads impassable; depth and flow thresholds prevent even emergency vehicle passage, severing referral networks.	Remote, semi-nomadic floodplain communities rely on limited transport modes; seasonal peak flooding and sparse facilities create extreme physical isolation.

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Reference	Event	Methods	Country / Period	Population	Outcome	Effect size	Mechanisms	Context
Orderud et al. (2022)	Flood	Quantitative: Multilevel linear probability models.	Bangladesh (2007, 2011, 2014, 2017–2018)	Ever-married women (15–49) with births between 2004 and 2018.	Antenatal care (ANC) and skilled delivery.	No significant average effect; small 2.1 pp decline in any ANC 25–36 months post-flood. Among non-migrants, exposure 10–36 months pre-birth reduced ANC, skilled attendance, and facility delivery by 4–6 pp ($\approx 10\text{--}15\%$).	Flood-related infrastructure and transport disruption, plus medium-term income shocks in rural agricultural areas, reduce affordability and access.	Flood-prone, rural, low-education regions with high out-of-pocket costs and weak health infrastructure; exposed areas have lower baseline service use.
Peyravi, Rahnavard e Marzaleh (2020)	Flood	Qualitative: Field observations and expert interviews.	Iran (Mar–Apr 2019)	24 experts and policymakers across health, environmental, and disaster fields.	MCH service provision.	Not reported.	Infrastructure damage, staff and supply shortages, limited coordination, and overcrowding disrupt maternal and child services; heightened risks of infection, malnutrition, and mental health problems.	Golestan Province experienced mass displacement to overcrowded camps; weak health infrastructure and poverty amplified service disruption.
Purno et al. (2023)	Flood	Qualitative: Retrospective case study.	Bangladesh (Aug 2022–Jan 2023)	28,729 women receiving SRHR services at union centers.	SRHR services (family planning, ANC, EmOC).	Not reported; intervention increased service uptake during flood response.	Floods damaged facilities, disrupted supplies, and increased travel barriers; crisis heightened GBV and adolescent pregnancy; midwife deployment and outreach mitigated access gaps.	Remote, climate-vulnerable northeast Bangladesh with pre-existing SRHR gaps; under-resourced rural facilities lacked EmOC prior to floods.

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Reference	Event	Methods	Country / Period	Population	Outcome	Effect size	Mechanisms	Context
Quilty et al. (2024)	Heatwave	Quantitative: Poisson and negative binomial regression.	Australia (Feb 2018–Dec 2019)	9,414 aeromedical retrieval patients in Northern Territory.	Aeromedical retrievals for obstetric emergencies.	In humid zones, obstetric retrievals increased 49% in hottest quartile vs. coldest; no effect in arid zone.	Heat increases preterm birth and obstetric complications, driving urgent retrievals; high-priority cases may strain emergency capacity.	Remote, climate-extreme regions with limited infrastructure; Indigenous communities rely heavily on aeromedical services; adaptation varies by climatic zone.
Ray-Bennett et al. (2021)	Flood	Mixed-methods: Facility assessments, interviews, FGDs, descriptive statistics.	Bangladesh (Oct 2015–Jan 2018)	370 married women (15–49) receiving menstrual regulation or PAC during 2016 floods.	Post-abortion care (PAC).	RHCC intervention improved PAC quality: follow-up advice 44.6%→69%; FP counseling 48.1%→82.8%; return for contraception 11.2%→55.2%.	Floods damage infrastructure and transport; stigma, confidentiality concerns, and staff shortages reduce PAC access; preparedness interventions improve continuity and quality.	Flood-prone rural unions with under-resourced facilities and cultural stigma around menstrual regulation; intervention pre-positioned kits and trained staff.
Sadia et al. (2016)	Flood	Quantitative: Descriptive statistics.	Pakistan (Sep 2013–Jan 2014)	386 women in Nowshera; 153 pregnant or recently delivered during 2010 floods.	ANC, PNC, and skilled delivery.	Coverage below national averages: ANC 45%, PNC 39%, skilled attendance 26%, facility delivery 19.7%.	Flood damage to facilities and transport barriers forced long travel; lack of information and exclusion from preparedness reduced utilization.	Rural, patriarchal, flood-prone district with under-resourced facilities; economic hardship, low female education, and gender norms constrained care-seeking.

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Reference	Event	Methods	Country / Period	Population	Outcome	Effect size	Mechanisms	Context
Saulnier et al. (2020)	Flood	Qualitative: FGDs, semi-structured interviews, thematic analysis.	Cambodia (Nov–Dec 2018)	Pregnant women, recent mothers, partners, and community members in eight flood-affected villages.	Delivery care.	Not reported.	Flooded roads and scarce transport delay facility access; staff shortages constrain service availability; exclusion of traditional birth attendants reduces community support; households rely on self-financed coping (savings, relocation, transport arrangements).	Rural districts with differing flood frequency; hierarchical leadership and centralized health system limit flexibility; frequent-flood communities more accustomed to coping strategies.
Saulnier et al. (2022)	Flood	Qualitative: Semi-structured interviews; thematic analysis.	Cambodia (Nov–Dec 2018)	23 public health professionals involved in antenatal and childbirth services.	ANC and childbirth service provision.	Not reported; minimal observed service interruption.	Flooding perceived as manageable strain; pre-existing contingency plans, inter-level coordination, and hierarchical decision-making maintained service continuity.	Service delivery more constrained by infrastructure gaps, staff shortages, weak referrals, and low trust in public care than by flooding itself.
Shah, Miller e Mothabbir (2019)	Flood	Mixed-methods: Trend analysis, interviews, FGDs.	Bangladesh (Jan 2015–Feb 2016)	Under-five children treated by community providers (CHCPs, VDs) and iCCM stakeholders.	iCCM service delivery.	During floods: 30% decline in treated children; 48% rise in referrals; commodity availability fell to 82%; supervision rates declined.	Clinic submersion, road damage, medicine shortages, and reduced supervision disrupted service delivery; informal coping (temporary clinics, boat transport, adjusted hours) partially mitigated gaps.	Rural, flood-prone Bhola District with weak infrastructure and limited preparedness; service continuity relied on local improvisation.

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Reference	Event	Methods	Country / Period	Population	Outcome	Effect size	Mechanisms	Context
Smith et al. (2016)	Heatwave	Quantitative: Multilevel mixed-effects models; incidence rate ratios.	England (2011–2014)	Syndromic surveillance data (GP in/out-of-hours; 18 EDs); focus on children 0–4.	ED visits for heat illness (0–4 years).	Incidence rate ratio 3.0 during 2013 heatwave (vs. non-heatwave years); ED visits peaked early; not statistically significant due to wide CI.	Immature thermoregulation and dehydration risk increase susceptibility; rapid physiological response to extreme heat drives early ED use.	Moderate-climate context; caregiver behavior, indoor heat exposure, and awareness shape observed care-seeking patterns.
Stowell et al. (2023)	Heatwave	Quantitative: Case-crossover; conditional logistic regression.	United States (2016–2019)	1.2 million ED visits (children 0–5) across 2,489 counties.	ED visits for heat-related illness (0–5 years).	Extreme heat associated with 6% higher odds of heat-related ED visits (OR 1.06; 95% CI: 0.93–1.20; non-significant); no association for all-cause visits.	Physiological vulnerability (immature thermoregulation) and caregiver-dependent protection influence heat-related morbidity; dehydration and metabolic stress key pathways.	Privately insured population likely underestimates vulnerability; geographic, housing, and cooling-access heterogeneity shape exposure and adaptation.
Surekha, Anbazhagan e Kiran (2020)	Flood	Qualitative: FGDs, in-depth interviews, cause–outcome analysis.	India (Aug–Sep 2017)	Community members, local leaders, Anganwadi workers, teachers, PHC staff.	MCH vulnerabilities and service disruptions.	Not reported.	Floods damaged roads and facilities, limiting access for pregnant women and children; medicine shortages and service interruptions reduced treatment availability; increased injuries, infectious disease, malnutrition, and psychological stress compounded maternal–child risks.	Rural Bihar characterized by poverty, weak infrastructure, overcrowding, limited early warning systems, and cultural barriers to preparedness, heightening vulnerability.

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Reference	Event	Methods	Country / Period	Population	Outcome	Effect size	Mechanisms	Context
Wilunda et al. (2016)	Flood	Qualitative: FGDs, key informant interviews, inductive content analysis.	South Sudan (Mar 2015)	Recent mothers, husbands, community actors in Rumbek North villages.	Delivery care.	Only 5.9% delivered in health facilities.	Flooding damaged roads and isolated villages; ambulances and referrals collapsed due to lack of transport and communication; women carried on foot, causing severe delays and preventable maternal/neonatal deaths.	Conflict-affected, semi-nomadic rural setting with weak infrastructure, low literacy, limited female autonomy, cultural preference for home births, and scarce under-resourced facilities.
Wilunda et al. (2017)	Flood	Qualitative: FGDs, key informant interviews, inductive content analysis.	South Sudan (Mar 2015)	Women (18+) with recent births, husbands, CHWs, local health actors.	Antenatal care.	Only 11.3% completed ≥ 4 ANC visits; 1.7% delivered in facilities.	Flooded and damaged roads limited physical access; supply-chain disruptions reduced service availability; increased travel time raised opportunity costs amid household responsibilities.	Patriarchal norms, low awareness of ANC benefits, hidden costs, transport scarcity, and male-dominated decision-making further constrained utilization in a fragile health system.
Xu et al. (2014)	Heatwave	Quantitative: Quasi-Poisson GLM with distributed lag non-linear model.	Australia (2003–2009)	131,249 ED attendances (children 0–14) in Brisbane.	Paediatric emergency admissions.	High temperatures ($\geq 26.5^{\circ}\text{C}$) associated with 27% increase in ED admissions; effects persisted 2–3 weeks; 9–11 excess EDAs/day for chronic respiratory disease during heatwaves.	Heat exacerbates respiratory, metabolic, neurological, and intestinal conditions; immature thermoregulation in young children increases vulnerability; dehydration and inflammatory responses likely pathways.	Subtropical urban setting with frequent heat exposure; adjustments for pollution and influenza; findings may not generalize to colder or rural regions.

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Reference	Event	Methods	Country / Period	Population	Outcome	Effect size	Mechanisms	Context
Xu et al. (2017)	Heatwave	Quantitative: Quasi-Poisson generalized additive model.	Australia (2005–2015)	Infants (<1 year) hospitalized in 11 Brisbane hospitals.	Infant hospital admissions.	Only intense heatwaves (≥ 97 th percentile) increased admissions; RR 1.05 (2-day), RR 1.13 (3-day), RR 1.18 (4-day; 95% CI: 1.05–1.32); dose-response by duration and intensity.	Immature thermoregulation and immune systems increase infant susceptibility; heat may exacerbate perinatal vulnerabilities; dependence on caregiver-mediated protection influences exposure.	Subtropical climate with frequent heat; no effect modification by gender, indigenous status, or SES; robust controls applied, but context-specific to Brisbane.

APPENDIX 2.C – LIST OF EXCLUDED STUDIES

Not related to heatwaves or floods

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Lack of discussion of mechanisms or contextual factors linking exposure to outcomes

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APPENDIX 2.D – MIXED METHODS APPRAISAL TOOL

Qualitative Studies

Table 2.D.1 – MMAT (2018) appraisal of qualitative studies

MMAT Criteria (2018)	Abdullah et al. (2019)	Firoz et al. (2016)	Peyravi, Rahnavard e Marzaleh (2020)	Purno et al. (2023)	Saulnier et al. (2020)	Saulnier et al. (2022)	Surekha, Anbazhagan e Kiran (2020)	Wilunda et al. (2016)	Wilunda et al. (2017)
S1. Are there clear research questions?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
S2. Do the collected data allow to address the research questions?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1.1. Is the qualitative approach appropriate to answer the research question?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1.2. Are the qualitative data collection methods adequate to address the research question?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1.3. Are the findings adequately derived from the data?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1.4. Is the interpretation of results sufficiently substantiated by data?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1.5. Is there coherence between qualitative data sources, collection, analysis and interpretation?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Quantitative Nonrandomized

Table 2.D.2 – MMAT (2018) appraisal of quantitative nonrandomized studies – Part I

MMAT Criteria (2018)	Barman, Sarif e Saha (2023)	Baten et al. (2020)	Bishwakarma (2022)	Campbell et al. (2019)	Chandir et al. (2023)	Chen et al. (2022)	Cil e Kim (2022)	Davis et al. (2020)	Kakkad et al. (2014)
S1. Are there clear research questions?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
S2. Do the collected data allow to address the research questions?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
3.1. Are the participants representative of the target population?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
3.2. Are measurements appropriate regarding both the outcome and intervention (or exposure)?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
3.3. Are there complete outcome data?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
3.4. Are the confounders accounted for in the design and analysis?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
3.5. During the study period, is the intervention administered (or exposure occurred) as intended?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 2.D.3 – MMAT (2018) appraisal of quantitative nonrandomized studies – Part II

MMAT Criteria (2018)	Kim, Lee e Rossin- Slater (2021)	Makanga et al. (2017)	Mroz et al. (2023)	Orderud et al. (2022)	Quilty et al. (2024)	Smith et al. (2016)	Stowell et al. (2023)	Xu et al. (2014)	Xu et al. (2017)
S1. Are there clear research questions?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
S2. Do the collected data allow to address the research questions?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
3.1. Are the participants representative of the target population?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
3.2. Are measurements appropriate regarding both the outcome and intervention (or exposure)?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
3.3. Are there complete outcome data?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
3.4. Are the confounders accounted for in the design and analysis?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
3.5. During the study period, is the intervention administered (or exposure occurred) as intended?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Quantitative Descriptive

Table 2.D.4 – MMAT (2018) appraisal of quantitative descriptive studies

MMAT Criteria (2018)	Amoo et al. (2018)	Haque, Parr e Muhidin (2019)	Haque, Parr e Muhidin (2020a)	Haque, Parr e Muhidin (2020b)	Jacquet et al. (2016)	Sadia et al. (2016)
S1. Are there clear research questions?	Yes	Yes	Yes	Yes	Yes	Yes
S2. Do the collected data allow to address the research questions?	Yes	Yes	Yes	Yes	Yes	Yes
4.1. Is the sampling strategy relevant to address the research question?	Yes	Yes	Yes	Yes	Yes	Yes
4.2. Is the sample representative of the target population?	Yes	Yes	Yes	Yes	Yes	Yes
4.3. Are the measurements appropriate?	Yes	Yes	Yes	Yes	Yes	Yes
4.4. Is the risk of nonresponse bias low?	Cannot tell	Yes	Yes	Yes	Yes	Yes
4.5. Is the statistical analysis appropriate to answer the research question?	Yes	Yes	Yes	Yes	Yes	Yes

Mixed-Methods

Table 2.D.5 – MMAT (2018) appraisal of mixed-methods studies

MMAT Criteria (2018)	Codjoe et al. (2020)	Ray-Bennett et al. (2021)	Shah, Miller e Mothabbir (2019)
S1. Are there clear research questions?	Yes	Yes	Yes
S2. Do the collected data allow to address the research questions?	Yes	Yes	Yes
5.1. Is there an adequate rationale for using a mixed methods design to address the research question?	Yes	Yes	Yes
5.2. Are the different components of the study effectively integrated to answer the research question?	Yes	Yes	Yes
5.3. Are the outputs of the integration of qualitative and quantitative components adequately interpreted?	Yes	Yes	Yes
5.4. Are divergences and inconsistencies between quantitative and qualitative results adequately addressed?	Yes	Yes	Yes
5.5. Do the different components of the study adhere to the quality criteria of each tradition of the methods involved?	Yes	Yes	Yes

3 CAUSAL LOOP DIAGRAM

ABSTRACT

Background: Floods and heatwaves increasingly threaten maternal and child health (MCH) by disrupting access to, and delivery of, essential services such as antenatal care, facility delivery, postnatal care, and immunisation. However, the mechanisms through which these hazards translate into changes in MCH utilisation remain insufficiently understood, limiting the design of context-sensitive resilience policies.

Methods: Building on a realist-informed systematic review (Chapter 2), we developed a Causal Loop Diagram (CLD) to synthesise heterogeneous evidence into an explanatory system map. We extracted recurrent Context–Mechanism–Outcome (CMO) patterns, consolidated them into six interacting mechanisms: (M1) impaired mobility, (M2) displacement, (M3) income loss, (M4) increased health risks, (M5) service disruption, and (M6) energy/productivity/supply-chain disruption. We translated these pathways into signed causal links and feedback structures. Contextual variation was made explicit through classification of studies by rurality, national income level, Köppen–Geiger climate zone, and health system type.

Results: The CLD identifies eighteen components distributed across the six mechanisms and reveals multiple reinforcing feedback loops concentrated in livelihood and displacement pathways. Flood impacts most consistently reduce routine MCH utilisation through concurrent failures in transport, livelihoods, facility functionality, and supply chains, particularly in rural and low-income settings. Heat evidence is more context-specific and often reflects a shift toward acute utilisation (e.g., emergency care) and operational stress via energy dependence and workforce strain. Across hazards, economic contraction and mobility constraints emerge as key upstream drivers that can trigger cascading reductions in service use.

Conclusion: A realist-informed CLD provides a transparent, mechanism-based account of how extreme weather events propagate through demand- and supply-side channels to shape MCH utilisation. The model highlights leverage points beyond the health sector—particularly income protection, transport continuity, and energy/logistics resilience—while underscoring that causal strength and loop dominance remain uncertain without empirical validation. Future work should validate the CLD with stakeholders and translate it into a stock-and-flow model to quantify dynamics, delays, and policy impacts under single and compound hazards.

Keywords: Climate change. Floods. Heatwaves. Maternal and child health. Healthcare utilisation. Realist synthesis. Systems thinking. Causal loop diagram. Health system resilience.

JEL: I10 (Health: General); I12 (Health Behavior, Health Management, Health Insurance); I18 (Government Policy: Regulation, Public Health); Q54 (Climate; Natural Disasters; Global Warming).

HIGHLIGHTS

- Develops a realist-informed Causal Loop Diagram to explain how floods and heatwaves affect MCH utilisation.
- Identifies six interacting mechanisms spanning mobility, displacement, income, health risks, service disruption, and infrastructure stress.
- Reveals reinforcing feedback loops concentrated in livelihood and displacement pathways that can amplify utilisation losses.
- Shows hazard-specific signatures: floods predominantly suppress routine care; heat more often induces operational stress and acute demand.
- Points to cross-sector leverage points (income protection, transport, energy and logistics resilience) to strengthen MCH system resilience.

3.1 INTRODUCTION

Climate change is increasing the frequency and intensity of extreme weather events, posing significant risks to health systems by disrupting healthcare services, straining limited resources, and exacerbating morbidity and mortality (Codjoe et al., 2020; Ebi et al., 2021; Chersich et al., 2023). Floods and heatwaves are among the most frequent and devastating natural disasters, disproportionately affecting low-income populations and threatening the continuity of essential health services. Pregnant women and children are particularly vulnerable to this disruption. Despite increasing recognition of these risks, there remains a persistent gap in understanding the specific mechanisms through which floods and heatwaves disrupt maternal and child healthcare (MCH) utilization. This gap limits the development of effective, context-sensitive policies capable of reducing adverse outcomes such as maternal and neonatal mortality, missed antenatal care, and complications during childbirth (Meherali et al., 2024).

This study addresses that gap by integrating systems thinking into disaster-health analysis. Our aim is to uncover how extreme weather events impact the delivery and use of MCH services, by identifying key causal mechanisms and feedback structures that shape these disruptions and proposing evidence-based strategies to improve MCH system resilience.

The central research question guiding this study is: Through which mechanisms do flood and heatwaves impact access to and use of maternal and child health services?

MCH utilisation is the extent to which healthcare services designed to improve the health outcomes of mothers and children - such as antenatal care, skilled birth attendance, postnatal care, and immunizations - are accessed and used by the target population (Sarikhani, Najibi e Razavi, 2024).

We hypothesize six main mechanisms that mediate this relationship:

- M1 - Impaired ability to travel.

- M2 - Displacement and migration of affected populations.
- M3 - Household income loss, reducing demand for services.
- M4 - Increased health risks and competing healthcare needs.
- M5 - Strain on healthcare supply and service availability.
- M6 - Disruption to energy, productivity and supply chain.

These mechanisms do not operate in isolation. Rather, they interact dynamically, sometimes reinforcing one another through feedback loops, which may either exacerbate or mitigate the effects of flooding on healthcare utilization. To explore these dynamics, we employ Causal Loop Diagrams (CLDs), a qualitative systems thinking tool used to visually map causal relationships. CLDs represent variables as nodes and use directional links to denote either reinforcing (positive) or balancing (negative) interactions (Stermann, 2000; Strelkovskii e Rovenskaya, 2021).

For example, flood-induced road damage (M1) may reduce patients' ability to reach care, decreasing service utilization. This drop in demand may result in service contraction (M5), strengthening the original access barrier and creating a self-reinforcing negative loop. Alternatively, investment in mobile units or resilient infrastructure could create positive feedback loops, helping restore or even improve service utilization. CLDs thus offer a structured lens to examine system interactions and identify strategic points of intervention.

Our analytical framework draws inspiration from realist evaluation (Pawson e Tilley, 1997), which emphasizes context-mechanism-outcome (CMO) configurations. While not aiming to test program effectiveness, this study seeks to refine the theoretical understanding of system behaviour under environmental stress. By examining where demand- and supply-side disruptions intersect, we aim to identify system vulnerabilities, leverage points, and potential tipping points.

The relevance of this study extends beyond public health, offering important contributions to the field of health economics. It sheds light on how climate-related shocks disrupt service delivery, reduce productivity, and strain resource allocation. These disruptions generate multidimensional economic costs (from lost labor days to increased healthcare expenditures) with consequences for both local resilience and national policy planning. By identifying the transmission mechanisms through a systems thinking lens, this study provides actionable insights for developing adaptive policies that mitigate long-term welfare losses. In doing so, it introduces a novel CLD-based framework to the health economics literature, emphasizing the complex economic pathways linking environmental shocks to maternal and child healthcare utilization.

Complex systems are sets of interconnected elements that interact dynamically and nonlinearly, resulting in unpredictable, emergent behaviors that cannot be explained solely by examining the isolated parts. Examples of such systems include ecosystems, financial markets, or the human body. They are considered adaptive, meaning they can change and evolve in response to the environment or internal interactions. Understanding these systems requires an approach

that considers the relationships among components and the patterns that emerge from these interactions (Mitchell, 2009).

Health systems are increasingly viewed as complex adaptive systems (CAS), networks of interdependent actors, institutions, and resources that evolve in response to shocks (Borghi et al., 2022). These systems exhibit nonlinear dynamics, emergent behaviors, and feedback sensitivity, which complicate efforts to manage them through linear policy responses (Mitchell, 2009; Khan et al., 2018). As McNab et al. (2020) and Nason (2023) highlight, designing effective responses to crises like floods requires recognizing and responding to the interconnectedness of social, institutional, and environmental influences within the system.

System mapping, through tools like CLDs, offers a valuable approach to navigating this complexity. It enables policymakers and practitioners to visualize dependencies, anticipate unintended consequences, and align interventions across sectors. As Pype et al. (2018) note, understanding the perceptions and interactions among health system actors enhances coordination and decision-making. When combined with evidence-based design principles (Engelseth et al., 2021), system mapping strengthens strategic planning and supports more resilient and equitable maternal and child healthcare in disaster-prone settings.

Based on this conceptual foundation, this study has three specific objectives: (i) to identify and analyze the key mechanisms linking flood exposure to changes in maternal and child healthcare utilization; (ii) to develop a Causal Loop Diagram that maps interactions, feedback loops, and leverage points within the system; (iii) to propose evidence-informed strategies to strengthen the resilience of MCH services in flood-prone contexts.

Beyond this Introduction (Section 3.1), the remainder of the essay is organized as follows. Section 3.2 describes the methodological approach, detailing the rationale for using Causal Loop Diagrams (CLDs). Section 3.3 presents the findings, including the development and interpretation of the causal model. Section 3.4 discusses the main results in light of the existing literature, highlighting policy implications and avenues for future research. Finally, Section 3.5 concludes the essay.

3.2 METHODS

This section describes the methods used to develop an evidence-informed causal model of how floods and heatwaves affect maternal and child healthcare utilisation. It first explains how realist synthesis was used to identify and refine context–mechanism–outcome configurations and then outlines how these findings were translated into a causal loop diagram to represent the interactions, feedback structures, and pathways operating within the health system.

3.2.1 Realist synthesis

Realist synthesis (also referred to as realist review) is a theory-driven approach to evidence synthesis rooted in scientific realism, designed to explain how and why outcomes

occur by uncovering the generative mechanisms that are triggered (or suppressed) in particular contexts (Pawson e Tilley, 1997; Pawson, 2006). In contrast to conventional effectiveness-focused reviews, realist synthesis asks: what works, for whom, in what circumstances, and through what mechanisms? It conceptualises causality through Context–Mechanism–Outcome (CMO) configurations, where outcomes arise from the interaction between contextual conditions (e.g., geography, institutions, socioeconomic vulnerability) and mechanisms (e.g., behavioural responses, resource constraints, organisational adaptation).

This approach is particularly appropriate for analysing extreme weather events (EWEs) and maternal and child health (MCH) utilisation because floods and heatwaves function as complex exogenous shocks, not discrete interventions. They perturb multiple system domains simultaneously (mobility and access, household livelihoods, disease risks, facility functionality, workforce capacity, and supply chains) often producing nonlinear effects and feedback (e.g., temporary access barriers translating into longer-term service contraction or risk escalation). Recent syntheses of EWEs and healthcare utilisation in rural/remote contexts highlight precisely this multi-pathway disruption and the importance of examining mechanisms and context-dependence rather than relying solely on average effects (Dewi et al., 2024). Given that health systems are increasingly analysed as complex adaptive systems (Borghi et al., 2022), realist synthesis is well suited to explaining why the same hazard can generate different utilisation outcomes across places and populations, depending on pre-existing vulnerability, institutional buffering, and adaptive capacity.

In this thesis, the realist-informed synthesis (Chapter 2) provided the evidentiary foundation for the CLD by (i) extracting and refining mechanism-based explanations from heterogeneous study designs and (ii) translating recurrent CMO patterns into a coherent causal architecture. The review followed structured searching and screening procedures consistent with contemporary systematic-review reporting norms (PRISMA 2020), while preserving the realist principle of iterative theory refinement (i.e., searching and interpretation continued until the emerging programme theory adequately explained observed patterns) (Akl et al., 2024). Evidence was not ranked using a traditional hierarchy. Instead, documents were appraised for their relevance (capacity to inform theory building/testing), richness (depth of causal explanation), and rigour (credibility of the methods generating the specific data used for theory refinement), in line with updated realist methodological guidance (Dada et al., 2023).

Data extraction was organised to identify and compare (a) Contexts (C): structural and situational features shaping vulnerability and response (e.g., rurality and remoteness, baseline infrastructure, facility density, governance coordination, poverty and informality, social protection coverage, existing service quality) (Greenhalgh e Manzano, 2022; Renmans, Holvoet e Criel, 2020); (b) Mechanisms (M): underlying processes activated by the hazard (e.g., reduced ability to travel; displacement; income loss and reallocation of time; heightened clinical risk and competing care needs; reduced service availability through facility damage/overcrowding/workforce depletion; energy and supply-chain disruption) (Shaw et al., 2018; Dalkin et al., 2015); and (c)

Outcomes (O): observable changes in MCH utilisation (e.g., missed or delayed antenatal care, reduced facility delivery or skilled attendance, disruptions to postnatal care and immunisation, shifts from routine to emergency care) (Sarikhani, Najibi e Razavi, 2024; Shanto et al., 2023; Tiruneh et al., 2022). This structure also aligned with established access/utilisation conceptualisations that separate demand-side constraints from supply-side capacity, while recognising their interaction (e.g., perceived need, affordability, acceptability, and availability) (Andersen, 2008). The synthesis iteratively compared CMOs across settings to identify recurring explanatory patterns and to consolidate them into higher-order mechanisms that were sufficiently general to travel across contexts, but specific enough to support causal mapping.

The resulting programme theory, grounded in the reviewed evidence, was summarised into six interacting pathways linking floods and heatwaves to MCH utilisation: (M1) impaired mobility and physical access; (M2) displacement and migration; (M3) income loss and economic insecurity; (M4) increased health risks and competing healthcare needs; (M5) strain on health-service supply and availability; and (M6) disruption of energy, productivity, and supply chains. Importantly, the realist synthesis emphasised that these pathways are seldom independent: they co-occur and may generate reinforcing dynamics (e.g., facility disruption lowers service quality and trust, which further reduces utilisation even after physical access is restored) or balancing dynamics (e.g., emergency mobilisation temporarily expands outreach capacity). This is precisely where realist synthesis becomes methodologically complementary to systems thinking: realist synthesis specifies what mechanisms are plausible and evidenced, while system mapping formalises how those mechanisms interact over time and through feedback.

Accordingly, the final stage of the realist synthesis consisted of translation from programme theory to system structure. Mechanisms and their associated variables were converted into CLD nodes and signed causal links based on the directionality supported by the evidence. This integration is consistent with recent methodological work that explicitly combines realist theory-building with causal loop diagramming as an approach to unpack complexity in health systems research, including in LMIC settings (Cassidy et al., 2022; Ehteshami et al., 2024). The CLD thereby functions as a structured representation of the realist programme theory: CMOs become connected causal chains; recurring CMOs form feedback loops; and leverage points emerge where interventions may interrupt reinforcing cycles of disruption or strengthen stabilising responses. The end result is an evidence-informed explanatory model of how climate-related hazards propagate through demand- and supply-side channels to shape MCH utilisation, providing a transparent basis for identifying resilience strategies that are sensitive to context and mechanism rather than “one-size-fits-all” prescriptions.

3.2.2 Causal loop diagrams

Causal loop diagrams (CLDs) are a foundational tool in systems thinking used to represent and analyze the dynamic cause-and-effect relationships within complex systems. Originating in the system dynamics tradition (Forrester, 2012), CLDs enable researchers to visualize how

variables interact, influence one another over time, and generate feedback loops (either reinforcing or balancing) that shape system behavior (Sterman, 2000).

In CLDs, variables are connected by arrows that indicate the direction and nature of influence: positive links denote changes in the same direction, while negative links imply opposing effects (Strelkovskii e Rovenskaya, 2021). These diagrams are particularly useful for identifying feedback structures and leverage points in systems where interventions often produce non-linear and unintended consequences.

Since the 1970s, CLDs have been applied in public health to model complex problems such as drug addiction (Levin, Roberts e Hirsch, 1975), infectious disease dynamics (Ritchie-Dunham e Galván, 1999), and healthcare policy design (Royston et al., 1999). These early studies demonstrated how feedback mechanisms explain persistent challenges in health systems and highlighted the importance of visualizing systemic interdependencies for policy planning.

From the mid-2000s onward, public health scholars increasingly advocated for systems thinking, and CLDs gained traction as tools to improve health systems analysis (Homer e Hirsch, 2006; Leischow e Milstein, 2006; Savigny e Adam, 2009). More recently, CLDs have been widely applied to explore maternal and child health issues across diverse contexts. For instance, Zafar et al. (2023) mapped disruptions to MCH services during the COVID-19 pandemic in Pakistan, while Cassidy et al. (2021) examined how pay-for-performance schemes influenced MCH coverage in Tanzania. In Malawi, Hamon et al. (2022) used CLDs to analyze the integration of family planning with child immunization services. Lemke et al. (2023) collaborated with communities in Texas to investigate how structural racism contributes to maternal morbidity among Black women. Similarly, Brereton e Jagals (2021) explored the complex causal relationships linking poverty, environmental factors, and child health outcomes in low- and middle-income countries.

Recent studies have increasingly integrated realist evaluation and systematic reviews with Causal Loop Diagrams (CLDs) to better understand how contextual factors and mechanisms interact to produce outcomes in complex health systems. For example, Ehteshami et al. (2024) combined realist logic with CLDs to model care pathways for diabetes and hypertension in low- and middle-income countries (LMICs), while Mukumbang et al. (2022) employed critical realist synthesis to uncover generative mechanisms in community-based care. Poix e Elmusharaf (2023) used meta-analyses to construct CLDs linking preconception care to maternal and child health outcomes. Similarly, Renmans, Holvoet e Criel (2020) and Singh et al. (2021) applied realist-informed CLDs to analyze performance-based financing schemes, demonstrating how interventions operate differently across settings depending on local context.

In Brazil, Causal Loop Diagrams (CLDs) have been increasingly used to examine complex dynamics within the Sistema Único de Saúde (SUS). Carpes (2024) developed a CLD based on interviews and process data to investigate failures in surgical scheduling, highlighting how issues related to communication, training, and technology contribute to poor data quality, which in turn reinforces ineffective decision-making. Similarly, Bergiante et al. (2024) mapped national-level causal links in the organ transplant system, revealing systemic inefficiencies. By combining CLD

analysis with Data Envelopment Analysis (DEA), their study identified mismatches between resources and outcomes across states, demonstrating the value of integrating qualitative systems mapping with quantitative performance evaluation to uncover feedback-driven bottlenecks in health service delivery.

According to Mueller (2015), the use of complex adaptive systems modelling techniques in the public policy cycle in Brazil remains incipient, despite growing academic interest in the topic. He argues that this gap reflects, in part, the relatively early diffusion of complexity ideas and, above all, the tension between the dominant policy-making logic (characterised by central planning, target-setting, control, and a strong reliance on information) and the nature of complex systems, in which heterogeneous agents interact locally under bounded rationality, producing emergent outcomes that are difficult to anticipate and control. In this context, Mueller (2015) contends that complex systems modelling can help reorient Brazilian public policies by shifting the focus from the “precise control” of outcomes toward the capacity to influence systemic properties (e.g., interdependence, connectivity, diversity, and adaptability). This is particularly relevant for public health and for Brazil’s Unified Health System (SUS), a prototypically complex domain with multiple levels of governance, service networks, budget constraints, and adaptive responses by professionals and users, where linear interventions and single metrics often generate unexpected effects and, at times, perverse incentives (Mueller, 2015).

The CLD presented in this study was developed based on the evidence synthesized in the Realist-Informed Systematic Review (Chapter 2), which assessed the impacts of flood events on maternal and child healthcare utilization. That review provided a comprehensive understanding of how floods affect both the demand for and supply of health services, shaping MCH outcomes through complex, interacting mechanisms.

Drawing from the literature, we identified key variables and causal processes that influence maternal and child healthcare (MCH) during extreme weather events. These processes, which we refer to as mechanisms or pathways, represent the chains of cause and effect linking exposure to floods and heatwaves with changes in MCH utilization. These insights informed the structure of our Causal Loop Diagram (CLD), allowing us to map dynamic interactions, feedback loops, and potential policy leverage points. The CLD was developed using Kumu (2026), a free, web-based platform for building interactive systems maps. It visually represents the components and connections that shape how floods affect healthcare access and delivery.

3.3 RESULTS

Based on the previously conducted systematic literature review (Chapter 2), we identified eighteen components distributed in six main mechanisms that link exposure to floods and heatwaves to outcomes related to the utilization of maternal and child health (MCH) services. These mechanisms provide a detailed view of the dynamics that will be presented in the complete CLD: (M1) restriction to travel, (M2) displacement and migration, (M3) income loss, (M4) increased health risks, (M5) health service disruption and (M6) general infrastructure damage.

Furthermore, we highlight the interactions among these mechanisms, which play a crucial role in understanding the overall system dynamics. Every component definition is presented in Appendix 3.A.

Table 3.1 synthesises the empirical literature by aligning each study's extreme weather event and observed service outcomes with the mechanisms (M1–M6) identified in this thesis, while explicitly locating those CMO patterns within a structured context framework. This table therefore operationalises the realist logic of “what happens, through which mechanisms, and under what contextual conditions”, and provides the bridge from evidence synthesis to system mapping (realist programme theory to CLD structure), consistent with realist reporting standards and mechanism-oriented synthesis (Wong et al., 2013; Page et al., 2021).

A central implication of realist synthesis is that the same shock (e.g., flooding) can generate different utilisation outcomes because contexts condition whether and how mechanisms fire. For climate-related shocks, this is especially important: hazards interact with settlement patterns, economic vulnerability, climatic baselines, and institutional arrangements in ways that shape both demand-side and supply-side disruption, including infrastructure failures, workforce strain, and supply-chain interruptions (World Health Organization, 2022; Hartinger et al., 2025).

To make these contextual contingencies analytically transparent (and comparable across studies), we classify each study along four dimensions:

- a) **Population context (Rural/Urban):** Rurality is a first-order determinant of physical access because it proxies longer travel times, greater reliance on vulnerable transport routes, and thinner service networks (Dewi et al., 2024; Valente et al., 2026). In Table 3.1, rural and remote settings are where mobility constraints (M1) and displacement (M2) most consistently translate into declines in routine MCH utilisation (ANC, PNC, facility delivery). This aligns with broader evidence that extreme events disproportionately impair healthcare utilisation and outcomes in rural/remote populations via transport fragility and service scarcity.
- b) **Economic context (High/Middle/Low income):** Income level is a compact indicator of baseline resilience: household buffers (savings, insurance coverage, social protection), infrastructure quality, and surge capacity. Lower-income contexts tend to amplify income shocks (M3) and supply strain (M5), increasing the probability that disruptions propagate from short-term barriers to prolonged service interruption. Climate–health assessments emphasise that vulnerability and impacts are uneven and strongly conditioned by development and institutional capacity (Dagneau, Ehgartner e Gulis, 2025; Hartinger et al., 2025).
- c) **Climate (Köppen–Geiger classification):** Using the Köppen–Geiger system allows us to place each extreme weather event within a comparable climatic regime defined by long-term temperature and precipitation patterns. This matters because impacts depend on what populations and health systems are accustomed to. Seasonality shapes adaptation:

historically hot regions are more likely to have cooling infrastructure and preparedness for heat stress, whereas colder regions may be more vulnerable as temperatures rise and heatwaves become more frequent. Likewise, changing precipitation patterns can make previously low-risk areas flood-prone, where infrastructure and emergency systems were not designed for such events. Classifying studies by Köppen zones therefore captures both baseline climate conditions and the degree of preparedness embedded in local systems, helping explain why similar hazards trigger different mechanisms and vulnerabilities across settings (Günsche et al., 2026; Lima et al., 2023).

- d) **Health system (Universal/Mixed/Private):** Health system type conditions both who can access care under stress and how services adapt. Mixed systems (where government services operate alongside private markets) are common in LMICs and are explicitly discussed in the mixed-systems literature; they may create fragmented surge response, uneven financial protection, and heterogeneous service availability across providers. Conversely, universal/publicly administered arrangements may offer stronger baseline financial protection and coordinated response capacity, while market-oriented systems can exhibit utilisation shifts that are more visible in emergency settings and administrative datasets. Comparative work on system typologies and public–private configuration supports the analytical value of classifying systems along this axis when interpreting access and utilisation patterns (Reibling, Ariaans e Wendt, 2019; Molander, 2025).

These four dimensions define the context space within which CMOs are interpreted. They also guide the CLD: the same mechanism (e.g., M1 travel impairment) can yield different feedback dynamics depending on rurality, poverty, climate regime, and system financing/provision structure.

Table 3.1 – Context-Mechanism-Outcome Configurations

Reference	EWE	Setting	Population	Economy	Climate	Health System	Mechanism	Outcomes	Effect
Abdullah et al. (2019)	Riverine flooding	Netrakona, Bangladesh	Rural	Low-income	Tropical monsoon	Mixed	M1, M4, M5	Institutional deliveries / Obstetric emergencies	Decrease / Increase
Amoo et al. (2018)	Riverine flooding	Edo, Kogi and Benue, Nigeria	Rural flood-displaced persons	Low-income	Tropical (humid and savanna)	Mixed	M2, M3	ANC	Decrease
Barman, Sarife Saha (2023)	Riverine flooding	India	Urban and rural hazard-prone districts	Lower-middle income	Tropical and subtropical monsoon	Mixed	M2	PNC	Decrease
Baten et al. (2020)	Riverine flooding	Bangladesh	Urban and rural districts	Low-income	Tropical monsoon	Mixed	M2, M3, M4, M5	ANC / PNC / Institutional delivery	Decrease
Bishwakarma (2022)	High-temperature days	Nepal	Rural remote areas	Low-income	Tropical, subtropical, temperate, and alpine	Mixed	M1, M3	ANC	Decrease
Campbell et al. (2019)	Heatwave	Tasmania, Australia	Densely populated urban regions	High-income	Temperate maritime	Universal	M4, M5	U5 ED visits	Increase
Chandir et al. (2023)	Riverine flooding	Sindh, Pakistan	Urban and rural districts	Low-income	Hot desert and semi-arid	Mixed	M5, M6	Childhood immunization	Decrease

Continued on next page

Table 3.1 continued

Reference	EWE	Setting	Population	Economy	Climate	Health System	Mechanism	Outcomes	Effect
Chen et al. (2022)	High-temperature days	Guangzhou, China	Mostly urban	Upper-middle income	Humid subtropical monsoon	Near universal	M3, M4	Newborn hospitalization costs	Increase
Cil e Kim (2022)	Hot days	United States	All states	High-income	Highly diverse	Market oriented	M4	NICU admissions	Increase
Codjoe et al. (2020)	Extreme heat / Flash flooding	Accra and Tamale, Ghana	Urban communities	Low-income	Tropical savanna	Mixed	M1, M3, M4, M5, M6	Routine MCH care	Decrease*
Davis et al. (2020)	Heatwave	Roanoke (Virginia), United States	Moderately populated urban area	High-income	Humid subtropical	Market oriented	M4, M5	U5 ED visits	Increase
Firoz et al. (2016)	Riverine flooding	Gaza and Maputo, Mozambique	Rural areas	Low-income	Tropical savanna	Public	M1, M4, M5	Routine MCH care	Decrease*
Haque, Parr e Muhidin (2019)	Riverine flooding	Sirajganj and Rajbari, Bangladesh	Rural households	Low-income	Tropical monsoon	Mixed	M2	Routine MCH care	Decrease
Haque, Parr e Muhidin (2020a)	Riverine flooding	Sirajganj and Rajbari, Bangladesh	Rural households	Low-income	Tropical monsoon	Mixed	M2	ANC	Decrease
Haque, Parr e Muhidin (2020b)	Riverine flooding	Sirajganj and Rajbari, Bangladesh	Rural households	Low-income	Tropical monsoon	Mixed	M2	Institutional delivery / PNC	Decrease

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Table 3.1 continued

Reference	EWE	Setting	Population	Economy	Climate	Health System	Mechanism	Outcomes	Effect
Jacquet et al. (2016)	Riverine flooding	Balochistan, Khyber Pakhtunkhwa, Punjab, and Sindh, Pakistan	Rural	Low-income	Arid to monsoon subtropical	Mixed	M2, M3	Institutional delivery / Skilled birth attendance	Decrease
Kakkad et al. (2014)	Heatwave	Ahmedabad, India	Urban	Low-income	Hot semi-arid	Mixed	M4, M5	NICU admissions	Increase
Kim, Lee e Rossin-Slater (2021)	High-temperature days	Arizona, New York, and Washington, United States	Urban areas	High-income	Arid and semi-arid (AZ); humid (NY); temperate marine (DC)	Market oriented	M4, M5	Maternal hospitalization	Increase
Makanga et al. (2017)	Riverine flooding	Gaza and Maputo, Mozambique	Rural remote areas	Low-income	Tropical savanna	Public	M1	Maternal care access	Decrease
Mroz et al. (2023)	Riverine flooding	Barotse Floodplain, Zambia	Remote semi-nomadic	Low-income	Tropical savanna	Public	M1	Institutional delivery / Obstetric emergencies	Decrease
Orderud et al. (2022)	Riverine flooding	Bangladesh	Rural and urban districts	Low-income	Tropical monsoon	Mixed	M2, M3, M6	ANC / Institutional delivery	Null
Peyravi, Rahnavard e Marzaleh (2020)	Flash flooding	Golestan, Iran	Rural and urban regions	Lower-middle income	Mediterranean/temperate to semi-arid	Mixed	M4, M6	Routine MCH care	Decrease*

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Table 3.1 continued

Reference	EWE	Setting	Population	Economy	Climate	Health System	Mechanism	Outcomes	Effect
Purno et al. (2023)	Riverine flooding	Sylhet, Sunamganj, and Netrakona, Bangladesh	Rural	Low-income	Tropical monsoon	Mixed	M2, M5	SRHR services (Family planning, ANC, emergency obstetric)	Maintained (post-intervention)*
Quilty et al. (2024)	High-temperature days	Northern Territory, Australia	Remote northern territory	High-income	Tropical savanna to arid	Universal	M1, M4, M5	Obstetric emergencies	Increase
Ray-Bennett et al. (2021)	Riverine flooding	Belkuchi, Bangladesh	Rural area	Low-income	Tropical monsoon	Mixed	M5	Post-abortion care (PAC)	Maintained (post-intervention)*
Sadia et al. (2016)	Riverine flooding	Nowshehra, Pakistan	Under-resourced rural area	Low-income	Semi-arid with monsoon	Mixed	M1, M2, M3, M5	ANC / PNC / Institutional delivery	Decrease
Saulnier et al. (2020)	Riverine flooding	Cambodia	Rural villages	Low-income	Tropical monsoon	Mixed	M5	Institutional delivery	Decrease*
Saulnier et al. (2022)	Riverine flooding	Cambodia	Rural villages	Low-income	Tropical monsoon	Mixed	M5	ANC	Decrease*
Shah, Miller e Mothabbir (2019)	Coastal/riverine flooding	Bhola, Bangladesh	Rural district	Low-income	Tropical monsoon	Mixed	M1, M6	Routine MCH care	Decrease
Smith et al. (2016)	Heatwave	England	All regions	High-income	Temperate oceanic	Universal	M4	U5 ED visits	Non-significant

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Table 3.1 continued

Reference	EWE	Setting	Population	Economy	Climate	Health System	Mechanism	Outcomes	Effect
Stowell et al. (2023)	High-temperature days	United States	Children with commercial health insurance	High-income	Highly diverse	Market oriented	M4	U5 ED visits	Non-significant
Surekha, Anbazhagan e Kiran (2020)	Riverine flooding	Bihar, India	Rural	Low-income	Humid subtropical	Mixed	M2, M4	Routine MCH care	Decrease*
Wilunda et al. (2016)	Riverine flooding	Rumbek North, South Sudan	Semi-nomadic (pastoralism)	Low-income	Tropical savanna	Fragile, aid-dependent	M1, M5, M6	Institutional delivery	Decrease*
Wilunda et al. (2017)	Riverine flooding	Rumbek North, South Sudan	Semi-nomadic (pastoralism)	Low-income	Tropical savanna	Fragile, aid-dependent	M1, M5, M6	ANC	Decrease*
Xu et al. (2014)	Heatwave	Brisbane, Australia	Urban	High-income	Subtropical	Universal	M4	U5 ED visits	Increase
Xu et al. (2017)	Heatwave	Brisbane, Australia	Urban	High-income	Subtropical	Universal	M4	U5 ED visits	Increase

Note: M1 denotes restriction to travel; M2, population displacement; M3, income loss; M4, increased health risks; M5, health service disruption; and M6, general infrastructure disruption. U5 ED refers to emergency department visits among children under five years of age. Economy classification follows World Bank income groups at the time of the study. Health system typology reflects the dominant financing and coverage structure. **Quantitative estimates were not reported; effect classifications are based solely on qualitative evidence documenting service disruption, access barriers, or reduced care delivery.*

3.3.1 Restrictions to travel (M1)

Mechanism 1, illustrated in Figure 3.1, links floods and heatwaves to MCH service utilisation by damaging transportation infrastructure, constraining physical mobility, and reducing willingness to travel under extreme conditions. Together, these factors shape whether, when, and what type of care women and children can access. Travel restrictions affect MCH services asymmetrically, lowering routine service use while increasing the likelihood that care (if accessed at all) occurs late and through emergency pathways.

Figure 3.1 – Restriction to Travel Mechanism (M1)



Source: Prepared by the author.

3.3.1.1 Floods

Floods frequently damage roads and bridges, and disrupt transportation systems, cutting off access to healthcare services by preventing ambulances or other vehicles from traveling (Wilunda et al., 2016). These challenges are particularly acute for pregnant women. In obstetric emergencies, when vehicles cannot reach communities due to flooding, women often rely on improvised transport, such as being carried by family members (Wilunda et al., 2016). In seasonal flooding during the rainy season in Rumbek North County, South Sudan, water levels can rise to shoulder height, making it physically impossible for pregnant women to walk or swim to safety and rendering access to health facilities infeasible due to submerged terrain, washed-away roads, and unsafe travel conditions (Wilunda et al., 2017). Whereas urban regions may offer multiple routes and have more health facilities, rural populations are left with limited alternatives and therefore face heightened risks of maternal and neonatal morbidity and mortality due to geographic isolation (Mroz et al., 2023).

For example, in rural southern Mozambique, seasonal flooding transforms already deteriorated roads into impassable terrain, leaving women with few transport options and often forcing long, difficult journeys on foot or reliance on costly emergency travel. In these settings, community members frequently step in to offer accompaniment or lend vehicles, which underscores the critical role of social support (Firoz et al., 2016). Modelling work further shows

that floods can drastically extend travel times and, in some cases, completely isolate communities for weeks or even months, denying pregnant women both routine and emergency care when it is needed most (Makanga et al., 2017). Similarly, during the major 2010 Pakistan floods, widespread road damage and inundated clinics severely disrupted service availability and imposed long travel distances (often more than five kilometres), while limited financial resources prevented families from accessing private care, leaving pregnant women and young children dependent on strained public facilities that were themselves difficult to reach (Sadia et al., 2016).

Economic conditions critically shape whether families can overcome flood-related transport barriers to reach care. In Bangladesh, for example, emergency boat travel may be available during floods, but it is often prohibitively expensive as boatmen demand a high amount of money for the emergency boat transportation (Abdullah et al., 2019). Hence poorer households are especially vulnerable to these constraints. Submerged roads and damaged transport networks can force clinics to reduce hours or relocate, deplete essential supplies, increase referrals, and ultimately leave many pregnant women and young children unable to complete those referrals because alternative transport options are too costly, creating major financial barriers to accessing essential maternal and child healthcare (Shah, Miller e Mothabbir, 2019).

3.3.1.2 Heatwaves

Bishwakarma (2022) suggests that in rural Nepal, where women often walk long distances or rely on limited transport, high temperatures may discourage ANC attendance by increasing physical discomfort during travel. However, the evidence remains tentative. The study finds an inverse association between high-temperature days during gestation and total ANC visits but cannot determine whether heat causes a sustained reduction in care or merely postpones visits to cooler periods. Because it relies on aggregate counts without examining timing or substitution, the persistence, clinical relevance, and underlying mechanism of the observed effect remain unclear.

Quilty et al. (2024) examine the association between high temperatures and aeromedical retrievals in the Northern Territory of Australia, a highly specific setting characterised by vast distances, limited local clinical capacity, and reliance on air transport for emergency care. The study finds a dose-response increase in obstetric retrievals during hotter periods in the humid zone and in surgical retrievals in the arid zone, pointing to heightened heat-related morbidity rather than direct disruption of care. Although retrieval services are reported to operate at or near capacity year-round, the evidence does not demonstrate observed service overload, delays, or unmet need; moreover, because heat exposure is operationalised using a 3-day lagged mean temperature and local quartiles, the analysis does not distinguish discrete heatwave events from broader seasonal heat. The authors therefore infer that increased demand for high-urgency obstetric retrievals may constrain capacity for other cases, indicating a potential system-level vulnerability in emergency transport logistics under extreme heat, rather than demonstrating observed delays or denial of care.

3.3.2 Population displacement (M2)

To date, Mechanism 2 is unique to floods. As illustrated in Figure 3.2, it links flooding to MCH service utilisation through displacement, understood as forced evacuation, temporary sheltering, or longer-term migration following the destruction of homes, land, and livelihoods. Displacement affects MCH service use by disrupting continuity of care, straining health system functioning, and altering women and children's ability and willingness to seek services. In this way, Mechanism 2 captures the direct impacts of flooding on households and property and their downstream consequences for MCH service utilisation.

Figure 3.2 – Population Displacement Mechanism (M2)



Source: Prepared by the author.

Floods can result in widespread destruction of properties and homes, leaving people homeless and without livelihoods. Loss of or damage to housing forces evacuations (Jacquet et al., 2016). After the 2022 Bangladesh floods, many community members took shelter inside health facilities, which further disrupted service delivery and contributed to temporary loss of routine SRHR and emergency perinatal services (Purno et al., 2023).

In flood prone regions, the increased frequency and intensity of floods, along with riverbank erosion, led to the complete destruction of land and homes, making it impossible for many families to return to their homes (Haque, Parr e Muhidin, 2019). This phenomenon is linked to migration, driven by factors such as food insecurity, social marginalization, and lack of access to institutional services, including healthcare and transportation (Haque, Parr e Muhidin, 2020b).

Migration resulting from displacement has a direct impact on the utilisation of MCH services. Migrant women, especially those moving from rural to urban areas or living in precarious settlements, face significant barriers to accessing antenatal care. These include a perceived low need for care, lower sociocultural status, reduced social networks, lack of awareness of available health services, and socioeconomic instability (Haque, Parr e Muhidin, 2020a). Moreover, newly migrated women tend to use healthcare services less frequently than local resident populations, with lower rates of antenatal visits among rural migrants (Haque, Parr e Muhidin, 2020a).

Infrastructure damage further intensifies displacement pressures. Floods can destroy roads, bridges, and communication networks, halting economic activity and prolonging disruption long after waters recede, often forcing households to relocate in search of basic services (Amoo et al., 2018). Economic resources also shape mobility options: wealthier families can sometimes evacuate early or employ protective strategies, whereas poorer households are more likely to be permanently displaced without adequate access to care (Barman, Sarif e Saha, 2023). Affluent

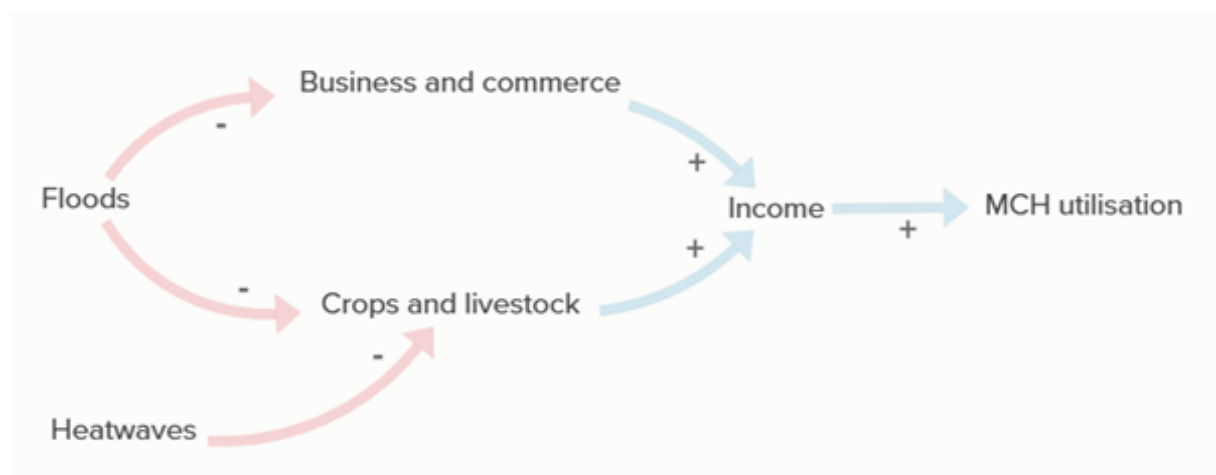
populations often avoid flood-prone areas, whereas more vulnerable communities remain in these high-risk regions, increasing their exposure to disasters and their consequences (Orderud et al., 2022).

Disaster preparedness directly affects whether pregnant women and children can travel to safety in an emergency. In flood-prone areas with no prior evacuation plans, roads can become impassable or unsafe, preventing timely relocation before or during a flood (Surekha, Anbazhagan e Kiran, 2020). Baten et al. (2020) emphasize early evacuation for pregnant women to minimize exposure to severe events, a strategy that inherently depends on having reliable transportation routes and methods. However, as Sadia et al. (2016) note, women are often excluded from practical drills and skill-building activities, such as swimming lessons or training on navigating collapsed structures.

3.3.3 Household income loss (M3)

Mechanism 3, shown in Figure 3.3, captures how floods and heatwaves affect agriculture, livestock, businesses, and property, thereby reducing household income and limiting families' capacity to access MCH services. Income loss influences MCH utilisation by lowering the ability to pay for care, shifting household spending away from preventive services, and encouraging delays or substitution with lower-cost alternatives, resulting in fewer routine contacts and greater reliance on late, informal, or emergency care, effects that are most clearly documented for floods and suggested more indirectly for heat.

Figure 3.3 – Income Loss Mechanism (M3)



Source: Prepared by the author.

3.3.3.1 Floods

Floods often devastate crops and livestock in farming communities, leaving rural populations who rely on the land for their livelihoods in extreme vulnerability and in greater need of humanitarian assistance (Jacquet et al., 2016). This economic impact is aggravated by the

collapse of essential infrastructure, such as roads and bridges, and by the shutdown of agricultural and industrial activities; resulting in a significant loss of jobs and income as noted in studies where up to 70% of people lost their means of subsistence after floods (Amoo et al., 2018).

The reduction in post-flood income directly affects families' financial capacity to access health services. In many cases, the cost of medical care is identified as the main barrier to obtaining required services. For example, over half of flood-affected families reported having to pay for health services even when their monthly income post-disaster was extremely low, rendering those costs unfeasible within many families' economic context (Jacquet et al., 2016). In the case of pregnant women, a lack of financial resources was cited as the primary reason for not seeking antenatal care in affected communities (Amoo et al., 2018). Orderud et al. (2022) indicate that households affected by flooding often prioritize obtaining sufficient food and water, leading to fewer resources available for recommended maternal healthcare, which may lead to worse health outcomes.

Post-flood income has been identified as one of the main predictors of adequate access to health services, demonstrating that families with higher post-disaster income are more likely to obtain care, while those with lower incomes face significant barriers (Jacquet et al., 2016). During the rainy season, declining household income often drives families to seek care from lower-cost traditional healers who offer credit, but these less effective treatments can delay formal care and increase complications when patients eventually reach health facilities (Codjoe et al., 2020).

Floods intensify poverty in ways that disproportionately harm women and children. Families often withdraw children (especially girls) from school due to financial strain, increasing risks of early marriage, teenage pregnancy, and child labour (Sadia et al., 2016). At the same time, reliance on public health facilities that are frequently damaged or disrupted leaves most families unable to access essential MCH services, further heightening vulnerability during and after floods (Sadia et al., 2016).

The effects of floods on household income also affect nutrition. Baten et al. (2020) highlight that floods make food both scarce and expensive, which then hurts household finances and, over time, productivity. When income levels are stable (or can be supplemented by loans and social networks, as discussed by Saulnier et al. (2020)), households are more likely to secure sufficient nutrition, which in turn helps preserve their health and work capacity.

3.3.3.2 Heatwaves

Evidence that heat exposure affects maternal health service use though income and financial strain remains limited and indirect. In rural Nepal, Bishwakarma (2022) showed that high temperatures are associated with reduced ANC use, likely operating through strained household resources. In settings where households depend on agriculture and informal labour, hot days increase the prices of key foods due to reduced yields for some crops and faster spoilage, which gradually erodes purchasing power and constrain spending. This financial pressure may

lead women to delay or forgo preventive ANC, although the study does not directly observe income losses or disentangle these effects from broader seasonal responses to heat (Bishwakarma, 2022).

Chen et al. (2022) highlighted a related, but downstream, mechanism in China: exposure to extreme heat late in pregnancy increases newborn hospitalization costs by nearly 2% of total health expenses. While this reflects higher expenditures rather than lost income, it similarly reduces household financial capacity and exacerbates inequalities, with greater effects on less-educated mothers.

3.3.4 Increased health risks (M4)

Mechanism 4, as illustrated in Figure 3.4, links to MCH utilisation through two main routes that can operate at the same time: (i) demand shifts (more acute illness and complications increase the need for care) and (ii) effective access falls (the health system and households have less capacity to deliver or attend MCH services).

Figure 3.4 – Increased Health Risks Mechanism (M4)



Source: Prepared by the author.

3.3.4.1 Floods

The evidence suggests that exposed populations may suffer physical injuries and infections caused by contaminated water, such as diarrhoea, cholera, typhoid fever, and other gastrointestinal illnesses, which become more prevalent in areas with inadequate sanitation, overcrowding, and displacement after floods (Surekha, Anbazhagan e Kiran, 2020; Baten et al., 2020). These disease outbreaks, often aggravated by lack of potable water and contamination of water sources, undermine the health of affected populations and increase the demand for both emergency and routine care.

The destruction of crops and livestock also reduces the availability of food, leading to widespread hunger and malnutrition in affected communities. One study reported that flooding

forced millions of people to consume raw and contaminated foods, resulting in widespread illness and malnutrition (Surekha, Anbazhagan e Kiran, 2020). Poor breastfeeding practices and inadequate complementary feeding for children are more likely during floods, worsening malnutrition rates among infants (Baten et al., 2020).

Furthermore, in rural Bangladesh, Abdullah et al. (2019) show that prolonged labour, haemorrhage, and other preventable complications often escalate during floods because women have limited ANC, complications are not promptly recognized, and skilled birth attendants are unavailable. When emergencies arise, families first seek help from untrained providers, causing critical delays in decision-making. Even when referral is attempted, transportation by boat is slow, risky, and costly (M1), frequently resulting in women delivering or dying en-route. These compounding delays (in recognition, decision to seek care, and transportation) heighten maternal mortality and morbidity risk in flood-affected settings where timely access to emergency obstetric care is limited.

In rural southern Mozambique, floods were perceived to heighten maternal and newborn health risks by promoting mosquito-borne diseases such as malaria and disrupting access to clean water and sanitation, thereby increasing vulnerability to harmful infections (Firoz et al., 2016). In northern Iran, floods increased psychological and physical stress for mothers, disrupting fertility, and heightening risks to pregnancy and early life (Peyravi, Rahnavard e Marzaleh, 2020).

3.3.4.2 Heatwaves

Heat exposure during pregnancy may trigger physiological stress responses, such as dehydration, reduced placental blood flow, and thermoregulatory strain, which can increase risks of hypertensive disorders and preterm labor (Chen et al., 2022; Kim, Lee e Rossin-Slater, 2021). These pathways lead to adverse birth outcomes including low birth weight, stillbirth, and preterm delivery, which elevate neonatal intensive care use (Cil e Kim, 2022). During extreme heat events, acute obstetric complications can escalate rapidly, driving greater demand for emergency care (Quilty et al., 2024). As a result, heatwaves disrupt both routine and emergency maternity services through increased admissions and visits, with socially vulnerable groups, including Black women, disproportionately affected by these disruptions (Kim, Lee e Rossin-Slater, 2021).

Young children face health risks during extreme heat due to their immature thermoregulatory systems and higher fluid needs. Studies consistently show increases in pediatric emergency visits during heatwaves, particularly among children under-5, driven by dehydration, respiratory distress, heat illness, and infections (Campbell et al., 2019; Davis et al., 2020; Smith et al., 2016). Heat exposure has also been linked to rises in intestinal infections, endocrine and metabolic disorders, nervous system conditions, and chronic respiratory diseases among this age group, reflecting combined effects of dehydration, bacterial growth, and thermal stress on multiple organ systems (Xu et al., 2014). Xu et al. (2017) shows that infants' hospital admissions only rose when heatwaves were very intense (mean temperature ≥ 97 th percentile for ≥ 2 days). The effect did not differ by sex, indigenous status or neighbourhood socioeconomic level; indicating that all infants

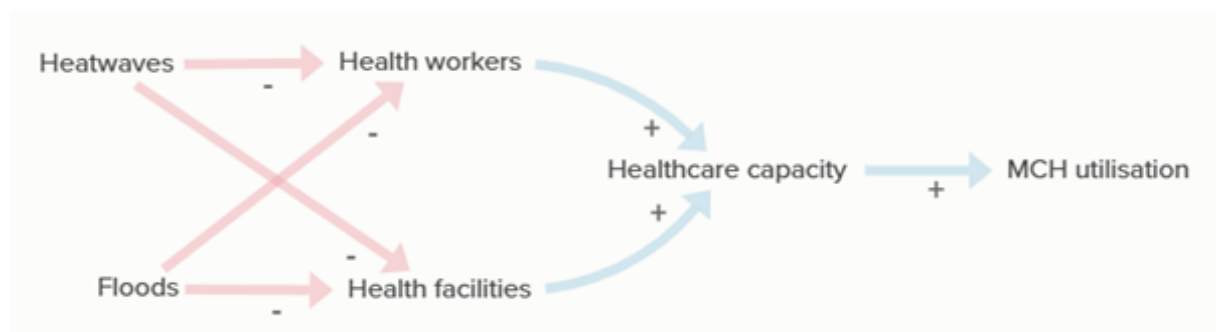
are at risk.

Structural vulnerabilities in health facilities further intensify health risks associated with heatwaves. Poor ventilation, unreliable power for cooling and vaccine storage, and overcrowded wards create unsafe conditions and may compromise treatment quality (Codjoe et al., 2020). In settings with widespread undernutrition, heat episodes additionally elevate the risk of neonatal hyperthermia, as seen in India, where poor nutritional status reduces the body's ability to cope with thermal strain (Kakkad et al., 2014). Overall, heatwaves threaten children and pregnant women through both direct physiological impacts and indirect disruptions to the quality and safety of care environments.

3.3.5 Health service disruption (M5)

Mechanism 5, illustrated in Figure 3.5, captures how floods and heatwaves disrupt the supply of health services by reducing the availability of health professionals and the operational capacity of health facilities, thereby limiting the delivery of MCH services. Health service disruption affects MCH service use by altering the feasibility and quality of care on the supply side, not merely people's need or willingness to seek services. Floods and heatwaves create a supply shock through staff shortages, reduced functional space and equipment, and interrupted supplies and referral pathways – often alongside a simultaneous surge in demand from injuries, infections, dehydration, and pregnancy complications. Together, these pressures reduce routine service provision and shift utilisation toward delayed or emergency care.

Figure 3.5 – Health Service Disruption Mechanism (M5)



Source: Prepared by the author.

3.3.5.1 Floods

Flooding disrupts multiple components of the health system, reducing workforce availability, constraining facility capacity, and ultimately limiting MCH service utilisation. A key disruption relates to health worker mobility: in flood-prone districts, staff struggle to commute due to road blockages and unsafe travel conditions, resulting in delayed arrivals or absenteeism, as reported in Cambodia and Mozambique (Saulnier et al., 2020; Firoz et al., 2016). In addition, some health professionals are themselves displaced following floods, further reducing the availability

of skilled providers for essential services such as antenatal care and skilled delivery (Abdullah et al., 2019). In setting such as Pakistan, where baseline shortages of skilled staff and uneven community health worker coverage already constrain service delivery, flood-related closures and understaffing leave pregnant women with few viable alternatives (Sadia et al., 2016).

Floods also disrupt the availability and functionality of healthcare infrastructure. Health facilities may be flooded, damaged, or destroyed, with losses of medical equipment, supplies, and usable clinical space (Abdullah et al., 2019; Codjoe et al., 2020; Orderud et al., 2022). Even where facilities remain operational, care is often delivered under precarious conditions, including muddy walkways, water entering wards, unusable beds, compromised sanitation and electrical systems, which slow clinical workflows and reduce the safety and quality of care (Codjoe et al., 2020). Repeated flooding in Bangladesh and Pakistan has caused recurrent infrastructure losses and slow recovery timelines, increasing vulnerability to cascading failures when subsequent events occur (Baten et al., 2020; Sadia et al., 2016).

These infrastructure and workforce disruptions are compounded by weakened system coordination. Flooding reduces supervision of community health workers, undermining outreach and continuity of care (Shah, Miller e Mothabbir, 2019). Limited public investment and poor geographical targeting further contribute to chronically under-resourced services in flood-prone areas, constraining adaptive capacity even outside disaster periods (Orderud et al., 2022). Blocked roads and rising transport costs also impede inter-facility referrals, while hospitals report recurring “no-bed syndrome,” with patients transferred repeatedly in search of available space (Codjoe et al., 2020). In contexts heavily reliant on public hospitals, such disruptions push women toward informal providers and increase the likelihood of home delivery without skilled attendance (Sadia et al., 2016).

Despite these challenges, some health systems demonstrate adaptive capacity. Social networks and community-based responses can mitigate service disruption: in Cambodia and Bangladesh, community volunteers facilitated emergency transport, supported communication, and enabled health workers to reach patients despite flooding (Saulnier et al., 2022; Ray-Bennett et al., 2021). Targeted deployment of midwives, supportive supervision, and restoration of medical supplies have proven effective in maintaining outreach and rebuilding service continuity in flood-affected settings (Purno et al., 2023).

Flooding often leads to significant logistical challenges that affect supply chains overall and disrupt the delivery of medical supplies. Floods frequently disrupt power systems and supply chains, rendering medical equipment such as emergency generators and vaccine refrigerators inoperable. In facilities located in flood-prone areas, the loss of medications, supplies, and equipment due to inundation is common (Codjoe et al., 2020). Supply chain interruptions and delays in transporting pharmaceuticals contribute to stock-outs and further restrict service provision (Wilunda et al., 2017; Chandir et al., 2023).

For instance, in some regions, essential commodities could not be readily transported from sub-district hospitals to community clinics due to submerged roads, forcing healthcare workers

to carry medicines manually, and ultimately causing stock-outs when demand increased (Shah, Miller e Mothabbir, 2019). Elsewhere, critical road infrastructure was washed away, preventing delivery of drugs and supplies to health facilities and rendering them unable to function (Wilunda et al., 2017). In addition, flooding hindered both access to healthcare centers and the delivery of vaccines, temporarily closing certain facilities and leaving carers unable to reach the services they needed (Chandir et al., 2023). In line with these disruptions, Orderud et al. (2022) emphasize that healthcare infrastructure may be flooded or damaged, and medical supply deliveries can be delayed.

3.3.5.2 Heatwaves

In low-resource settings, health workers often deliver care in overcrowded, poorly ventilated environments where high indoor temperatures, power outages and compromised refrigeration place both staff and patients at risk. In Ghana, Codjoe et al. (2020) show that extreme heat causes health worker fatigue, increases infection risks during procedures and contributes to delays in deliveries and outpatient care, while also interrupting vaccine and medicine cold chains.

Heatwaves also drive sharp surges in emergency care use among children. In Tasmania, severe and extreme heatwaves increased paediatric ED visits for heatstroke and sunburn, overwhelming routine capacity and limiting responsiveness to other urgent needs (Campbell et al., 2019). Similar patterns were observed in Roanoke, Virginia, where heatwaves “overtaxed” ED systems, leading to crowding, poorer quality of care and increasing reliance on costly emergency pathways (Davis et al., 2020).

Maternal care is similarly strained during extreme heat. In the United States, unplanned emergency admissions for pregnancy complications rise on extremely hot days, increasing workload for obstetric teams and reducing availability for scheduled care (Kim, Lee e Rossin-Slater, 2021). In remote tropical regions such as northern Australia, extreme heat leads to more emergency obstetric retrievals when health systems are already operating at capacity. This can delay transfer of non-obstetric patients and place additional pressure on aeromedical teams and tertiary hospitals (Quilty et al., 2024).

Importantly, some evidence indicates that targeted, context-specific adaptations can mitigate heat-related risks and ease pressure on health systems. In a resource-constrained hospital setting in India, relocating a maternity ward from an overheated upper floor to a cooler ground level during a heatwave reduced neonatal intensive care admissions by 64%, demonstrating how simple structural adjustments can lower neonatal risk and reduce staff burden under extreme heat (Kakkad et al., 2014). More broadly, the literature suggests that anticipatory measures, such as heatwave forecasting, surge staffing plans, and caregiver-focused prevention campaigns, may help limit avoidable heat-related demand, although their effectiveness depends on local capacity, infrastructure, and governance contexts (Campbell et al., 2019; Davis et al., 2020).

Extreme heat places substantial strain on health service infrastructure and delivery, with particularly severe consequences in low-resource settings. In Ghana, poorly ventilated maternity

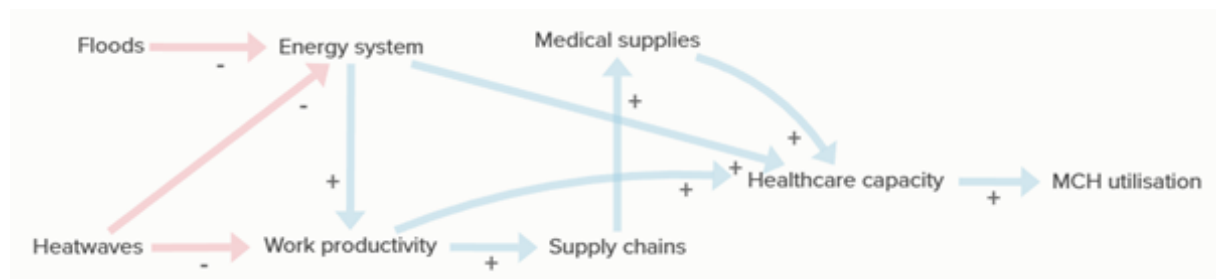
and primary care facilities have recorded indoor temperatures up to 6°C higher than outdoors, while frequent power outages disrupt water supply, infection control, and vaccine refrigeration, compromising the safety of deliveries and essential procedures and increasing staff fatigue; together, these failures in basic infrastructure undermine maternal and neonatal care and, in some cases, push patients toward costly private providers (Codjoe et al., 2020).

Similar pressures are observed in higher-income settings. During the 2013 heatwave in England, rapid surges in health service use overwhelmed emergency and primary care, reducing workforce productivity and straining energy systems that support equipment and cold-chain reliability (Smith et al., 2016). Across contexts, extreme heat drives spikes in emergency visits among children, overloading facilities and increasing dependence on energy-intensive cooling, which heightens risks of power instability and supply chain disruption (Stowell et al., 2023).

3.3.6 General infrastructure disruption (M6)

Mechanism 6, as illustrated in Figure 3.6, links floods and heatwaves to MCH service utilisation through disruption of energy systems, transport and supply chains, and workforce productivity, which together undermine the operational foundations required to deliver care. When infrastructure outside the health sector fails, these services may remain nominally available but become functionally inaccessible, unsafe, or unreliable, reducing effective utilisation.

Figure 3.6 – General Infrastructure Disruption Mechanism (M6)



Source: Prepared by the author.

3.3.6.1 Floods

Across flood-affected settings, the evidence shows that disruptions to general infrastructure are sufficient to reduce effective MCH service utilisation even when facilities remain physically intact. In Pakistan's 2022 floods, Chandir et al. (2023) document a sharp and persistent decline in immunization delivery in affected districts, with substantial shortfalls remaining months after floodwaters receded despite intensive catch-up campaigns. This pattern is consistent with an “operational foundations” mechanism: when routine outreach schedules, workforce deployment, and local service organisation are disrupted at scale, services do not simply pause and rebound, and missed care may not be fully recovered.

Evidence from Ghana (Codjoe et al., 2020) makes the energy and productivity pathway explicit. Flooding disrupted electricity supply not only through facility-level damage but because utilities pre-emptively disconnected power to prevent electrocution, while fuel supply interruptions rendered generators unusable. These energy disruptions constrained refrigeration, lighting, emergency procedures, and basic operations. At the same time, muddy and waterlogged environments reduced staff mobility within and between facilities, increased time spent on cleaning and recovery tasks, and diverted workforce effort away from routine care, directly lowering productivity even without structural facility collapse.

Supply-chain fragility further amplifies these effects. In rural Bangladesh, Shah, Miller e Mothabbir (2019) report increased stock-outs of essential commodities and reduced supervision during flooding months, alongside an approximate 30% decline in child treatments delivered by community providers. Their findings highlight how flooded transport routes interrupt medicine distribution, delay resupply, and weaken supervisory and reporting systems. Similarly, Orderud et al. (2022) synthesise evidence that floods damage transport and logistics networks, hinder medical supply deliveries, and disrupt referral systems, even if average utilisation effects are muted in chronically flood-prone areas where partial adaptation has occurred. In northern Iran, Peyravi, Rahnavard e Marzaleh (2020) describe the immediate post-flood period as dominated by logistical and coordination challenges, as the emergency health service continuity depends on functioning distribution systems, communication networks, and energy supplies.

Finally, qualitative evidence from South Sudan (Wilunda et al., 2016; Wilunda et al., 2017) shows that when roads are impassable for extended periods, referral and resupply systems stall, forcing improvised solutions and placing additional physical burdens on communities and health workers. This erosion of general infrastructure constrains utilisation and can generate both immediate service declines and incomplete recovery over time.

3.3.6.2 Heatwaves

The available evidence linking heatwaves to disruptions in general infrastructure remains limited and context-specific. In our review, the most detailed account comes from Ghana (Codjoe et al., 2020).

In the Ghanaian context, Codjoe et al. (2020) describe how extreme heat increases electricity demand for cooling, both in households and facilities, at the same time that national power systems face structural fragilities. Ghana's grid is characterized by periodic instability due to generation constraints, transmission losses, reliance on hydropower sensitive to rainfall variability, and limited reserve margins. During heat events, heightened demand for fans and air conditioning can trigger voltage fluctuations, localized outages, or load shedding. Health facilities dependent on prepaid electricity meters or diesel generators are particularly vulnerable: rising fuel costs and supply constraints may force rationing of power to essential units, interrupting lighting, refrigeration, and routine operations.

Heat also exposes weaknesses in temperature-sensitive supply chains. According to

Codjoe et al. (2020), vaccine refrigeration becomes unreliable when power outages coincide with peak heat, as freezers struggle to maintain stable internal temperatures and ice packs fail to refreeze adequately. Even where generators are available, fuel constraints limit continuous use. These disruptions reflect vulnerabilities in upstream energy and logistics systems rather than intrinsic failures of clinical care.

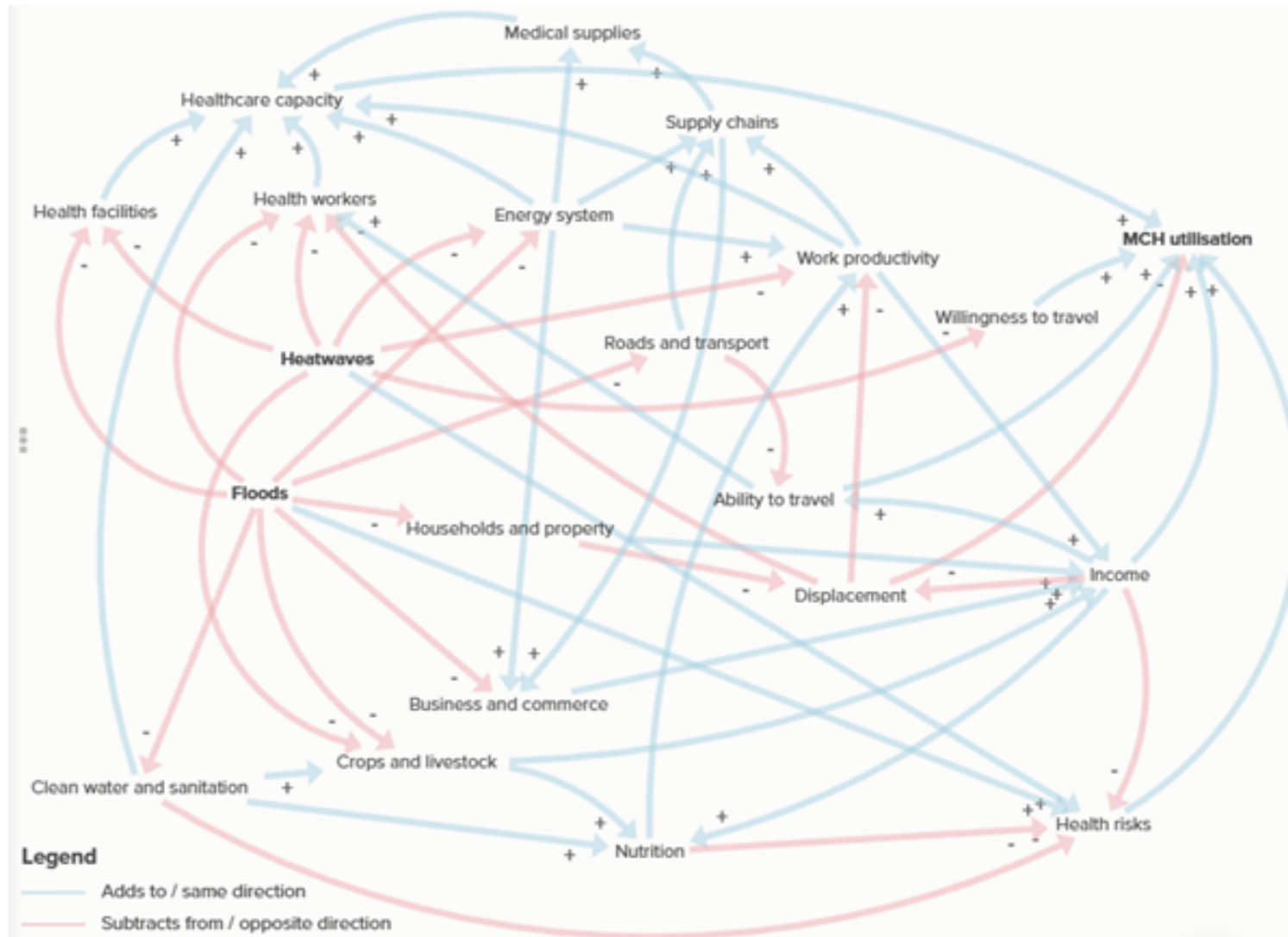
Finally, Codjoe et al. (2020) report that extreme heat reduces workforce productivity both inside and outside health facilities. High temperatures impair sleep, increase fatigue, and complicate clinical procedures due to excessive perspiration. Beyond the health sector, heat reduces working hours in informal and manual labor markets, lowering household income and constraining spending on healthcare. In settings where economic buffers are thin, this productivity loss may further weaken supply chains and service demand.

Overall, the evidence for an “operational stress” pathway under heatwaves is suggestive rather than definitive. It is grounded primarily in the Ghanaian case, where power systems are susceptible to demand surges and structural instability. While similar vulnerabilities likely exist in other LMIC contexts with fragile grids and limited backup capacity, further comparative research is needed to determine how broadly this mechanism applies to other settings.

3.3.7 Causal loop diagram

The mechanisms presented were initially described in a largely linear manner for analytical clarity. In practice, however, these mechanisms do not operate in isolation. Instead, they are interconnected and mutually reinforcing, such that changes in one mechanism can trigger responses in others. When these interactions are integrated into a single system representation, they give rise to multiple feedback loops and reflect the complexity of dynamic system behaviour, as illustrated in Figure 3.7.

Figure 3.7 – Causal Loop Diagram

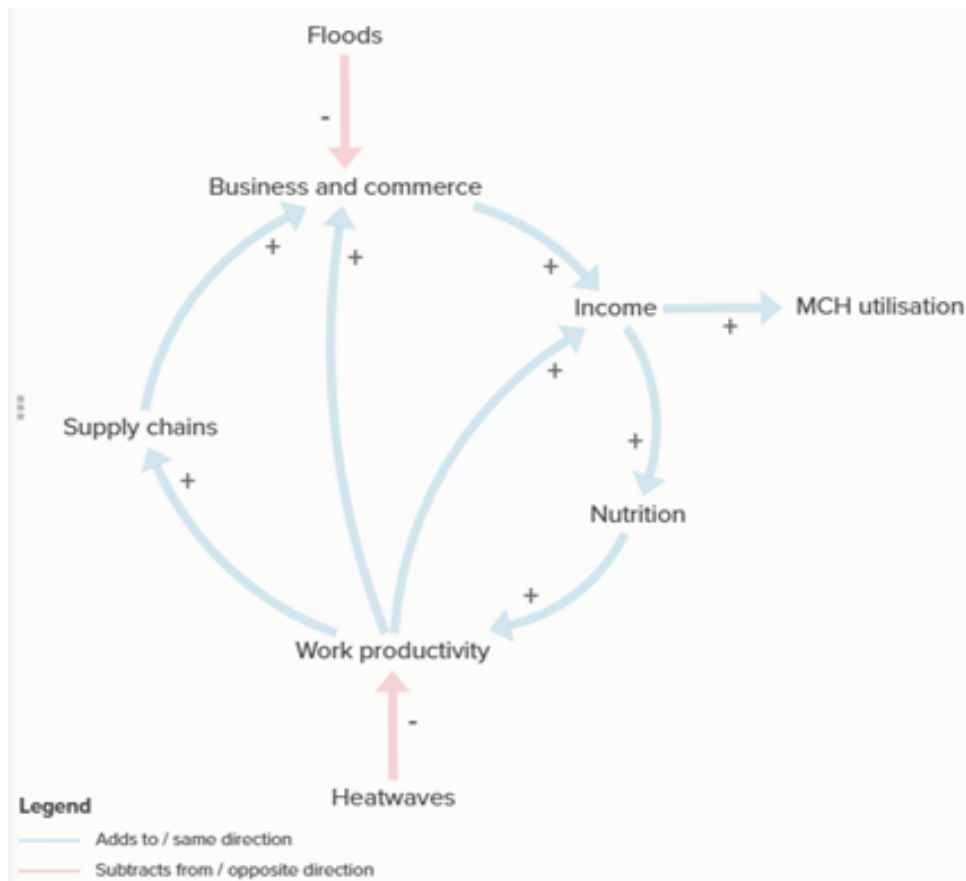


Note: Online version available at <https://kumu.io/ivan-silva/reach-v3>.

In the CLD, arrows indicate the direction of causal influence, rather than simple association or correlation. Blue arrows represent positive links from element X to Y, meaning that an increase (or decrease) in X tends to produce a change in Y in the same direction, while red arrows represent negative links, indicating that changes in X tend to produce an opposite change in Y. When chains of such causal links form closed paths, they generate feedback loops, which capture how system behaviour evolves over time. Reinforcing loops amplify change, potentially intensifying disruptions or inequalities, whereas balancing loops counteract change and can stabilise the system.

Figure 3.8 shows three nested reinforcing feedback loops linking work productivity, economic activity, and nutrition.

Figure 3.8 – Reinforcing Loops R1, R2 and R3



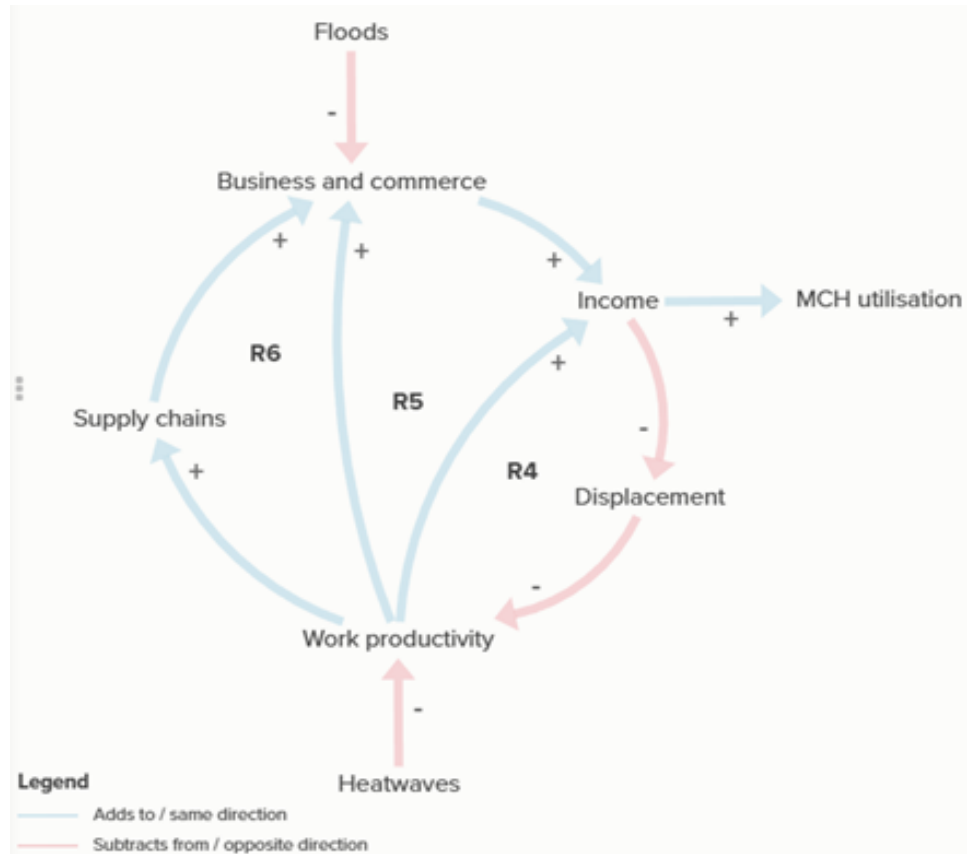
Source: Prepared by the author.

R1 captures a core biological–economic mechanism in which productivity increases income, income improves nutrition, and better nutrition sustains productivity. R2 extends this pathway through business and commerce, while R3 further embeds both loops within a supply-chain-mediated system that amplifies income and nutritional gains. In this structure, heatwaves directly reduce work productivity, activating all three loops, whereas floods primarily disrupt business and commerce, weakening income and its downstream effects. Across the three loops,

income loss directly constrains MCH utilisation, identifying income as a potential leverage point for containing impacts within the system.

Figure 3.9 presents a similar structure of three nested reinforcing loops, but with displacement instead of nutrition, with an otherwise analogous interpretation to Figure 3.8.

Figure 3.9 – Reinforcing Loops R4, R5 and R6

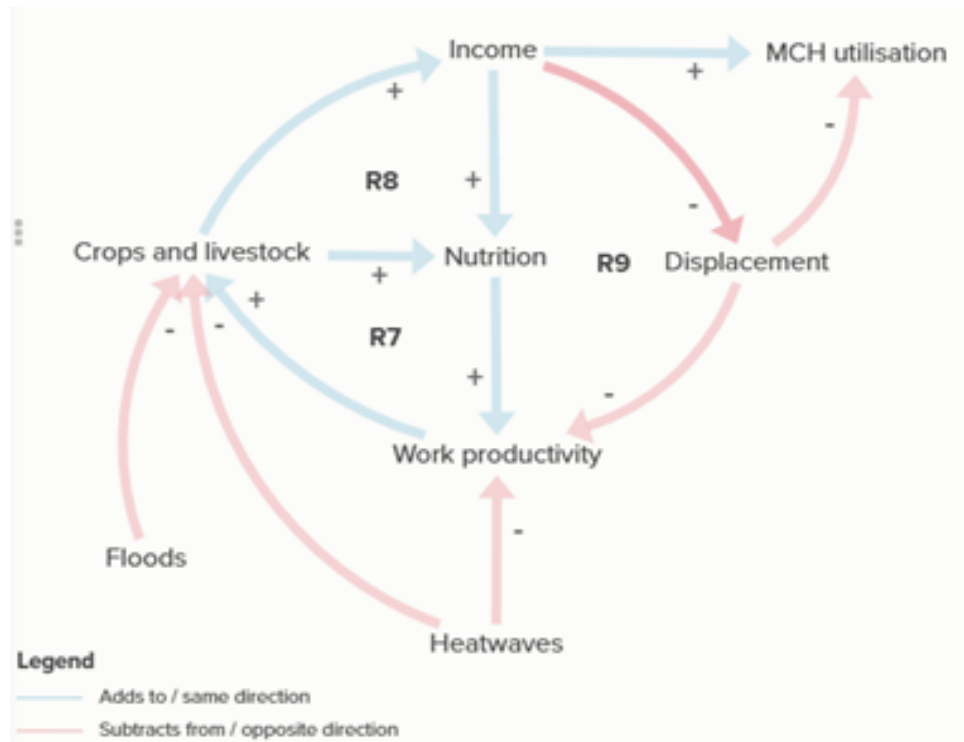


Source: Prepared by the author.

R4 captures the displacement pathway: income supports MCH utilisation, but shocks that reduce income increase displacement, which in turn lowers work productivity, further depressing income. This reinforcing loop shows how economic instability triggered by extreme events can translate into mobility pressures that weaken productive capacity and reduce service use. R5 represents a productivity–income–business loop, in which higher work productivity strengthens business and commerce, increasing income and further sustaining productivity. When heatwaves reduce productivity, or floods disrupt commercial activity, this loop contracts, amplifying income losses and downstream effects on MCH utilisation. R6 embeds productivity within supply-chain dynamics: productive activity sustains supply chains, which support business and commerce and reinforce productivity. Flood-induced supply-chain disruption weakens business activity and income generation, creating cascading economic effects that indirectly constrain MCH utilisation.

Figure 3.10 presents a related but distinct configuration, comprising three interconnected reinforcing pathways.

Figure 3.10 – Reinforcing Loops R7, R8 and R9



Source: Prepared by the author.

R7 links work productivity, crops and livestock, and nutrition. Although this loop does not connect directly to MCH utilisation, it strengthens internal system dynamics by reinforcing productivity and nutritional status, thereby potentiating the operation of the other two loops. R8 extends this pathway through income and affects MCH utilisation directly via income. R9 follows a parallel structure but incorporates displacement, influencing MCH utilisation through both income constraints and displacement-related barriers to access. Importantly, heatwaves directly reduce both work productivity and agricultural output, while floods primarily affect crops and livestock, initiating these reinforcing dynamics. Finally, the income–displacement–productivity loop identified earlier (R4) is also embedded within this structure but is not shown here again for clarity.

It is important to note that all feedback loops identified in this analysis are reinforcing, reflecting cascading dynamics that tend to intensify the impacts of floods and heatwaves. This configuration is consistent with the purpose of the review, which focuses on mapping impact pathways rather than modelling interventions that might counterbalance or stabilise these effects. For analytical clarity, the loops are presented in isolation; however, they are embedded within the broader causal structure shown in Figure 9, and a full traversal of the entire network would be impractical. The reinforcing dynamics are primarily concentrated in economic and livelihood-related variables, such as income, work productivity, business and agricultural activity, as well as in displacement-related pathways. In contrast, health system variables (e.g. healthcare capacity, health workers, and health facilities) lie outside the feedback loops: they are affected by upstream

dynamics but do not feed back into the system, a structural position that does not diminish their relevance or importance for the analysis.

3.4 DISCUSSION

A central contribution of this study is the development of a realist-informed Causal Loop Diagram to synthesise and organise heterogeneous evidence on extreme weather events and maternal and child health utilisation. By combining realist logic with systems thinking, the review moves beyond linear cause–effect descriptions and instead maps how interacting mechanisms generate reinforcing or dampening dynamics within health systems. This approach aligns with broader calls in health systems research to adopt complexity-informed methods capable of capturing feedback, spillovers, and unintended consequences (Adam e Savigny, 2012; Cassidy et al., 2022).

The methodological positioning of this CLD is consistent with emerging guidance on the use of CLDs in LMIC settings. Cassidy et al. (2022) emphasise that CLDs are particularly valuable where primary data may be limited and where researchers seek to integrate multiple forms of evidence (including systematic or realist reviews) into a coherent system representation. Our ex-post development approach, grounded in secondary evidence synthesis, mirrors this recommended pathway. However, as highlighted by Cassidy et al. (2022), validation through stakeholder engagement remains critical. The present CLD therefore represents a theoretically grounded synthesis rather than a fully validated system model.

Conceptually, our CLD also aligns with prior health sector CLDs that examine service delivery constraints, immunisation uptake, and financing reforms in LMICs (Rwashana, Williams e Neema, 2009; Paina e Peters, 2011; Cassidy et al., 2021). Similar to payment-for-performance or immunisation CLDs, our model reveals reinforcing loops linking supply-side readiness, demand-side income dynamics, and broader system foundations such as logistics and workforce capacity. However, this study extends the literature by explicitly modelling climate shocks as exogenous stressors that activate multiple loops simultaneously. While earlier CLDs often focus on policy interventions internal to the health sector, our diagram embeds the health system within energy systems, labour markets, transport networks, and climatic regimes. In doing so, it contributes to a growing body of work applying systems thinking to health system resilience and shock responsiveness (Ozawa, Paina e Qiu, 2016; Mudd et al., 2024).

One conceptual challenge concerns the treatment of context. Realist methodology treats context as dynamically interacting with mechanisms, whereas CLDs often operationalise context as an external boundary condition. In this study, contextual dimensions (population structure, economic level, climate regime, and health system type) were not embedded as endogenous variables but instead interpreted as conditions shaping which mechanisms activate and how strongly. This choice preserves diagram clarity but may underrepresent the co-evolution of context and mechanism over time. Future modelling could endogenise selected contextual variables (e.g., adaptive capacity, infrastructure investment) to capture longer-term system transformation under

repeated shocks.

The synthesis reveals clear hazard-specific patterns. Flood studies (predominantly conducted in low-income, tropical monsoon or savanna climates with mixed health systems) most frequently activate mechanisms related to impaired mobility (M1), displacement (M2), income loss (M3), and service strain (M5), often coupled with supply-chain or energy disruptions (M6). The dominant outcome is reduced utilisation of routine MCH services. These patterns are consistent with resilience frameworks that highlight infrastructure, logistics, and governance as core determinants of shock propagation within health systems (World Health Organization, 2022). In systems terms, floods appear to trigger reinforcing loops linking infrastructure breakdown, income contraction, and service contraction.

In contrast, heatwave studies (largely from high-income and urban settings) tend to activate clinical risk pathways (M4) and occasionally operational strain (M5), with outcomes often characterised by increased emergency utilisation rather than decreased routine care. This suggests that heat may shift utilisation patterns toward acute episodes rather than uniformly suppressing service access. The geographic distribution of this evidence also reflects data availability: high-income settings more often capture emergency department visits, hospitalisations, and cost outcomes. Importantly, climate regime and baseline adaptation capacity appear to condition these effects, reinforcing the analytical value of incorporating Köppen–Geiger classifications and system capacity variables.

The presence of null or maintained effects in a subset of flood studies further supports the realist premise that outcomes depend on contextual activation. In settings with prior exposure, established outreach systems, or rapid adaptive responses (e.g., mobile clinics, community boats, temporary service reorganisation), shock-to-utilisation translation may be dampened. From a systems perspective, these represent balancing loops that counteract reinforcing decline. Such findings resonate with resilience-oriented CLDs that emphasise absorptive and adaptive capacities as buffers against cascading failure (Ozawa, Paina e Qiu, 2016).

Several limitations warrant consideration. First, the CLD is derived from secondary evidence only; empirical validation through stakeholder engagement or group model-building, as recommended by Cassidy et al. (2022), would strengthen causal confidence and contextual specificity. Second, although contextual variation was systematically incorporated in the realist synthesis, the diagram itself represents a generalised system architecture; regional adaptations may be required to reflect differences in governance structures, financing arrangements, or climate exposure. Third, leverage-point identification remains preliminary. Applying Meadows (1999) hierarchy of leverage points could help distinguish shallow parameter changes (e.g., increasing emergency supplies) from deeper structural interventions (e.g., redesigning information flows, infrastructure standards, or system goals).

Finally, the CLD should not be interpreted as a predictive model. While many links are empirically grounded, their relative strengths, time delays, and nonlinear thresholds remain unquantified. Translating the diagram into a stock-and-flow simulation model would enable

estimation of loop dominance, exploration of sequential or compound hazards, and assessment of policy scalability. Such integration of realist synthesis with quantitative system dynamics would represent a next methodological frontier, deepening understanding of how climate-related shocks propagate through maternal and child health systems.

3.5 CONCLUSION

This essay used a realist-informed CLD to synthesise evidence on the mechanisms linking floods and heatwaves to MCH utilisation and to represent their interdependencies as a coherent system. The resulting model suggests that utilisation losses are frequently driven by cascading interactions between mobility constraints, household economic shocks, and disruptions to facility operations and supply chains, with reinforcing loops concentrated outside the health sector. While the CLD offers a structured basis for identifying potential leverage points (especially income buffering, transport continuity, and energy/logistics reliability) it remains an explanatory synthesis rather than a predictive tool. Priority next steps include stakeholder validation and quantitative system dynamics modelling to estimate loop dominance, time delays, and the likely effectiveness of resilience policies under repeated and compound climate hazards.

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APPENDIX 3.A – GLOSSARY

MCH Utilisation: The extent to which healthcare services designed to improve the health outcomes of mothers and children (such as antenatal care, skilled birth attendance, postnatal care, and immunizations) are accessed and used by the target population.

Routine MCH Services: Regular maternal and child health services aimed at promoting health and preventing disease, including antenatal, intranatal, and postnatal care; child immunization and growth monitoring; family planning; and health education.

Emergency MCH Services: Specialized maternal and child health interventions provided during emergencies or disasters to ensure continuity of essential care for pregnant and postpartum women, newborns, and young children. These services include safe delivery, emergency obstetric and neonatal care, access to medications and supplies, and measures to protect vulnerable populations in crisis situations.

Mechanism: The underlying process or reasoning that explains how and why an intervention produces outcomes. It refers to the responses, resources, and changes triggered in participants when the intervention interacts with their context.

Context: The set of conditions or circumstances in which an intervention is implemented, including social, cultural, organizational, and environmental factors. Context influences whether and how mechanisms are activated and outcomes occur.

Road Transport: The movement of goods and people from one location to another via roads. A road is a designated route (either paved or improved) to facilitate transportation using both motorized and non-motorized vehicles. This mode of transport can be categorized into two main types: the transportation of goods and materials, and the transportation of passengers. In many cases, road transport is the only viable option for reaching remote rural areas that lack access to rail, water, or air transportation.

Ability to Travel: The overall capacity (both physical and logistical) of an individual or entity to move from one location to another. Physical mobility refers to the ability to move and encompasses the quality of different travel modes such as walking, bicycling, taxis, public transportation, and air travel, evaluated by their availability, speed, frequency, comfort, and other similar factors.

Displacement: The forced movement of people from their homes or usual places of residence, often due to conflict, violence, disasters, or other crises. It can be internal (within a country, involving Internally Displaced Persons or IDPs) or external (across international borders, involving refugees).

Household: A social unit made up of individuals who live together and often share financial and domestic responsibilities. It is typically defined as a single housing unit occupied by one or more people. A household can consist of a family, roommates, or other living arrangements, serving as a fundamental unit for economic and demographic analysis.

Property: Any asset, tangible (such as real estate, vehicles, and personal belongings) or intangible (such as intellectual property), over which an individual or legal entity holds ownership

rights.

Business: Any activity undertaken with the goal of generating profit. It encompasses not only the sale of goods and services but also the entire process of creating, distributing, and delivering products to consumers.

Commerce: A subset of business that specifically refers to the exchange of goods or services between two or more parties. Unlike other aspects of business such as sourcing, manufacturing, transportation, or marketing, commerce focuses on the buying and selling of finished or unfinished products.

Income (Personal): Total earnings received by individuals from all sources, including wages, salaries, dividends, interest, business ownership, rental income, and government transfers (e.g., social security or unemployment benefits).

Crops: The cultivated plants such as cereals, legumes, fruits, vegetables, and fibre crops that are grown for direct human consumption, animal feed, or industrial uses.

Livestock: Domesticated animals (for example, cattle, pigs, sheep, goats, and poultry) that are reared primarily for products such as meat, milk, eggs, wool, and leather, as well as for draft power and other services.

Health Risks: A broad array of hazards, from biological, behavioural, socio-economic, environmental, to health system factors, that elevate the likelihood of adverse outcomes. In maternal and child health, health risks encompass obstetric complications (including hemorrhage, hypertensive disorders, and infections), preterm birth or low birth weight, nutritional deficiencies, and inadequate access to quality prenatal, perinatal, and postnatal care.

Water, Sanitation, and Hygiene (WASH): Reliable access to safe, sufficient, and affordable water for drinking, cooking, personal hygiene, and in health care facilities, alongside effective sanitation systems that manage human waste in a hygienic and environmentally sound manner.

Nutrition (Good): A well-balanced, high-quality diet that provides all essential macro- and micronutrients to prevent malnutrition, enhance immune function, support cognitive development, and lower the risk of chronic diseases.

Energy System: Integrated network of physical and institutional components responsible for the production, conversion, transmission, distribution, and consumption of energy to deliver essential energy services. It encompasses not only the technologies and infrastructure (such as power plants, grids, and renewable installations) but also the economic, regulatory, and social factors that shape how energy is managed, optimized, and delivered to end users.

Health Facilities: Locations where healthcare services are delivered, ranging from hospitals and clinics to outpatient and specialized centres such as birthing and psychiatric care centres.

Medical Supplies: Consumable products and equipment used in the delivery of health care services to diagnose, treat, manage, or prevent disease and injury. They include a broad range of items, from disposable products like gloves, bandages, syringes, and diagnostic reagents

to non-disposable items such as certain surgical instruments and durable materials, that support both routine care and emergency medical interventions.

Supply Chains: Integrated network of producers, suppliers, distributors, and retailers involved in transforming raw materials into finished goods and delivering them to end consumers.

Work Productivity: The efficiency with which an individual, team, or organization converts inputs such as time, labour, and capital into valuable outputs, including goods, services, or measurable results. This concept is typically measured as the ratio of output to input (for example, output per hour worked) and is a critical indicator of economic performance, competitiveness, and overall organizational effectiveness. In addition to quantitative output, modern frameworks also consider quality, innovation, and employee well-being as integral components of work productivity.

Health Workers: All individuals (both clinical and non-clinical) whose primary purpose is to promote, maintain, or restore health. This broad category includes physicians, nurses, midwives, allied health professionals, community health workers, laboratory and pharmacy staff, as well as support, administrative, and managerial personnel who contribute to health service delivery.

Healthcare Capacity: The combined ability of a health system to effectively deliver health services to a population. It encompasses the physical infrastructure (such as hospitals, clinics, and medical equipment), the availability and competency of human resources (health workers and support staff), and the organizational and logistical systems (including supply chains, information systems, and financing mechanisms) required to meet healthcare needs. This capacity is measured in terms of both volume (e.g., number of hospital beds or healthcare professionals) and functional performance (e.g., quality, efficiency, and responsiveness), and is critical for ensuring universal health coverage, timely emergency response, and the overall improvement of health outcomes.

4 FLOODS AND ANTENATAL CARE UTILISATION: EVIDENCE FROM NORTH-EASTERN BRAZIL

ABSTRACT

Background: Floods can disrupt maternal healthcare delivery through infrastructure damage, access constraints, and health system strain. Yet little causal evidence exists on how extreme floods affect routine antenatal care (ANC) utilisation in middle-income countries.

Methods: We examine the impact of the June 2010 floods in Pernambuco and Alagoas on prenatal care utilisation using municipality-month panel data from 2008–2012. Municipalities are classified as treated if officially affected by hydrological disasters during June 17–20, 2010, and we implement strict exclusions to avoid contamination from other hydrological shocks and potential spatial spillovers. Outcomes are municipality-level shares of live births by ANC intensity (0, 1–3, 4–6, and ≥ 7 visits) constructed from SINASC. We triangulate across two quasi-experimental strategies: controlled interrupted time series (CITS), and modern event-study difference-in-differences (DiD) estimators, while controlling for time-varying socioeconomic, demographic, and health-system covariates and clustering standard errors at the municipality level.

Results: Both CITS and event-study DiD estimates indicate a clear post-flood compositional shift in ANC intensity. Flood exposure reduced the share of births preceded by ≥ 7 ANC visits and increased the share with 4–6 visits, with little evidence of effects on 0 visits or 1–3 visits. The CITS estimates show an immediate relative decline of about 6 percentage points in ≥ 7 visits and an offsetting rise of roughly 4 percentage points in 4–6 visits at the intervention point, with no meaningful differential post-flood slope changes. Aggregated event-study DiD estimates are consistent with this pattern, indicating a statistically significant increase of 8.9 percentage points in 4–6 visits and a decline of 10.5 percentage points in ≥ 7 visits, while effects on 0 visits and 1–3 visits are small and statistically non-significant. Dynamic estimates show no systematic pre-trends and suggest persistent post-treatment differences over subsequent months.

Conclusion: The June 2010 floods did not meaningfully increase complete non-utilisation of antenatal care, but they reduced the likelihood that women completed high-intensity ANC schedules, shifting utilisation from ≥ 7 visits toward intermediate care (4–6 visits). These findings suggest that in a high-coverage universal system, floods may primarily erode continuity and schedule completion rather than eliminate contact with care. Climate-resilient health-system policies should therefore protect not only basic access but also the capacity to sustain recommended care trajectories during and after hydrological shocks.

Keywords: Floods; Antenatal care; Health systems; Difference-in-differences; Brazil

JEL: I12 (Health Behavior); I18 (Government Policy; Regulation; Public Health); Q54 (Climate; Natural Disasters; Global Warming).

HIGHLIGHTS

- June 2010 floods in Northeast Brazil shifted ANC utilisation away from high-intensity schedules.
- ≥ 7 ANC visits fell, while 4–6 visits rose; 0 visits and 1–3 visits were largely unchanged.
- Controlled ITS suggests effects operate mainly through an immediate level shift at the shock.
- Event-study DiD shows no pre-trends and a persistent post-flood “downshifting” in intensity.
- Flood resilience should protect continuity and completion of recommended ANC schedules, not only basic access.

4.1 INTRODUCTION

In June 2010, an extreme episode of heavy rainfall affected the northeastern region of Brazil, producing widespread flooding across the states of Alagoas and Pernambuco. Torrential rains over several days caused riverine overflow, infrastructure collapse (including bridges and roads), and forced tens of thousands of people from their homes, with official reports noting dozens of fatalities and emergency declarations across multiple municipalities (Moura e Landivar, 2010; Freire, Bonfim e Natenzon, 2014). Such hydrological disasters disrupt daily life, isolate communities, and strain local public services, including primary healthcare facilities.

Floods represent the most common type of natural disaster worldwide, affecting approximately 102 million people each year, with higher incidence in developing countries and densely populated urban areas (Freitas e Ximenes, 2012). In Brazil, hydrological disasters - that is, those resulting in flooding and inundations - accounted for about one-third (5,568 occurrences, 34.9%) of natural disasters between 2000 and 2015 (15,950 occurrences in total). However, in the same period the average cost of hidrological disasters to healthcare facilities (R\$ 599,411,75) was proportionally higher than meteorological (R\$ 292,416,43) and geological (R\$ 199,228,61) disasters (Freitas et al., 2020).

Prenatal services represent a cornerstone of maternal and child health systems worldwide, offering preventive care, risk screening, and continuity of support throughout pregnancy. Antenatal care (ANC) utilisation is widely considered not merely a measure of maternal health behaviours but also a proxy for broader health service accessibility and functioning, particularly in settings with universal or near-universal coverage.

At the international level, the WORLD HEALTH ORGANIZATION (2016) currently recommends a minimum of eight contacts with antenatal care providers during pregnancy to optimize maternal and perinatal outcomes and ensure a positive pregnancy experience. In Brazil, complementary national guidelines and practices emphasise early initiation of care and completion

of an adequate number of visits, often operationalized locally as six or more consultations, reflecting clinical priorities embedded in the Brazilian public health system (BRASIL, 2000).

Brazil's Unified Health System (SUS) provides universal, publicly-funded healthcare, with maternal and child health services delivered primarily at the municipal level through primary health care networks (Padilha et al., 2025). The Family Health Strategy (Estratégia Saúde da Família, FHS/ESF) is the principal model of primary care delivery, characterised by multidisciplinary teams assigned to defined catchment populations. Prenatal care under SUS is typically initiated and maintained within this primary care framework, with municipalities exercising administrative autonomy while being supported by federal and state coordination (Andrade et al., 2017). The decentralised structure of SUS implies that external shocks such as floods operate at the local level and have the potential to disrupt routine service availability, utilization patterns, and continuity of care.

Despite extensive research on the health impacts of disasters, the literature on how flood events affect routine maternal health services remains limited, particularly in lower-middle-income contexts (Chandra et al., 2025; Ummah, 2024; Orderud et al., 2022; Baten et al., 2020; Haque, Parr e Muhidin, 2020a). Most existing studies on floods and maternal healthcare focus on conditional probabilities of care or cross-sectional associations, and evidence using causal inference frameworks is scarce. This gap is notable given that climate change is projected to increase the frequency and intensity of extreme weather events globally, with significant implications for health systems in LMICs.

This study aims to address this gap by investigating the causal effect of the June 2010 floods on antenatal care utilisation across municipalities in Pernambuco and Alagoas, Brazil. Our primary research question is: ***Did exposure to the June 2010 floods cause measurable changes in antenatal care utilisation, and if so, how did the distribution of utilisation intensity shift following the event?***

To estimate these effects, we exploit a large municipality–month panel (2008–2012) and use two complementary quasi-experimental identification strategies: (i) a controlled interrupted time series analysis that contrasts pre- and post-shock trends between exposed and unexposed municipalities; and (ii) modern event-study difference-in-differences estimators designed for staggered treatment settings, which allow for dynamic measurement of impacts relative to the timing of exposure. By combining these strategies, we aim to strengthen causal inference in a setting where randomized experiments are infeasible and where conventional DiD approaches face challenges related to treatment timing and panel imbalance.

Beyond this Introduction (Section 4.1), the remainder of the essay is structured as follows. Section 4.2 outlines the empirical strategy, explaining the rationale for employing both a controlled interrupted time series design and a difference-in-differences approach. Section 4.3 presents the main findings. Section 4.4 situates these results within the broader literature, drawing out their policy implications and identifying directions for future research. Finally, Section 4.5 concludes the essay.

4.2 METHODS

This section describes the data, outcome measures, and empirical strategies used to estimate the effects of the June 2010 floods on prenatal care utilisation. After defining the treated and control municipalities and the study variables, it presents two complementary quasi-experimental approaches: a controlled interrupted time-series analysis and an event-study difference-in-differences model. All statistical analyses were performed in R (version 4.4.2). For transparency and replicability, all data processing and estimation scripts are made available in the Appendix 4.A.

4.2.1 Data and measurement

In 2010, the study area consisted of 287 municipalities, of which 102 (35.54%) were located in Alagoas and 185 (64.45%) in Pernambuco.

For this study, municipalities were classified as treated ($n = 80$; 27.87%) if they were officially affected by any hydrological disasters - namely floods, heavy rainfall, flash floods, inundations, or mass movements - between June 17 and 20, 2010. Treatment status was defined based on official records from the Integrated Disaster Information System (S2iD), accessed through the Brazilian Digital Atlas of Disasters.

To construct a credible control group and minimize confounding effects, several exclusion criteria were applied. First, municipalities affected by other hydrological disasters during the study period (2008–2012) were excluded ($n = 98$; 34.15%), in order to avoid contamination of the counterfactual by additional disaster shocks.

Second, municipalities not officially recorded as affected by the June 2010 floods but geographically adjacent (queen contiguity) to treated municipalities were also excluded ($n = 45$; 15.68%). This exclusion aimed to account for potential spatial spillover effects and to mitigate concerns regarding possible underreporting in the official disaster database - i.e., municipalities that may have been affected but were not formally registered.

Finally, the municipality of Fernando de Noronha ($n = 1$; 0.35%) was also excluded due to its insular geographic condition and markedly distinct socioeconomic and infrastructural characteristics relative to mainland municipalities.

In total, 144 municipalities (50.18%) were excluded from the analysis. The remaining municipalities ($n = 63$; 21.95%) constituted the control group.

We construct a municipality–month panel using SINASC microdata aggregated at the municipality level, covering 60 monthly periods from January 2008 to December 2012. This time window spans approximately five years around the June 2010 flood event, including roughly two and a half years before and two and a half years after exposure.

The primary outcomes of interest are municipality-level shares of live births by number of prenatal visits. Let $Y_{it}^{(k)}$ denote the share of live births in municipality i at time t that fall into prenatal care category k , where:

$$Y_{it}^{(1)} = \text{Share of live births with 7 or more prenatal visits,} \quad (4.1)$$

$$Y_{it}^{(2)} = \text{Share of live births with 4 to 6 prenatal visits,} \quad (4.2)$$

$$Y_{it}^{(3)} = \text{Share of live births with 1 to 3 prenatal visits,} \quad (4.3)$$

$$Y_{it}^{(4)} = \text{Share of live births with 0 prenatal visits.} \quad (4.4)$$

Table 4.1 presents the control variables included in this study.

Table 4.1 – Control Variables

Variable	Unit	Source
Real GDP per capita at the municipal level (IPCA-deflated, December 2012 prices)	R\$	Instituto Brasileiro de Geografia e Estatística (2026)
Rural population share (municipal resident population)	%	Instituto Brasileiro de Geografia e Estatística (2026)
Family Health Strategy (FHS) coverage rate	%	Ministério da Saúde (2026a)
Births to adolescent mothers (< 20 years)	%	Ministério da Saúde (2026b)
Births to primiparous mothers (first birth)	%	Ministério da Saúde (2026b)
Births to mothers with low educational attainment (< 8 years of schooling)	%	Ministério da Saúde (2026b)
Births to mothers without a partner	%	Ministério da Saúde (2026b)
Preterm births (< 37 weeks of gestation)	%	Ministério da Saúde (2026b)
Births occurring in the mother's municipality of residence	%	Ministério da Saúde (2026b)

Municipal GDP per capita is included as a proxy for local economic development and fiscal capacity. It captures time-varying differences in municipalities' labor markets, infrastructure, public resources, and households' ability to bear the direct and indirect costs of prenatal care, such as transportation and time away from work. Within the health services utilisation framework, such socioeconomic “enabling resources” are central determinants of care-seeking behaviour and may confound estimated flood impacts if economic trajectories differ between treated and control municipalities around 2010. Empirical evidence consistently shows that income, wealth, and broader socioeconomic status are strongly associated with antenatal care utilisation and adequacy (Aboagye et al., 2025; Basha, 2019; Viellas et al., 2014).

The rural population share controls for geographic remoteness and physical accessibility to health services. Rural municipalities often face greater distances to facilities, lower road

density, and more limited service availability, factors that directly affect antenatal care utilisation. These structural characteristics may also shape both exposure vulnerability and recovery capacity following floods. Systematic reviews in LMIC contexts identify rural residence and access barriers (e.g., travel time and distance) as key determinants of antenatal care use, supporting its inclusion as an important baseline municipal characteristic (Jafari et al., 2026; Habeeb et al., 2025).

Family Health Strategy (FHS) coverage captures local primary care supply and outreach capacity. The FHS plays a central role in early pregnancy identification, continuity of follow-up, and reduction of access barriers through community-based teams. Because FHS expansion evolved over time and varied across municipalities, controlling for coverage rates helps isolate the effect of flood exposure from concurrent changes in primary care availability. Brazilian evidence documents the relevance of the FHS for primary-care-sensitive outcomes and maternal health indicators, including prenatal care performance (Andrade et al., 2017; Bastos et al., 2017).

Maternal age composition is included because adolescent pregnancy is associated with distinct patterns of antenatal care utilisation, reflecting socioeconomic disadvantage, lower autonomy, stigma, and differential risk perception. Municipalities with higher shares of adolescent mothers may exhibit systematically different prenatal care profiles. If flood exposure correlates with such demographic composition (or if shocks alter fertility patterns) failure to adjust for maternal age structure could bias difference-in-differences estimates by conflating behavioural responses with compositional changes. Brazilian studies on prenatal adequacy consistently document strong associations between maternal age and utilisation patterns (Leal et al., 2020; Anjos e Boing, 2016; Domingues et al., 2015; Viellas et al., 2014).

Parity is included because prenatal care utilisation often differs by birth order. First-time mothers may seek more intensive follow-up due to uncertainty and clinical guidance, whereas multiparous mothers may reduce visits owing to prior experience or competing time constraints. Shifts in parity composition at the municipal level can therefore generate changes in outcome shares unrelated to flood exposure. The antenatal care determinants literature routinely identifies parity as an important predisposing factor influencing utilisation (Morón-Duarte et al., 2021; Aboagye et al., 2025; Wulandari, Laksono e Nantabah, 2020; Viellas et al., 2014).

Maternal education is one of the most robust predictors of antenatal care utilisation and adequacy. It proxies health literacy, information-processing capacity, bargaining power within the household, and broader socioeconomic position. Education levels are spatially clustered and may correlate with both vulnerability to disaster impacts and the ability to recover from shocks. Brazilian evidence shows persistent educational gradients in prenatal care adequacy, reinforcing the importance of adjusting for compositional differences across municipalities (Morón-Duarte et al., 2021; Leal et al., 2020; Anjos e Boing, 2016; Domingues et al., 2015; Viellas et al., 2014).

Marital or partner status is included as a proxy for social support and household resources. Having a partner may facilitate access to transportation, financial support, and childcare, thereby affecting antenatal care attendance. Partner status may also shape households' resilience and

coping strategies following shocks. Brazilian studies on prenatal care inequalities document associations between marital status and adequacy indicators, suggesting that differences in family structure may confound estimated treatment effects if not accounted for (Morón-Duarte et al., 2021; Leal et al., 2020; Domingues et al., 2015).

The share of preterm births is included to account for time-varying obstetric risk and case-mix at the municipal level. Higher-risk pregnancies may require more intensive clinical follow-up, potentially influencing the distribution of prenatal visit categories. At the same time, prematurity may reflect underlying environmental and health-system conditions that correlate with flood exposure and local service strain. Brazilian research highlights associations between prenatal care processes and adverse birth outcomes, including prematurity, underscoring the importance of adjusting for risk composition when modelling prenatal utilisation patterns (Oliveira et al., 2022; Andrade et al., 2022; Anjos e Boing, 2016).

Finally, the share of births occurring in the mother's municipality of residence serves as a proxy for geographic access to obstetric services and local maternity care capacity. High levels of cross-municipality delivery may signal limited local service availability and greater reliance on referral networks. Mobility patterns are documented as common in Brazil and are associated with inequalities in maternal health access. Because floods may disrupt transportation networks and service provision, controlling for this measure helps account for baseline differences in local service infrastructure and spatial care-seeking behaviour (Albuquerque et al., 2025; Neto et al., 2024; Leite et al., 2023).

4.2.2 Empirical strategy

To estimate the causal effects of the June 2010 floods on prenatal care utilisation, we employ two complementary quasi-experimental strategies. A controlled interrupted time-series model identifies immediate changes in outcome levels and post-flood trends, while an event-study difference-in-differences approach provides a more flexible assessment of dynamic treatment effects and pre-intervention trends. Together, these methods allow us to evaluate both the timing and persistence of changes across prenatal care intensity categories.

4.2.2.1 Controlled interrupted time series

Interrupted time series (ITS) is widely regarded as one of the strongest quasi-experimental approaches for evaluating population-level interventions implemented at a clearly defined point in time, particularly when randomized assignment is infeasible and repeated outcome measurements are available before and after the intervention (Bernal, Cummins e Gasparrini, 2017; Wagner et al., 2002). Adding a contemporaneous control series strengthens internal validity by helping to account for secular trends and time-varying confounding that affects both treated and control units (e.g., national policies, seasonality, or macroeconomic shocks), thereby reducing bias relative to an uncontrolled ITS (Bernal, Cummins e Gasparrini, 2018).

In our setting, the intervention is defined as the June 2010 flood event, with July 2010 considered the first fully exposed month. We collapse the municipality–month panel into group-month means, where $g \in \{\text{treated}, \text{control}\}$ denotes the exposure group and t indexes months relative to the cutoff. Specifically, we define $t = 0$ for July 2010, $t < 0$ for pre-intervention months, and $t > 0$ for post-intervention months. For each prenatal care outcome Y_{gt} (mean share of live births in group g at month t in a given prenatal visit category), we estimate the following two-group segmented regression model (Bernal, Cummins e Gasparrini, 2017; Linden, 2015; Wagner et al., 2002):

$$\begin{aligned}
 Y_{gt} = & \beta_0 + \beta_1 t + \beta_2 Post_t + \beta_3 (t \times Post_t) \\
 & + \beta_4 Treated_g + \beta_5 (t \times Treated_g) \\
 & + \beta_6 (Post_t \times Treated_g) \\
 & + \beta_7 (t \times Post_t \times Treated_g) + \varepsilon_{gt},
 \end{aligned} \tag{4.5}$$

where $Post_t$ is an indicator equal to 1 for $t \geq 0$ (July 2010 onward) and 0 otherwise, $Treated_g$ is an indicator equal to 1 for the treated group and 0 for the control group, and t is the monthly time index centered at the cutoff.

This specification allows the treated and control groups to differ in baseline levels and pre-intervention trends, and it estimates both an immediate post-intervention level change and a post-intervention slope change in the treated group relative to the control group. The coefficients can be interpreted as follows:

- β_1 captures the baseline (pre-intervention) trend in the control group;
- β_2 captures the immediate level change at the cutoff in the control group;
- β_3 captures the change in slope in the control group after the cutoff;
- β_4 captures the baseline level difference between treated and control groups;
- β_5 captures the differential pre-intervention trend in the treated group relative to the control group;
- β_6 is the difference-in-differences estimate of the *immediate* level change at the cutoff attributable to flood exposure;
- β_7 is the difference-in-differences estimate of the *change in post-intervention slope* attributable to flood exposure.

The parameters of primary interest are β_6 and β_7 , which identify the causal impact of flood exposure on prenatal care utilisation under the assumption that, conditional on the modelled pre-intervention trends, treated and control groups would have followed parallel trajectories in the absence of the flood (Bernal, Cummins e Gasparrini, 2017; Bernal, Cummins e Gasparrini,

2018). Separate models are estimated for each prenatal care category (0 visits, 1–3 visits, 4–6 visits, and 7 or more visits). Consistent with our implementation, all models are estimated by ordinary least squares on the group-month mean outcomes.

4.2.2.2 Event-study difference-in-differences

While the controlled interrupted time series (CITS) framework provides a flexible way to estimate changes in levels and trends at the intervention cutoff, it imposes a parametric structure on post-treatment dynamics. In particular, CITS restricts treatment effects to operate through discrete level shifts and linear trend changes. However, disaster impacts may unfold nonlinearly over time, with potentially heterogeneous short-, medium-, and longer-term responses.

To allow for a more flexible dynamic response and to formally test for pre-treatment parallel trends, we complement the CITS approach with an event-study difference-in-differences (DiD) design. This framework estimates period-specific treatment effects relative to the intervention month, without imposing linearity in post-treatment trends.

We implement the group-time average treatment effect estimator proposed by Callaway e Sant’Anna (2021), which generalizes the traditional two-way fixed effects DiD model and provides consistent estimates under staggered or group-specific treatment timing. Although treatment in our setting occurs at a single common time (July 2010), the Callaway e Sant’Anna (2021) estimator offers two important advantages: (i) it avoids biases associated with conventional two-way fixed effects estimators under treatment effect heterogeneity (Goodman-Bacon, 2021; Sun e Abraham, 2021), and (ii) it allows for flexible aggregation of dynamic treatment effects across event time.

Let i index municipalities and t index calendar months. Define the treatment indicator:

$$D_i = \begin{cases} 1, & \text{if municipality } i \text{ was exposed to the June 2010 floods,} \\ 0, & \text{otherwise.} \end{cases} \quad (4.6)$$

Let T_0 denote July 2010, the first fully exposed month. We define event time as:

$$e = t - T_0, \quad (4.7)$$

such that $e = 0$ corresponds to July 2010, $e < 0$ to pre-treatment months, and $e > 0$ to post-treatment months.

All treated municipalities share the same treatment cohort $G_i = T_0$, while control municipalities are never treated ($G_i = 0$).

Let $Y_{it}(1)$ and $Y_{it}(0)$ denote potential outcomes under treatment and no treatment, respectively. The group-time average treatment effect for cohort g at time t is defined as:

$$ATT(g, t) = \mathbb{E} [Y_{it}(1) - Y_{it}(0) \mid G_i = g], \quad \text{for } t \geq g. \quad (4.8)$$

Identification relies on a conditional parallel trends assumption: in the absence of treatment, treated and control municipalities would have experienced similar trends in outcomes, conditional

on observed covariates X_{it} . Formally,

$$\mathbb{E} [Y_{it}(0) - Y_{i,t-1}(0) \mid G_i = g, X_{it}] = \mathbb{E} [Y_{it}(0) - Y_{i,t-1}(0) \mid G_i = 0, X_{it}]. \quad (4.9)$$

We additionally assume no anticipation (i.e., outcomes prior to July 2010 are unaffected by future treatment) and sufficient overlap in covariate distributions between treated and control municipalities.

To recover dynamic treatment effects, we estimate an event-study DiD model of the form:

$$Y_{it}^{(k)} = \alpha_i + \lambda_t + \sum_{e \neq -1} \beta_e^{(k)} (D_i \times \mathbb{I}\{t - T_0 = e\}) + X_{it}'\gamma + \varepsilon_{it}, \quad (4.10)$$

where:

- $Y_{it}^{(k)}$ denotes the share of live births in municipality i and month t in prenatal category k ;
- α_i are municipality fixed effects;
- λ_t are month fixed effects;
- $\mathbb{I}\{t - T_0 = e\}$ is an indicator for event time e ;
- X_{it} is a vector of time-varying covariates (health system coverage, socioeconomic indicators, and maternal composition variables);
- the period $e = -1$ is omitted and serves as the reference category.

The coefficients $\beta_e^{(k)}$ trace the dynamic causal effect of flood exposure on prenatal care utilisation relative to the month immediately preceding treatment. Pre-treatment coefficients ($e < 0$) provide a diagnostic test of the parallel trends assumption, while post-treatment coefficients ($e \geq 0$) capture the evolution of treatment effects over time.

We estimate group-time average treatment effects using the doubly robust procedure implemented in the `did` package in R (Callaway e Sant'Anna, 2021). Control municipalities are defined as never-treated units. Standard errors are clustered at the municipality level to account for serial correlation.

Dynamic treatment effects are aggregated across treated municipalities to obtain event-time average effects:

$$ATT(e) = \mathbb{E} [ATT(g, t) \mid t - g = e]. \quad (4.11)$$

In addition to the dynamic event-study profile, we report an overall post-treatment average effect:

$$ATT_{\text{post}} = \sum_{t \geq T_0} w_t ATT(g, t), \quad (4.12)$$

where w_t are aggregation weights proportional to the number of treated observations in each period.

Separate models are estimated for each prenatal care category (0 visits, 1–3 visits, 4–6 visits, and 7 or more visits).

The event-study estimates allow us to assess a) whether treated and control municipalities exhibited similar pre-treatment trends, and b) how the distribution of prenatal visits changed dynamically following the June 2010 floods. This flexible specification complements the CITS model by relaxing linear trend assumptions and by providing a transparent visualization of treatment effects over time.

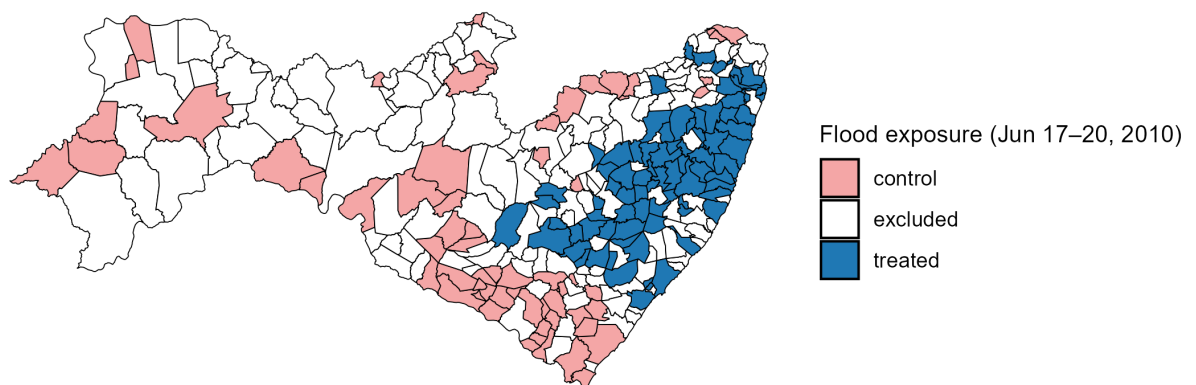
4.3 RESULTS

This section presents the empirical findings on the effects of the June 2010 floods on prenatal care utilisation. It first assesses the comparability of treated and control municipalities and then reports the results from the controlled interrupted time-series and event-study difference-in-differences analyses, focusing on changes across prenatal care intensity categories.

4.3.1 Control and treatment groups balance assessment

Before estimating the effects of the June 2010 floods, we examine the composition and pre-intervention comparability of the treatment and control groups. This assessment describes their spatial distribution and evaluates differences in baseline outcomes and covariates, providing evidence on the extent of observable imbalance and informing the interpretation of the subsequent quasi-experimental estimates. Figure 4.1 presents the spatial distribution of treated and control municipalities.

Figure 4.1 – Flood Exposure in Pernambuco and Alagoas, June 17–20, 2010



Within the treatment group, 21 municipalities were located in Alagoas (26.25%) and 59 in Pernambuco (73.75%). Within the control group, 34 municipalities were located in Alagoas (53.97%) and 29 in Pernambuco (46.03%).

To assess pre-intervention comparability between treated and control municipalities, we examine differences in pre-flood averages of outcomes and covariates using both two-sample t-tests and standardized mean differences (SMD). The t-test evaluates whether differences in means are statistically distinguishable from zero, while the SMD provides a scale-invariant measure of imbalance that is independent of sample size. Because statistical significance is sensitive to the number of observations, SMD is particularly informative in large panels and is commonly used in quasi-experimental designs to assess covariate balance.

Table 4.2 reports the pre-intervention means by group, associated t-test p-values, and SMDs.

Table 4.2 – Pre-intervention Balance Between Treated and Control Municipalities

Means	Treated	Control	T-Test (p-value)	SMD
Share of 0 ANC visits	0.020	0.015	0.006***	0.478
Share of 1–3 ANC visits	0.080	0.083	0.619	-0.085
Share of 4–6 ANC visits	0.429	0.529	0.000***	-0.990
Share of 7+ ANC visits	0.471	0.373	0.000***	0.878
Family health strategy (ESF) coverage	0.885	0.919	0.223	-0.198
Real GDP per capita	7528	5794	0.042**	0.324
Rural population share	30.4	45.9	0.000***	-0.831
Share of adolescent pregnancies	0.256	0.240	0.010**	0.423
Share of primiparous mothers	0.402	0.294	0.000***	1.050
Share of mothers with low education	0.602	0.658	0.005***	-0.479
Share of mothers without partner	0.745	0.637	0.000***	1.130
Preterm birth rate	0.044	0.037	0.129	0.283
Share of births outside municipality of residence	0.385	0.325	0.169	0.238

Standardized Mean Difference (SMD) reported in the last column.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Baseline comparisons reveal meaningful differences in levels between treated and control municipalities. Treated areas exhibit a higher share of 7+ ANC visits and a lower share of 4–6 visits, as well as higher proportions of primiparous mothers and mothers without a partner, and a lower rural population share. They also display somewhat higher real GDP per capita. However, not all covariates exhibit imbalance. Primary care coverage (ESF), preterm birth rates, and births outside municipality show relatively small differences and statistically insignificant t-tests.

Importantly, difference-in-differences and controlled interrupted time series designs do not require equality in baseline levels; identification relies on parallel pre-treatment trends rather than identical means. Event-study estimates show no systematic differential trends prior to the intervention, supporting the parallel trends assumption. In addition, all baseline covariates are included as controls in the DiD specifications, accounting for time-varying compositional differences, while municipality fixed effects absorb time-invariant structural disparities between treated and control areas. Accordingly, although treated municipalities differ in baseline levels of prenatal care intensity and maternal characteristics, the empirical strategy exploits within-

municipality changes over time, and the presence of parallel pre-trends indicates that these level differences do not compromise causal inference.

4.3.2 Interrupted time series analysis

This subsection presents the controlled interrupted time-series estimates of the effects of the June 2010 floods on prenatal care utilisation. It compares trends in treated and control municipalities before and after the flood across the four prenatal visit categories, distinguishing immediate level changes from shifts in post-intervention trends. The graphical evidence is presented first, followed by the corresponding regression estimates. Figure 4.2 shows the trends in the mean share of women receiving seven or more ANC visits.

Figure 4.2 – Controlled Interrupted Time Series, Mean Share of 7+ ANC Visits

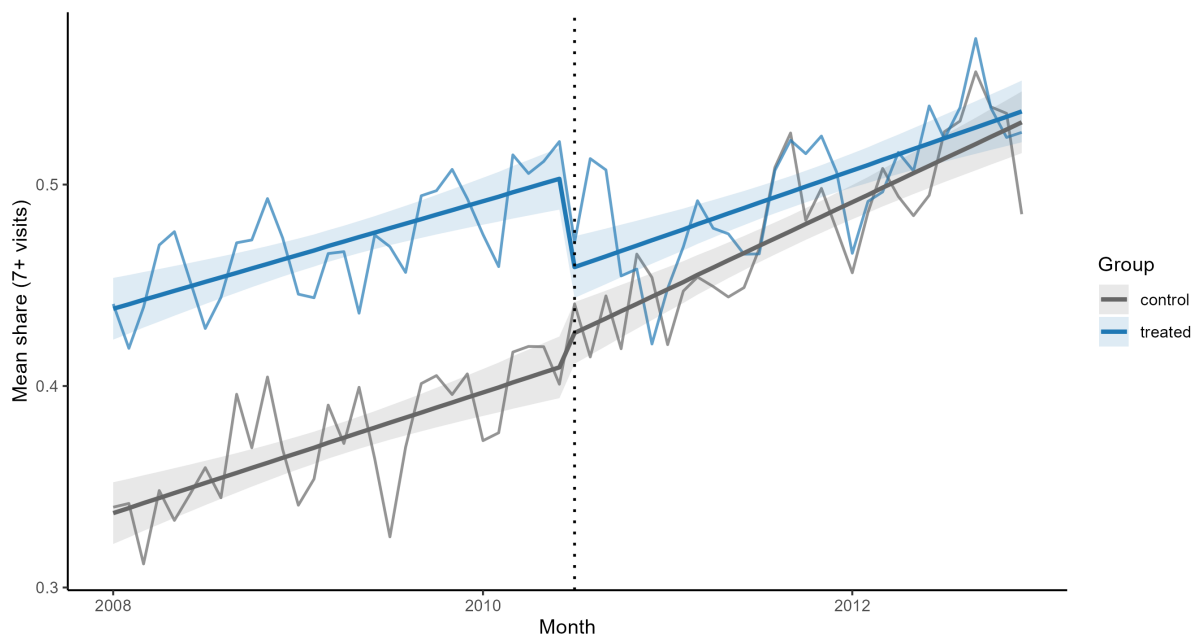


Figure 4.2 shows broadly parallel pre-intervention trends between treated and control municipalities in the share of women receiving 7 or more ANC visits, with no statistically significant differential pre-trend (β_5 not significant). At the intervention point, there is a statistically significant negative difference-in-differences level change ($\beta_6 = -0.0606$, $p < 0.001$), indicating an immediate relative decline of approximately 6 percentage points in the treated group. The post-intervention differential slope (β_7) is small and statistically insignificant, suggesting that the main effect operates through an immediate level shift rather than a persistent divergence in trend.

Figure 4.3 presents pre- and post-flood trends for the mean share of 7+ ANC visits.

Figure 4.3 – Controlled Interrupted Time Series, Mean Share of 4 to 6 ANC Visits

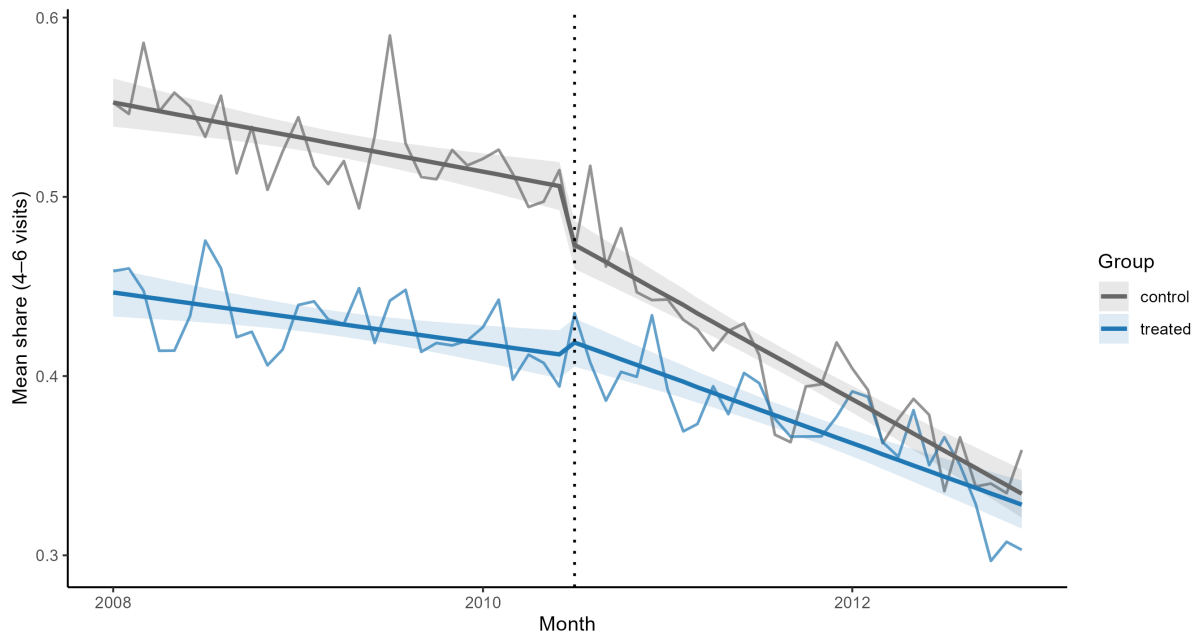
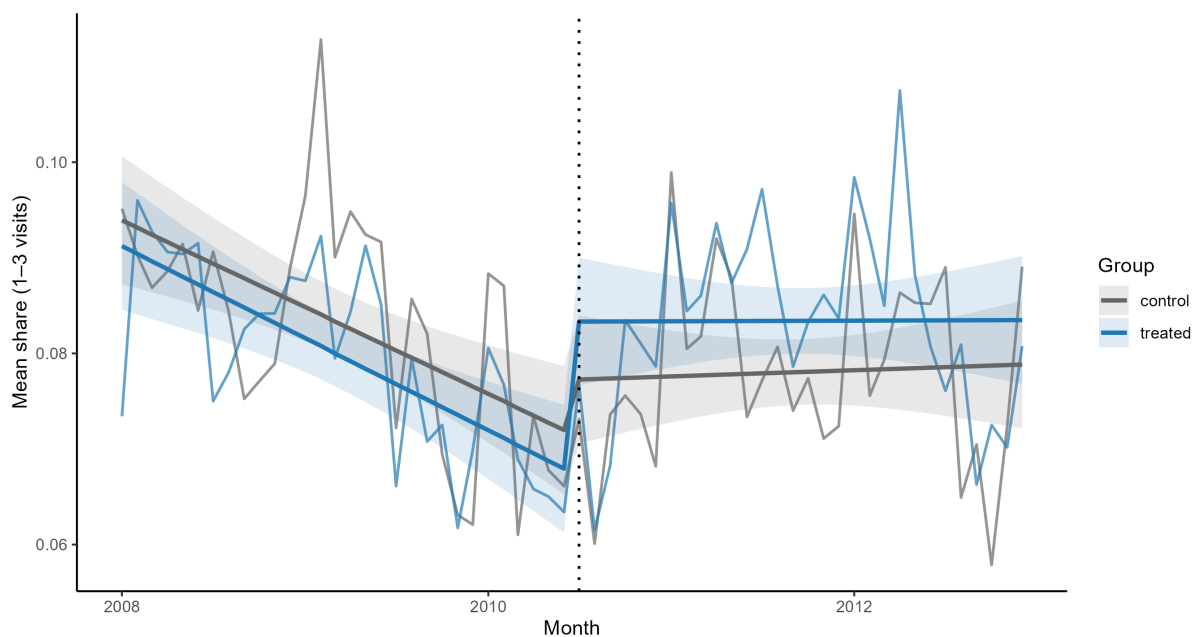


Figure 4.3 indicates similar pre-intervention dynamics between groups, with no significant differential pre-trend. At the time of the flood, treated municipalities exhibit a statistically significant positive difference-in-differences level effect ($\beta_6 = 0.0390$, $p = 0.006$), corresponding to an immediate relative increase of roughly 4 percentage points in the share of 4–6 visits. The triple interaction term (β_7) is not statistically significant, implying no additional post-intervention slope divergence. The results therefore point to a discrete upward shift in intermediate-intensity care in treated areas.

Figure 4.4 presents pre- and post-flood trends for the mean share of 7+ ANC visits.

Figure 4.4 – Controlled Interrupted Time Series, Mean Share of 1 to 3 ANC Visits



As shown in Figure 4.4, pre-flood trends are comparable across groups. Neither the difference-in-differences level effect at the intervention (β_6) nor the post-intervention slope differential (β_7) is statistically significant. Although the control group exhibits some change in post-intervention slope ($\beta_3 > 0$), the absence of significant interaction terms indicates that flood exposure did not materially alter the relative trajectory of 1–3 ANC visits.

Figure 4.5 presents pre- and post-flood trends for the mean share of 7+ ANC visits.

Figure 4.5 – Controlled Interrupted Time Series, Mean Share of 0 ANC Visits

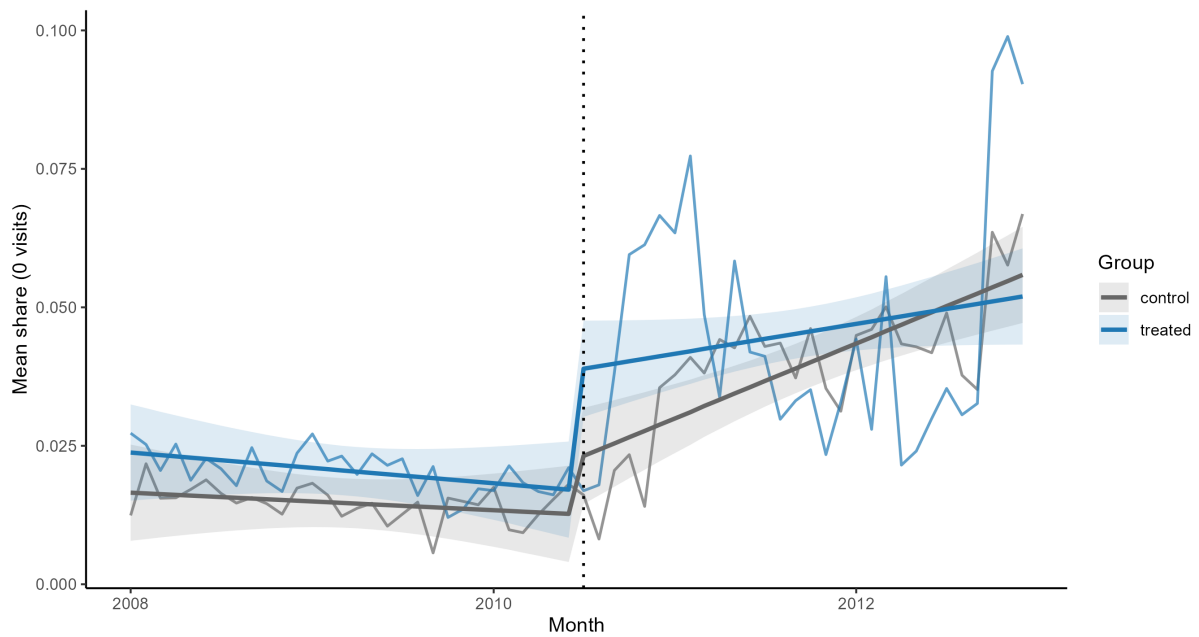


Figure 4.5 shows no evidence of significant differential pre-trends. At the intervention point, the level difference-in-differences effect (β_6) is not statistically significant, and the post-intervention slope differential (β_7) is also insignificant. While there is a modest positive change in slope for the control group after the intervention ($\beta_3 > 0$), treated municipalities do not display a statistically distinct response. Thus, the floods do not appear to have increased the probability of complete exclusion from prenatal care.

Table 4.3 reports controlled interrupted time series estimates.

Table 4.3 confirms the graphical evidence from the controlled interrupted time series analysis. The coefficients of primary interest, β_6 (difference-in-differences level change) and β_7 (difference-in-differences slope change), indicate that the June 2010 floods generated an immediate and statistically significant redistribution across prenatal care intensity categories.

For 7 or more visits, there is a large and statistically significant negative level effect ($\beta_6 = -0.0606$, $p < 0.001$), with no significant post-intervention slope differential, indicating an abrupt but not progressively widening relative decline. Conversely, for 4–6 visits, the level effect is positive and statistically significant ($\beta_6 = 0.0390$, $p < 0.01$), again without evidence of differential post-intervention slope change.

Table 4.3 – Controlled Interrupted Time Series (CITS) Estimates

	(1) 7+ visits	(2) 4–6 visits	(3) 1–3 visits	(4) 0 visits
β_0 (Intercept: baseline level, control, $t = 0$)	0.412*** (0.008)	0.504*** (0.007)	0.071*** (0.004)	0.013*** (0.005)
β_1 (t : pre-intervention trend, control)	0.0025*** (0.0005)	-0.0016*** (0.0004)	-0.0008*** (0.0002)	-0.0001 (0.0003)
β_2 ($Post_t$: level change at cutoff, control)	0.0146 (0.0112)	-0.0312** (0.0098)	0.0060 (0.0049)	0.0106 (0.0064)
β_4 ($Treated_g$: baseline level difference)	0.0933*** (0.0115)	-0.0935*** (0.0101)	-0.0041 (0.0050)	0.0043 (0.0065)
β_3 ($t \times Post_t$: slope change, control)	0.0011 (0.0006)	-0.0032*** (0.0006)	0.0008** (0.0003)	0.0013*** (0.0004)
β_5 ($t \times Treated_g$: differential pre-trend)	-0.0003 (0.0006)	0.0004 (0.0006)	-0.0000 (0.0003)	-0.0001 (0.0004)
β_6 ($Post_t \times Treated_g$: DiD level effect)	-0.0606*** (0.0158)	0.0390** (0.0139)	0.0101 (0.0069)	0.0115 (0.0090)
β_7 ($t \times Post_t \times Treated_g$: DiD slope effect)	-0.0007 (0.0009)	0.0013 (0.0008)	-0.0000 (0.0004)	-0.0006 (0.0005)

Standard errors in parentheses.

β_6 and β_7 identify the causal flood effect under parallel pre-trends.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

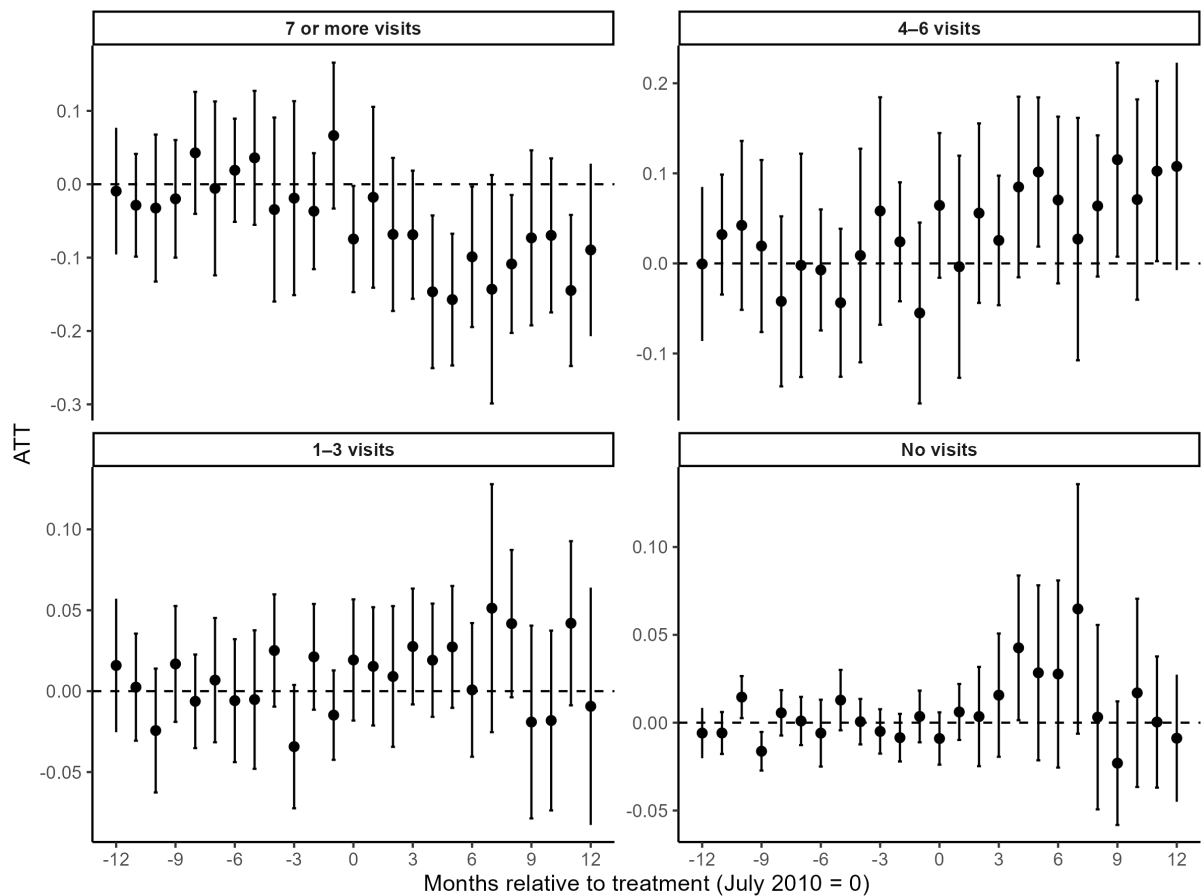
In contrast, for 1–3 visits and 0 visits, neither β_6 nor β_7 is statistically significant, implying no causal impact of flood exposure on low-level or absent prenatal care.

The CITS results indicate that the floods did not increase complete non-utilisation of prenatal care. Instead, they induced an immediate reallocation from the highest-intensity category (7+ visits) toward intermediate utilisation (4–6 visits), with effects operating primarily through level shifts at the intervention point rather than sustained changes in post-flood trends.

4.3.3 Difference-in-differences

This subsection presents the event-study difference-in-differences estimates of the effects of the June 2010 floods on prenatal care utilisation. It first examines pre-treatment trends and the evolution of effects over time, before reporting the aggregated average treatment effects for each prenatal visit category. Figure 4.6 shows the dynamic estimates by month relative to the flood.

Figure 4.6 – Event Study Estimates



Overall, the event-study estimates support the parallel trends assumption and point to a clear post-treatment reallocation across prenatal care intensity categories.

In the pre-treatment period ($t \leq -1$), coefficients for all outcomes fluctuate around zero and are generally statistically indistinguishable from zero. This pattern is especially evident in the 7+ visits and 4–6 visits panels, where there is no systematic upward or downward trend prior to July 2010. The absence of pronounced pre-trends reinforces the credibility of the identifying assumption.

After the shock ($t \geq 0$), a marked divergence emerges between higher- and intermediate-intensity care. The probability of receiving 7 or more visits declines persistently, with negative point estimates increasing in magnitude roughly between three and twelve months post-treatment. In contrast, the probability of receiving 4–6 visits rises over the same horizon, with positive and gradually increasing coefficients. This divergence is clearly visible in the upper panels of Figure 4.6 and is consistent with the aggregated ATT estimates.

For lower-intensity care, the effects are comparatively modest. Estimates for 1–3 visits remain close to zero throughout most of the post-treatment period and do not display a consistent dynamic pattern. Similarly, the probability of no visits exhibits small and imprecise fluctuations, with confidence intervals generally encompassing zero.

Figure 4.6 suggests that the shock did not substantially increase complete exclusion from

prenatal care. Rather, it induced a shift away from the highest-intensity category (7+ visits) toward intermediate utilisation (4–6 visits). The dynamic profile indicates a gradual and persistent adjustment, rather than a short-lived disruption. Complete event-study estimates for the full period are reported in Appendix 4.B.

Table 4.4 reports the aggregated difference-in-differences estimates. The results indicate a clear compositional shift in prenatal care intensity following the shock.

Table 4.4 – Difference-in-Differences Estimates

Outcome	ATT	Std. Error	95% CI	p-value
0 visits	0.0062	(0.0115)	[-0.0163, 0.0288]	0.587
1–3 visits	0.0091	(0.0154)	[-0.0210, 0.0393]	0.553
4–6 visits	0.0893**	(0.0380)	[0.0148, 0.1639]	0.019
7+ visits	-0.1047***	(0.0397)	[-0.1824, -0.0269]	0.008

Notes: Estimates from the Callaway e Sant’Anna (2021) difference-in-differences model. Standard errors clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

There is no statistically significant effect on the probability of receiving no prenatal visits (ATT = 0.0062, SE = 0.0115, $p = 0.587$) or 1–3 visits (ATT = 0.0091, SE = 0.0154, $p = 0.553$). In both cases, the confidence intervals include zero, suggesting that the shock did not meaningfully affect the lower tail of prenatal care utilisation.

In contrast, the probability of receiving 4–6 visits increased by 8.9 percentage points (ATT = 0.0893, SE = 0.0380, $p = 0.019$), statistically significant at the 5% level. At the same time, the probability of receiving 7 or more visits declined by 10.5 percentage points (ATT = -0.1047, SE = 0.0397, $p = 0.008$), significant at the 1% level.

These findings suggest a downward shift in prenatal care intensity among treated municipalities: while extreme underutilisation did not increase, women appear to have substituted away from higher-intensity prenatal care (7+ visits) toward intermediate levels (4–6 visits). This pattern is consistent with partial disruption of care continuity rather than complete exclusion from the health system.

4.4 DISCUSSION

This study provides quasi-experimental evidence that the June 2010 floods in Pernambuco and Alagoas affected the intensity of antenatal care utilisation, rather than generating widespread non-utilisation. Across both the controlled interrupted time series and the event-study DiD estimates, we find a clear post-shock compositional shift: the share of births preceded by ≥ 7 ANC visits declined, while the share with 4–6 visits increased, with no statistically meaningful effects on 0 visits or 1–3 visits. These results are consistent with a partial disruption of care continuity and/or scheduling regularity that reduced the probability of completing high-intensity ANC, but did not substantially push women into complete exclusion from care.

International quantitative evidence generally suggests that flooding and related hazards can depress maternal health-care utilisation, especially where shocks disrupt transport, facilities, and household livelihoods. For example, studies in Bangladesh report lower odds of receiving ANC from skilled providers and other MNH services in flood-affected areas, with stronger negative associations where flooding is recurrent rather than episodic (Baten et al., 2020; Orderud et al., 2022). Related work linking displacement due to extreme events to maternal care utilisation in rural Bangladesh finds that repeated displacement is associated with lower probabilities of receiving ANC and of completing ≥ 4 visits with a trained provider (Haque, Parr e Muhidin, 2020a). In India, recent analyses merging household survey data with flood-prone geographies similarly report worse ANC adequacy or higher probability of incomplete ANC in flood-prone districts, with particularly large disparities for socially marginalised groups (Barman et al., 2025; Chandra et al., 2025).

At the same time, not all settings exhibit strong average effects once area characteristics are accounted for. Evidence from Bangladesh using multi-survey data suggests that bivariate associations between flooding and maternal service use can attenuate substantially after adjusting for socioeconomic and demographic factors, implying that lower utilisation may partly reflect structural disadvantages of flood-prone areas rather than the causal effect of flood events per se (Orderud et al., 2022). Our findings sit between these two patterns. On one hand, we detect a meaningful post-flood change that is consistent with disruption mechanisms emphasised in the broader extreme-weather and maternal-health literature (Pappas, Kovats e Ranganathan, 2024; Mallett e Etzel, 2018). On the other hand, the absence of effects at the lower tail (0 visits) is consistent with the idea that, even under stress, some systems maintain a basic floor of access, particularly where baseline ANC coverage is high and primary care is geographically widespread.

A key substantive contribution of our results to this literature is what changes: rather than a uniform decline in utilisation, the shock primarily reduced ≥ 7 visits and increased 4–6 visits. This "downshifting" pattern is often not observable in studies that focus only on a single adequacy threshold (e.g., ≥ 4 visits) or binary outcomes. In practical terms, floods may not eliminate ANC use in a high-coverage system, but they can impair the ability to complete recommended schedules, potentially weakening the preventive, screening, and continuity functions of ANC.

The pattern we observe is also coherent with the broader determinants-of-ANC literature. Frameworks of health services utilisation emphasise that care-seeking depends on (i) enabling resources (income, transport, time), (ii) need/risk and perceived need, and (iii) supply-side access and quality (Andersen, 2008; Levesque, Harris e Russell, 2013). Flood shocks plausibly operate through multiple channels simultaneously: damage to roads and bridges raises travel costs; facility disruption and workforce displacement reduce appointment availability; and income losses tighten household budgets and time constraints. These mechanisms predict reductions in visit frequency even when some contact with the health system is maintained.

In Brazil, a large body of evidence documents persistent socioeconomic and geographic gradients in ANC adequacy/quality, including strong associations with maternal education, age,

parity, and partnership status, as well as with broader regional and access barriers (Domingues et al., 2015; Leal et al., 2020; Morón-Duarte et al., 2021; Anjos e Boing, 2016; Silva et al., 2019). This context supports the interpretation that flood exposure may interact with pre-existing disadvantages: municipalities with higher rurality, weaker transport connectivity, or more fragile service networks may find it harder to sustain high-intensity ANC schedules during and after a shock. In our setting, the finding that women did not move into non-utilisation on average is consistent with Brazil's near-universal ANC coverage reported in national surveys (Viellas et al., 2014), but the redistribution away from ≥ 7 visits is consistent with documented challenges in ensuring adequate and high-quality care even in non-disaster periods (Cunha et al., 2019; Ortelan et al., 2023; Esposti et al., 2020).

The CITS results suggest that the main effect operates through an immediate level shift rather than a sustained divergence in trends, while the event-study DiD indicates a persistent post-treatment difference over subsequent months. These two views are not contradictory. A shock can generate an abrupt drop in high-frequency visits (missed appointments, cancelled clinic days, interrupted follow-up), and then a new equilibrium may emerge as services "catch up" partially but not fully restore pre-shock intensity. In settings where disruptions last beyond the acute phase (because infrastructure repairs, rescheduling backlogs, and household recovery take time) a level shift can remain visible over an extended horizon.

The international literature offers supportive parallels. Work relating precipitation anomalies to disruptions in ANC provision in Malawi, for example, emphasises that climate-related disruption can be recurrent and cumulative rather than a short-lived acute interruption, particularly when transport and physical access are repeatedly compromised (Murray-Watson et al., 2025). Similarly, evidence from floodplain settings shows that seasonal flooding can sharply reduce geographic access to maternal services at particular times of year, indicating that even temporary access barriers can meaningfully reshape utilisation patterns (Mroz et al., 2023).

The WHO climate-resilient health systems agenda emphasises strengthening core health-system functions to manage climate risks, including service delivery, workforce, infrastructure, information systems, financing, governance, and community engagement (World Health Organization, 2022). Interpreted through a resilience lens (absorptive, adaptive, and transformative capacities), our results point to an important nuance: the maternal health-care system appears to have absorbed the shock sufficiently to prevent large increases in complete non-use (0 ANC visits), but did not fully preserve performance at the upper end of recommended contact. In the language of resilience trajectories, this resembles a system that maintains an "acceptable" level of functioning for most users (continued contact), while experiencing a measurable performance drop in more resource-intensive or continuity-dependent components (completion of ≥ 7 visits).

This distinction matters for policy. Absorptive capacity may reflect the existence of a broad primary-care network, continuity mechanisms, and baseline high coverage that can keep some ANC functioning even when infrastructure is stressed. Adaptive capacity may involve reorganisation of care pathways, outreach, prioritisation of higher-risk pregnancies, and temporary

alternatives (e.g., rebooking, redistribution of staff, or mobile care), which can sustain intermediate levels of ANC but may not fully restore high-frequency schedules immediately. Qualitative evidence from flood contexts highlights precisely these adaptive strategies (reorganising flows, itinerant care, and coordination across levels) as key to maintaining maternal care during flooding (Saulnier et al., 2022; Nunes et al., 2025). From this perspective, our "downshifting" pattern may represent an adaptive prioritisation under constraint: ensuring a minimum package of visits for many women, at the cost of fewer women completing the highest-intensity schedules.

This study contributes to the floods–maternal health literature in three ways. First, it provides causal, event-study DiD evidence in a large municipality–month panel around a well-defined extreme event, using estimators designed for staggered-treatment and modern DiD settings (Callaway e Sant’Anna, 2021). Second, it distinguishes impacts across the full distribution of ANC intensity (0, 1–3, 4–6, ≥ 7), showing that shocks may reallocate utilisation across categories rather than simply reduce use on average. Third, by combining CITS and event-study DiD evidence, it offers convergent validity: both approaches indicate that the key effect is a reallocation from high-intensity to intermediate-intensity care.

Several limitations should be considered. First, treatment status relies on official disaster records; misclassification (underreporting or heterogeneity in severity within municipalities) could attenuate estimated effects. Second, outcomes are municipality-level shares based on live births; while this design captures population-level patterns, it cannot separate behavioural changes from compositional shifts in who becomes pregnant or carries to term after the shock. Third, we do not observe the content or quality of ANC visits, only visit counts/categories; a stable number of visits does not necessarily imply stable quality. Fourth, despite excluding adjacent municipalities and other hydrological disasters to reduce contamination, some spillovers (e.g., cross-municipality service substitution) may persist and would typically bias estimates toward zero. Finally, external validity is bounded: the SUS context, baseline ANC coverage, and the institutional footprint of primary care in Northeast Brazil may differ from lower-coverage systems where floods can plausibly generate stronger effects on complete non-use.

The results suggest that flood preparedness and response policies should not focus solely on preventing complete care interruption, but also on protecting continuity and the ability to complete recommended ANC schedules. In practice, this points to resilience investments aligned with WHO’s climate-resilient health systems framework (World Health Organization, 2022): safeguarding facility functionality and transport connectivity, surge staffing and flexible scheduling, robust information systems for tracing missed appointments, and proactive outreach for pregnant women in affected areas. Future work should examine heterogeneity (e.g., by baseline rurality, poverty, or health-system capacity), incorporate severity measures of flood exposure, and, where possible, link utilisation shifts to maternal and neonatal outcomes to quantify welfare-relevant consequences of the observed "downshifting" in ANC intensity.

4.5 CONCLUSION

Using municipality–month panel data from 2008–2012 and triangulating across controlled interrupted time series and modern event-study difference-in-differences estimators, we find that the June 2010 floods in Pernambuco and Alagoas causally altered the intensity of antenatal care utilisation rather than producing widespread non-use. Flood exposure generated a clear redistribution from high-intensity care (≥ 7 visits) toward intermediate utilisation (4–6 visits), with no statistically meaningful changes in the shares receiving 0 visits or only 1–3 visits. Interpreted through a health-systems resilience lens, these results suggest that SUS and local primary care networks largely preserved a floor of access during the shock, but struggled to maintain continuity and schedule completion at the upper end of recommended care. Policy efforts to build climate-resilient maternal health services should therefore prioritise mechanisms that sustain continuity so that extreme floods do not translate into systematic shortfalls in recommended ANC trajectories.

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APPENDIX 4.A – R SCRIPT

```

1 #####
2 # Floods and ANC utilisation (SINASC + Atlas de Desastres)
3 # Pernambuco (PE) and Alagoas (AL), 2008-2012
4 #
5 # This script:
6 # 1) Downloads and processes SINASC microdata (PE/AL)
7 # 2) Builds municipal treatment/control/excluded groups using
8 #     the Atlas Digital de Desastres Naturais (hydrological events)
9 # 3) Builds a municipality-month panel of ANC utilisation and
10 #    covariates
11 # 4) Adds municipal controls (PIBpc real, rural population, AB/ESF
12 #    coverage)
13 # 5) Estimates (i) ITSA (two-group) and (ii) DiD event study (did
14 #    package)
15 # 6) Produces balance checks (t-test and SMD)
16 #
17 # Notes:
18 # - Dates are monthly at municipality of residence (CODMUNRES).
19 # - Treated municipalities: at least one hydrological disaster
20 #    recorded
21 #    between 2010-06-17 and 2010-06-20.
22 # - Excluded municipalities: hydrological events between 2008-2012
23 #    but not
24 #    in the treatment window, plus "neighbors" of treated
25 #    municipalities.
26 # - Controls: no hydrological event in 2008-2012 and not neighbor of
27 #    treated.
28 #####
29
30 # =====
31 # 0) Packages & global settings
32 # =====
33
34 suppressPackageStartupMessages({
35   library(tidyverse)
36   library(lubridate)
37   library(microdatasus)
38   library(geobr)
39   library(sf)

```

```

34 library(sidrar)
35 library(deflateBR)
36
37 library(httr)
38 library(jsonlite)
39
40 library(stringr)
41 library(stringi)
42
43 library(did)
44 })
45
46 options(scipen = 999)
47
48 # --- Study window (SINASC) ---
49 date_start <- as.Date("2008-01-01")
50 date_end   <- as.Date("2012-12-31")
51
52 # --- Treatment window (flood shock) ---
53 treat_start <- as.Date("2010-06-17")
54 treat_end   <- as.Date("2010-06-20")
55
56 # --- Intervention month for monthly panel models ---
57 cutoff <- as.Date("2010-07-01") # July/2010 = t=0 in monthly specs
58
59 # --- States in analysis ---
60 ufs <- c("PE", "AL")
61
62 # =====
63 # 1) SINASC: download + process (PE/AL) and keep 2008-2012
64 # =====
65
66 # Helper to fetch SINASC for one UF
67 fetch_sinasc_uf <- function(uf, year_start = 2007, year_end = 2013) {
68   microdatasus::fetch_datasus(
69     year_start = year_start,
70     year_end   = year_end,
71     month_start = 1,
72     month_end   = 12,
73     uf = uf,
74     information_system = "SINASC"
75   )

```

```

76 }
77
78 sinasc_raw <- ufs %>%
79   set_names() %>%
80   map(fetch_sinasc_uf) %>%
81   bind_rows()
82
83 sinasc <- sinasc_raw %>%
84   microdatasus::process_sinasc() %>%
85   mutate(
86     DTNASC = as.Date(DTNASC)
87   ) %>%
88   filter(
89     DTNASC >= date_start,
90     DTNASC <= date_end
91   )
92
93 # =====
94 # 2) Atlas de Desastres: hydrological events (PE/AL)
95 #   and municipal exposure classification
96 # =====
97
98 atlas_url <- "https://atlasdigital.mdr.gov.br/arquivos/BD_Atlas_1991_
99             2024_v1.0_2025.04.14_Consolidado.csv"
100
101 desastres <- readr::read_csv2(
102   atlas_url,
103   locale = locale(encoding = "Latin1")
104 )
105
106 hidrologicos_pe_al <- desastres %>%
107   mutate(
108     Data_Evento = dmy(Data_Evento)
109   ) %>%
110   filter(
111     Sigla_UF %in% ufs,
112     grupo_de_desastre == "Hidrol gico"
113   ) %>%
114   # Fix known municipality name inconsistencies (Atlas vs. geobr)
115   mutate(
116     Nome_Municipio = recode(
117       Nome_Municipio,

```

```

117     "Tanque_D'Arca" = "Tanque_dArca",
118     "Olho_D'Água_das_Flores" = "Olho_dÁgua_das_Flores",
119     "SÃço_Vicente_FÃlrrer" = "SÃço_Vicente_Ferrer"
120 )
121 )
122
123 # Flag municipalities:
124 # - treated: any event in the treatment window (Jun 17â€š20, 2010)
125 # - excluded: any event in the broad window (2008â€š2012) but not
    treated
126 # - control: no event in 2008â€š2012
127 municipios_exposure <- hidrologicos_pe_al %>%
128   mutate(Data_Evento = as.Date(Data_Evento)) %>%
129   group_by(Nome_Municipio, Cod_IBGE_Mun, Sigla_UF) %>%
130   summarise(
131     treated_flag = any(Data_Evento >= treat_start & Data_Evento <=
        treat_end, na.rm = TRUE),
132     window_flag = any(Data_Evento >= date_start & Data_Evento <=
        date_end, na.rm = TRUE),
133
134     # Audit strings (useful for appendix / traceability)
135     eventos_trat = paste(
136       sort(unique(Data_Evento[Data_Evento >= treat_start & Data_
        Evento <= treat_end])),
137       collapse = ", "
138     ),
139     eventos_window = paste(
140       sort(unique(Data_Evento[Data_Evento >= date_start & Data_Evento
        <= date_end])),
141       collapse = ", "
142     ),
143     eventos_fora_window = paste(
144       sort(unique(Data_Evento[!(Data_Evento >= date_start & Data_
        Evento <= date_end)])),
145       collapse = ", "
146     ),
147     eventos_todos = paste(sort(unique(Data_Evento)), collapse = ", ")
148     ,
149     n_eventos_distintos = n_distinct(Data_Evento),
150     .groups = "drop"
151   ) %>%

```

```

152 mutate(
153   Grupo = case_when(
154     treated_flag ~ "treated",
155     !treated_flag & window_flag ~ "excluded",
156     TRUE ~ "control"
157   )
158 ) %>%
159 select(
160   Nome_Municipio, Cod_IBGE_Mun, Sigla_UF,
161   Grupo,
162   n_eventos_distintos,
163   eventos_trat, eventos_window, eventos_fora_window, eventos_todos
164 )
165
166 # =====
167 # 3) Spatial layer (geobr): PE/AL municipalities
168 #   + merge exposure group + exclude neighbors of treated
169 # =====
170
171 # Read all municipalities (2010) and keep PE/AL
172 muni_br <- geobr::read_municipality(
173   year = 2010,
174   showProgress = FALSE
175 )
176
177 muni_pe_al <- muni_br %>%
178   filter(abbrev_state %in% ufs) %>%
179   filter(code_muni != 2605459) %>%           # remove Fernando de
180     Noronha
181   st_make_valid()
182
183 # Merge exposure group to shape (left join keeps all municipalities
184   in PE/AL shape)
185 muni_pe_al_join <- muni_pe_al %>%
186   left_join(
187     municipios_exposure %>%
188       select(code_muni = Cod_IBGE_Mun, Grupo),
189     by = "code_muni"
190   ) %>%
191   mutate(
192     Grupo = coalesce(Grupo, "control"),
193     # DataSUS municipal codes are often 6 digits (drop DV)

```

```

192   Cod_Mun_DataSUS = substr(as.character(code_muni), 1, 6)
193 )
194
195 # Identify "neighbors" of treated (sharing border OR touching
    corners)
196 # Using st_intersects = any intersection (corner or border touch)
197 treated_sf <- muni_pe_al_join %>% filter(Grupo == "treated")
198 control_sf <- muni_pe_al_join %>% filter(Grupo == "control")
199
200 inter_list <- st_intersects(control_sf, treated_sf, sparse = TRUE)
201
202 control_neighbors_flag <- control_sf %>%
203   st_drop_geometry() %>%
204   transmute(
205     code_muni,
206     flag_vizinho_tratado = lengths(inter_list) > 0
207   )
208
209 muni_reclass <- muni_pe_al_join %>%
210   left_join(control_neighbors_flag, by = "code_muni") %>%
211   mutate(
212     flag_vizinho_tratado = coalesce(flag_vizinho_tratado, FALSE),
213     Grupo = if_else(Grupo == "control" & flag_vizinho_tratado, "
        excluded", Grupo)
214   )
215
216 # Quick counts
217 muni_reclass %>% count(Grupo)
218
219 # Treated vs Control Municipalities Map
220 mapa_flood <- ggplot(muni_reclass) +
221   geom_sf(aes(fill = Grupo), color = "black", linewidth = 0.15) +
222   scale_fill_manual(
223     values = c(treated = "#1f78b4", control = "#F4A6A6", excluded = "
        white"),
224     name = "Flood_exposure_(Jun_17_20_2010)"
225   ) +
226   theme_void() +
227   theme(
228     legend.position = "right",
229     legend.title = element_text(size = 10),
230     legend.text = element_text(size = 9)

```

```

231   )
232
233   print(mapa_flood)
234
235   # =====
236   # 4) Build municipality-month panel from SINASC (ANC + covariates)
237   # =====
238
239   # --- Define coding sets once (readability + less repetition) ---
240   esc_baixa_levels <- c("Nenhum", "0", "1_a_3_anos", "4_a_7_anos")
241   esc_alta_levels  <- c("8_a_11_anos", "12_anos_ou_mais")
242
243   sem_parceiro_levels <- c("Solteira", "Separada_judicialmente", "
      Viãžva")
244
245   prematuro_levels <- c("22_a_27_semanas", "28_a_31_semanas", "32_a_36_
      semanas")
246
247   parto_estab_levels <- c("Hospital", "Outro_estabelecimento_de_saãžde"
      )
248
249   sinasc_panel <- sinasc %>%
250     mutate(
251       data      = ymd(DTNASC),
252       mes_ano = floor_date(data, unit = "month"),
253
254       # Robust numeric parsing (SINASC can come as character/factor)
255       idade_num   = readr::parse_number(as.character(IDADEMAE)),
256       paridade_num = readr::parse_number(as.character(QTDFILVIVO)),
257
258       # Binary indicators (individual-level, later aggregated to muni-
         month)
259       esc_baixa = ESCMAE %in% esc_baixa_levels,
260       esc_alta  = ESCMAE %in% esc_alta_levels,
261
262       sem_parceiro = ESTCIVMAE %in% sem_parceiro_levels,
263
264       parto_estab = LOCNASC %in% parto_estab_levels,
265
266       prematuro = GESTACAO %in% prematuro_levels,
267
268       cesareo = PARTO == "Cesãžreo",

```

```

269
270     # Birth occurred in the municipality of residence?
271     nasc_local = CODMUNNASC == CODMUNRES
272 ) %>%
273 filter(!is.na(mes_ano)) %>%
274 group_by(CODMUNRES, mes_ano) %>%
275 summarise(
276     # ANC counts (CONSULTAS is categorical)
277     n_total = sum(!is.na(CONSULTAS)),
278     n_0      = sum(CONSULTAS == "Nenhuma",      na.rm = TRUE),
279     n_1a3    = sum(CONSULTAS == "1_a_3_vezes",  na.rm = TRUE),
280     n_4a6    = sum(CONSULTAS == "4_a_6_vezes",  na.rm = TRUE),
281     n_7m     = sum(CONSULTAS == "7_ou_mais_vezes", na.rm = TRUE),
282
283     # Total births (denominator for covariate means)
284     n_nascimentos = n(),
285
286     # Maternal composition (muni-month)
287     idade_media    = mean(idade_num,      na.rm = TRUE),
288     paridade_media = mean(paridade_num, na.rm = TRUE),
289
290     prop_adolescente = mean(idade_num < 20, na.rm = TRUE),
291     prop_geriatrica  = mean(idade_num >= 35, na.rm = TRUE),
292
293     prop_primipara = mean(paridade_num < 1, na.rm = TRUE),
294
295     prop_esc_baixa = mean(esc_baixa, na.rm = TRUE),
296     prop_esc_alta  = mean(esc_alta,  na.rm = TRUE),
297
298     prop_sem_parceiro = mean(sem_parceiro, na.rm = TRUE),
299
300     prop_parto_estab = mean(parto_estab, na.rm = TRUE),
301     prop_prematuro    = mean(prematuro,  na.rm = TRUE),
302     prop_cesareo      = mean(cesareo,     na.rm = TRUE),
303
304     prop_nasc_local   = mean(nasc_local,  na.rm = TRUE),
305
306     .groups = "drop"
307 ) %>%
308 # Complete missing municipality-months to build a balanced calendar
309 complete(
310     CODMUNRES,

```

```

311   mes_ano = seq.Date(
312     from = as.Date("2008-01-01"),
313     to   = as.Date("2012-12-01"),
314     by   = "month"
315   )
316 ) %>%
317 # Fill counts with 0 where months were artificially created
318 mutate(
319   across(c(n_total, n_0, n_1a3, n_4a6, n_7m, n_nascimentos), ~
320     coalesce(., 0L)),
321
322   # Shares of ANC categories (set NA if no births w/ ANC info)
323   prenatal_0      = if_else(n_total > 0, n_0 / n_total, NA_real_),
324   prenatal_1a3    = if_else(n_total > 0, n_1a3 / n_total, NA_real_),
325   prenatal_4a6    = if_else(n_total > 0, n_4a6 / n_total, NA_real_),
326   prenatal_7mais = if_else(n_total > 0, n_7m / n_total, NA_real_)
327 )
328 # =====
329 # 5) Merge treatment group to panel
330 # =====
331
332 painel <- sinasc_panel %>%
333   left_join(muni_reclass %>% st_drop_geometry(),
334     by = c("CODMUNRES" = "Cod_Mun_DataSUS")) %>%
335   filter(!is.na(Grupo))
336
337 # =====
338 # 6) Interrupted Time Series (two-group ITSA) on group means
339 # =====
340
341 serie_long <- painel %>%
342   filter(Grupo %in% c("treated", "control")) %>%
343   group_by(Grupo, mes_ano) %>%
344   summarise(
345     prenatal_7mais = mean(prenatal_7mais, na.rm = TRUE),
346     prenatal_4a6   = mean(prenatal_4a6,   na.rm = TRUE),
347     prenatal_1a3   = mean(prenatal_1a3,   na.rm = TRUE),
348     prenatal_0     = mean(prenatal_0,     na.rm = TRUE),
349     .groups = "drop"
350   ) %>%
351   pivot_longer(

```

```

352     cols = starts_with("prenatal_"),
353     names_to = "outcome",
354     values_to = "y"
355 ) %>%
356 mutate(
357     treated = as.integer(Grupo == "treated"),
358     post     = as.integer(mes_ano >= cutoff),
359     # Monthly time index centered at cutoff (Jul/2010 = 0)
360     t = (year(mes_ano) * 12 + month(mes_ano)) - (year(cutoff) * 12 +
361           month(cutoff))
362 )
363 ylabs <- c(
364     prenatal_7mais = "Mean_share_(7+ANC_visits)",
365     prenatal_4a6   = "Mean_share_(4-6ANC_visits)",
366     prenatal_1a3   = "Mean_share_(1-3ANC_visits)",
367     prenatal_0     = "Mean_share_(0ANC_visits)"
368 )
369
370 # Containers
371 itsa_fits <- list()
372 itsa_tables <- list()
373 itsa_plots <- list()
374
375 for (o in unique(serie_long$outcome)) {
376     dat <- serie_long %>% filter(outcome == o)
377
378     # Two-group ITSA with interactions:
379     # y ~ time + post + time*post + treated + time*treated + post*
380         treated + time*post*treated
381     fit <- lm(
382         y ~ t + post + t:post + treated + t:treated + post:treated + t:
383             post:treated,
384         data = dat
385     )
386     itsa_fits[[o]] <- fit
387
388     tab <- as.data.frame(summary(fit)$coefficients)
389     tab$term <- rownames(tab)
390     rownames(tab) <- NULL
391     tab$outcome <- o
392     itsa_tables[[o]] <- tab

```

```

391
392 # Predictions with confidence intervals
393 pred <- predict(fit, newdata = dat, interval = "confidence")
394 datp <- dat %>%
395   mutate(
396     fitted = pred[, "fit"],
397     lwr     = pred[, "lwr"],
398     upr     = pred[, "upr"]
399   )
400
401 p <- ggplot(datp, aes(x = mes_ano, color = Grupo)) +
402   geom_line(aes(y = y), linewidth = 0.7, alpha = 0.7) +
403   geom_ribbon(aes(ymin = lwr, ymax = upr, fill = Grupo), alpha =
404     0.15, color = NA) +
405   geom_line(aes(y = fitted), linewidth = 1.1) +
406   geom_vline(xintercept = cutoff, linetype = "dotted", linewidth =
407     0.7) +
408   scale_color_manual(values = c(treated = "#1f78b4", control = "
409     grey40")) +
410   scale_fill_manual(values = c(treated = "#1f78b4", control = "
411     grey40")) +
412   theme_classic() +
413   labs(
414     x = "Month",
415     y = ylabs[[0]],
416     color = "Group",
417     fill = "Group"
418   )
419
420 itsa_plots[[0]] <- p
421 }
422
423 itsa_results_table <- bind_rows(itsa_tables)
424 itsa_results_table
425
426 walk(itsa_plots, print)
427
428 # =====
429 # 7) Municipal controls (SIDRA + deflation + rural share)
430 # =====
431
432 # --- GDP (PIB municipal), nominal (in thousand R$) ---

```

```

429 PIBm <- get_sidra(
430   x = 5938,
431   variable = 37,
432   period = "2008-2012",
433   geo = "City",
434   geo.filter = list(State = c(26, 27)), # PE=26, AL=27
435   header = TRUE,
436   format = 3
437 )
438
439 # --- Population estimates (annual) ---
440 pop_est <- get_sidra(
441   x = 6579,
442   variable = 9324,
443   period = "2008-2012",
444   geo = "City",
445   geo.filter = list(State = c(26, 27)),
446   header = TRUE,
447   format = 3
448 )
449
450 # --- Census population (2010), total / urban / rural ---
451 # Table 1378: population by situation of domicile (c1)
452 pop_censo <- get_sidra(
453   x = 1378,
454   variable = 93,
455   period = "2010",
456   geo = "City",
457   geo.filter = list(State = c(26, 27)),
458   classific = c("c1"),
459   category = list(0), # total
460   header = TRUE,
461   format = 3
462 )
463
464 pop_urbana <- get_sidra(
465   x = 1378,
466   variable = 1000093,
467   period = "2010",
468   geo = "City",
469   geo.filter = list(State = c(26, 27)),
470   classific = c("c1"),

```

```

471   category = list(1),      # urban
472   header = TRUE,
473   format = 3
474 )
475
476 pop_rural <- get_sidra(
477   x = 1378,
478   variable = 1000093,
479   period = "2010",
480   geo = "City",
481   geo.filter = list(State = c(26, 27)),
482   classific = c("c1"),
483   category = list(2),      # rural
484   header = TRUE,
485   format = 3
486 )
487
488 # Combine census total (2010) + annual estimates (2008-2012)
489 pop_total <- bind_rows(
490   pop_censo %>% select(intersect(names(pop_censo), names(pop_est))),
491   pop_est %>% select(intersect(names(pop_censo), names(pop_est)))
492 )
493
494 # Standardize PIB and population
495 pib <- PIBm %>%
496   transmute(
497     cod_mun = as.character('MunicÃpio (CÃdigo)'),
498     ano      = as.integer(Ano),
499     pib_nom  = as.numeric(Valor) * 1000 # convert from "thousand R$"
500     to R$
501   )
502
503 pop <- pop_total %>%
504   transmute(
505     cod_mun = as.character('MunicÃpio (CÃdigo)'),
506     ano      = as.integer(Ano),
507     pop      = as.numeric(Valor)
508   )
509
510 # Merge and deflate PIB to Dec/2012 (IPCA)
511 base_pib <- pib %>%
512   left_join(pop, by = c("cod_mun", "ano")) %>%

```

```

512   mutate(data_ref = as.Date(sprintf("%d-12-01", ano)))
513
514 PIBm_pc <- base_pib %>%
515   mutate(
516     pib_real = deflateBR::deflate(
517       nominal_values = pib_nom,
518       nominal_dates  = data_ref,
519       real_date      = "12/2012",
520       index          = "ipca"
521     ),
522     pib_pc_real = if_else(pop > 0, pib_real / pop, NA_real_)
523   ) %>%
524   select(cod_mun, ano, pib_pc_real)
525
526 # Add GDPpc to panel (annual merge via year)
527 painel <- painel %>%
528   mutate(
529     ano = year(mes_ano),
530     cod_mun = as.character(code_muni) # 7-digit IBGE code in geobr
531   ) %>%
532   left_join(PIBm_pc, by = c("cod_mun", "ano"))
533
534 # Add rural population (2010 only; time-invariant control)
535 painel <- painel %>%
536   left_join(
537     pop_rural %>%
538       transmute(
539         cod_mun = as.character('MunicÃpio (CÃdigo)'),
540         pop_rural = as.numeric(Valor)
541       ),
542     by = "cod_mun"
543   )
544
545 # =====
546 # 8) Primary care coverage (AB/ESF) from APS dashboard API
547 # =====
548
549 # This endpoint returns monthly coverage series by municipality for
550   NE region (coRegiao=2)
551 url_ab <- paste0(
552   "https://relatorioaps-prd.saude.gov.br/cobertura/ab?",

```

```

552   "unidadeGeografica=MUNICIPIO&coRegiao=2&nuCompInicio=200801&
      nuCompFim=201212"
553 )
554
555 res <- httr::GET(url_ab)
556 httr::stop_for_status(res)
557
558 json_txt <- httr::content(res, "text", encoding = "UTF-8")
559 df_ab <- jsonlite::fromJSON(json_txt, flatten = TRUE)
560
561 atencao_basica <- df_ab %>%
562   filter(sgUf %in% ufs) %>%
563   mutate(
564     code_muni = as.character(coMunicipioIbge),
565     cobertura_ab = as.numeric(pcCoberturaAb) / 100,
566     cobertura_esf = as.numeric(pcCoberturaSf) / 100,
567     mes_ano = ymd(paste0(nuComp, "01"))
568   )
569
570 # Merge to panel (municipality-month)
571 painel <- painel %>%
572   left_join(
573     atencao_basica %>%
574       select(CODMUNRES = coMunicipioIbge, mes_ano, cobertura_ab,
575             cobertura_esf),
576     by = c("CODMUNRES", "mes_ano")
577   )
578 # =====
579 # 9) DiD Event Study (Callaway & Sant'Anna): single shock in Jul/2010
580 # =====
581
582 ys <- c("prenatal_0", "prenatal_1a3", "prenatal_4a6", "prenatal_7mais")
583
584 controls <- c(
585   "cobertura_esf", "pib_pc_real", "pop_rural",
586   "prop_adolescente", "prop_primipara", "prop_esc_baixa",
587   "prop_sem_parceiro", "prop_prematuro", "prop_nasc_local"
588 )
589
590 xform <- as.formula(paste0("~", paste(controls, collapse = "+")))

```

```

591
592 # Optional window cap for event-study horizons (Inf = use full
      support)
593 win_cap <- Inf
594
595 # Reference month (first month observed)
596 t0 <- painel %>%
597   summarise(t0 = min(mes_ano, na.rm = TRUE)) %>%
598   pull(t0)
599
600 # Convert month (Date) to a sequential integer index (1..T)
601 month_index <- function(d, origin_month) {
602   as.integer((year(d) * 12 + month(d)) - (year(origin_month) * 12 +
      month(origin_month)) + 1)
603 }
604
605 g_treat_index <- month_index(cutoff, t0)
606
607 painel_did <- painel %>%
608   filter(Grupo %in% c("treated", "control")) %>%
609   mutate(
610     id_muni = as.integer(CODMUNRES),
611
612     # Índice de tempo mensal contínuo (1, 2, 3, ..., T)
613     t_index = month_index(mes_ano, t0),
614
615     # corte: tratados entram em Jul/2010 (Índice g_treat_index);
      controles = 0 (never-treated)
616     g = if_else(Grupo == "treated", g_treat_index, 0L),
617
618     # tempo relativo ao tratamento (Jul/2010 = 0)
619     rel_time = as.integer((year(mes_ano) * 12 + month(mes_ano)) -
      (year(cutoff) * 12 + month(cutoff)))
620   ) %>%
621   filter(n_total > 0) %>%                                     # remove meses artificiais
      (complete sem nascimentos)
622
623   arrange(id_muni, mes_ano) %>%
624   group_by(id_muni) %>%
625   fill(all_of(controls), .direction = "downup") %>%
626   ungroup()
627
628 # -----

```

```

629 # 2) Estimation function (no fallback)
630 # -----
631 run_eventstudy <- function(yvar, data, xformula, win_cap = Inf, g_
    treat_index) {
632
633   keep_cols <- c(
634     "id_muni", "t_index", "g", "mes_ano", "Grupo", "rel_time",
635     yvar, all.vars(xformula)
636   )
637
638   dat <- data %>%
639     select(all_of(keep_cols)) %>%
640     drop_na(all_of(yvar)) %>%           # exige Y observado
641     drop_na(all.vars(xformula))       # exige controles observados
642
643   # suporte real do outcome nos tratados (para definir min_e/max_e)
644   rel_treated <- dat %>%
645     filter(g == g_treat_index) %>%
646     pull(rel_time)
647
648   if (length(rel_treated) == 0 || all(is.na(rel_treated))) {
649     stop("Sem suporte nos tratados após limpeza de NA para o outcome
        :_", yvar)
650   }
651
652   min_e_ok <- suppressWarnings(min(rel_treated, na.rm = TRUE))
653   max_e_ok <- suppressWarnings(max(rel_treated, na.rm = TRUE))
654
655   # aplica teto opcional
656   min_e_ok <- max(min_e_ok, -win_cap)
657   max_e_ok <- min(max_e_ok, win_cap)
658
659   if (!is.finite(min_e_ok) || !is.finite(max_e_ok) || min_e_ok >= max
        _e_ok) {
660     stop(
661       "Janela inválida para o outcome:_", yvar,
662       "_(min_e_ok=", min_e_ok, ", _max_e_ok=", max_e_ok, ")"
663     )
664   }
665
666   # ---- att_gt: allow_unbalanced_panel não se a versão suportar
667   att_args <- list(

```

```

668   yname   = yvar,
669   tname   = "t_index",           # <<< usa Āndice de tempo
670   idname  = "id_muni",
671   gname   = "g",
672   data    = dat,
673   panel   = TRUE,
674   control_group = "nevertreated",
675   xformula = xformula,
676   clustervars = "id_muni"
677 )
678
679 if ("allow_unbalanced_panel" %in% names(formals(did::att_gt))) {
680   att_args$allow_unbalanced_panel <- TRUE
681 }
682
683 cs <- do.call(did::att_gt, att_args)
684
685 # ---- aggte: event study (dinĀmico)
686 es <- did::aggte(
687   cs,
688   type = "dynamic",
689   na.rm = TRUE,
690   min_e = min_e_ok,
691   max_e = max_e_ok
692 )
693
694 # ---- aggte: ATT agregado
695 simple <- did::aggte(
696   cs,
697   type = "simple",
698   na.rm = TRUE
699 )
700
701 es_tbl <- tibble(
702   outcome = yvar,
703   event_time = es$egt,
704   att = es$att.egt,
705   se = es$se.egt,
706   ci_low = es$att.egt - 1.96 * es$se.egt,
707   ci_high = es$att.egt + 1.96 * es$se.egt,
708   # mapeia event_time para mĀis calendĀrio (Ātil p/ plot/tabela)
709   calendar_month = cutoff %m+% months(es$egt)

```

```

710   )
711
712   simple_tbl <- tibble(
713     outcome = yvar,
714     att = simple$overall.att,
715     se = simple$overall.se,
716     ci_low = simple$overall.att - 1.96 * simple$overall.se,
717     ci_high = simple$overall.att + 1.96 * simple$overall.se,
718     p_value = 2 * stats::pnorm(abs(simple$overall.att / simple$
       overall.se), lower.tail = FALSE)
719   )
720
721   list(
722     cs = cs,
723     es = es,
724     simple = simple,
725     es_tbl = es_tbl,
726     simple_tbl = simple_tbl,
727     win_used = c(min_e_ok, max_e_ok)
728   )
729 }
730
731 res_did <- ys %>%
732   set_names() %>%
733   map(~ run_eventstudy(.x, painel_did, xform, win_cap = win_cap, g_
     treat_index = g_treat_index))
734
735 es_all <- map_dfr(res_did, "es_tbl")
736 main_all <- map_dfr(res_did, "simple_tbl")
737
738 win_used <- tibble(
739   outcome = names(res_did),
740   min_e = map_dbl(res_did, ~ .x$win_used[1]),
741   max_e = map_dbl(res_did, ~ .x$win_used[2])
742 )
743 win_used
744
745 plot_eventstudy_relative <- function(es_tbl, win_plot = 12, title =
  NULL) {
746
747   df <- es_tbl %>%
748     filter(event_time >= -win_plot, event_time <= win_plot) %>%

```

```

749   mutate(
750     outcome = factor(
751       outcome,
752       levels = c("prenatal_7mais", "prenatal_4a6", "prenatal_1a3",
753                 "prenatal_0"),
754       labels = c("7_or_more_visits", "4-6_visits", "1-3_visits",
755                 , "No_visits")
756     )
757   )
758
759   ggplot(df, aes(x = event_time, y = att)) +
760     geom_hline(yintercept = 0, linetype = "dashed") +
761     geom_point(size = 1.8) +
762     geom_errorbar(aes(ymin = ci_low, ymax = ci_high), width = 0.2) +
763     facet_wrap(~ outcome, scales = "free_y") +
764     scale_x_continuous(limits = c(-win_plot, win_plot),
765                       breaks = seq(-win_plot, win_plot, by = 3)) +
766     labs(
767       x = "Months_relative_to_treatment_(July_2010_=0)",
768       y = "ATT",
769       title = title
770     ) +
771     theme_classic() +
772     theme(
773       plot.title = element_text(face = "bold"),
774       strip.text = element_text(face = "bold")
775     )
776   }
777
778   p_es_rel <- plot_eventstudy_relative(es_all, win_plot = 12)
779   print(p_es_rel)
780
781   # Main (paper) table: aggregated ATT
782   main_did <- main_all %>%
783     mutate(
784       outcome = factor(
785         outcome,
786         levels = c("prenatal_7mais", "prenatal_4a6", "prenatal_1a3", "
787                   prenatal_0"),
788         labels = c("7+_visits", "4-6_visits", "1-3_visits", "0_
789                   visits")
790       ),

```

```

787     stars = case_when(
788       p_value < 0.01 ~ "****",
789       p_value < 0.05 ~ "***",
790       p_value < 0.10 ~ "**",
791       TRUE ~ ""
792     ),
793     att_fmt = sprintf("%.4f%s", att, stars),
794     se_fmt = sprintf("(%.4f)", se),
795     ci_fmt = sprintf("[%%.4f, %%.4f]", ci_low, ci_high),
796     p_fmt = sprintf("%.3f", p_value)
797   ) %>%
798   select(outcome, att_fmt, se_fmt, ci_fmt, p_fmt)
799
800 main_did
801
802 # Event-study table for appendix (wide)
803 es_tbl_wide <- es_all %>%
804   filter(event_time >= -39, event_time <= 39) %>%
805   mutate(
806     outcome = factor(
807       outcome,
808       levels = c("prenatal_7mais", "prenatal_4a6", "prenatal_1a3", "
809         prenatal_0"),
810       labels = c("7+ visits", "4 visits", "1 visit", "0
811         visits")
812     ),
813     month_lbl = format(calendar_month, "%Y-%m"),
814     cell = sprintf("%.4f (%.4f)", att, se)
815   ) %>%
816   select(month_lbl, outcome, cell) %>%
817   pivot_wider(names_from = outcome, values_from = cell) %>%
818   arrange(month_lbl)
819
820 es_tbl_wide %>% print(n = Inf)
821
822 # =====
823 # 10) Pre-trend balance: t-test and Standardized Mean Difference (SMD
824   )
825 # =====
826
827 vars_balance <- c(ys, controls)

```

```

826 dados_pre <- painel %>%
827   filter(mes_ano < cutoff) %>%
828   group_by(CODMUNRES, Grupo) %>%
829   summarise(across(all_of(vars_balance), ~ mean(.x, na.rm = TRUE)),
830             .groups = "drop") %>%
831   filter(Grupo %in% c("treated", "control"))
832
833 # t-test by variable (treated vs control)
834 resultado_ttest <- map_dfr(vars_balance, function(v) {
835
836   x_trat <- dados_pre %>% filter(Grupo == "treated") %>% pull(all_of(
837     v))
838   x_ctrl <- dados_pre %>% filter(Grupo == "control") %>% pull(all_of(
839     v))
840
841   teste <- t.test(x_trat, x_ctrl)
842
843   tibble(
844     Variavel = v,
845     Media_Treated = mean(x_trat, na.rm = TRUE),
846     Media_Control = mean(x_ctrl, na.rm = TRUE),
847     p_value = teste$p.value
848   )
849 })
850
851 resultado_ttest
852
853 # SMD by variable
854 resultado_smd <- map_dfr(vars_balance, function(v) {
855
856   x_trat <- dados_pre %>% filter(Grupo == "treated") %>% pull(all_of(
857     v))
858   x_ctrl <- dados_pre %>% filter(Grupo == "control") %>% pull(all_of(
859     v))
860
861   media_trat <- mean(x_trat, na.rm = TRUE)
862   media_ctrl <- mean(x_ctrl, na.rm = TRUE)
863
864   sd_trat <- sd(x_trat, na.rm = TRUE)
865   sd_ctrl <- sd(x_ctrl, na.rm = TRUE)
866
867   n_trat <- sum(!is.na(x_trat))

```

```
864 n_ctrl <- sum(!is.na(x_ctrl))
865
866 sd_pooled <- sqrt(((n_trat - 1) * sd_trat^2 + (n_ctrl - 1) * sd_
      ctrl^2) /
867                (n_trat + n_ctrl - 2))
868
869 smd <- (media_trat - media_ctrl) / sd_pooled
870
871 tibble(
872   Variavel = v,
873   Media_Treated = media_trat,
874   Media_Control = media_ctrl,
875   SMD = smd
876 )
877 })
878
879 resultado_smd
```

APPENDIX 4.B – EVENT STUDY ESTIMATES

Table 4.B.1 – Event-study estimates by calendar month (ATT
with standard errors in parentheses)

Month	0 visits	1–3 visits	4–6 visits	7+ visits
2008-02	-0.0154 (0.0080)	0.0273 (0.0186)	0.0103 (0.0428)	-0.0222 (0.0400)
2008-03	0.0022 (0.0083)	0.0290 (0.0175)	-0.0832 (0.0395)	0.0521 (0.0369)
2008-04	0.0078 (0.0077)	-0.0172 (0.0156)	0.0431 (0.0422)	-0.0337 (0.0352)
2008-05	-0.0040 (0.0080)	-0.0049 (0.0164)	-0.0074 (0.0400)	0.0163 (0.0345)
2008-06	-0.0052 (0.0079)	0.0025 (0.0151)	0.0306 (0.0339)	-0.0279 (0.0343)
2008-07	0.0027 (0.0092)	-0.0224 (0.0197)	0.0156 (0.0439)	0.0040 (0.0398)
2008-08	0.0002 (0.0051)	-0.0008 (0.0179)	-0.0150 (0.0350)	0.0156 (0.0342)
2008-09	0.0034 (0.0062)	0.0195 (0.0119)	-0.0380 (0.0409)	0.0151 (0.0433)
2008-10	-0.0035 (0.0085)	0.0014 (0.0178)	0.0230 (0.0347)	-0.0209 (0.0379)
2008-11	0.0008 (0.0098)	0.0112 (0.0191)	-0.0099 (0.0352)	-0.0021 (0.0440)
2008-12	-0.0047 (0.0102)	-0.0063 (0.0164)	-0.0324 (0.0393)	0.0434 (0.0385)
2009-01	-0.0074 (0.0146)	-0.0301 (0.0226)	0.0060 (0.0374)	0.0315 (0.0498)
2009-02	0.0157 (0.0195)	-0.0104 (0.0186)	0.0207 (0.0336)	-0.0260 (0.0391)
2009-03	0.0004 (0.0135)	0.0278 (0.0153)	0.0053 (0.0295)	-0.0335 (0.0333)
2009-04	-0.0061 (0.0079)	-0.0014 (0.0215)	-0.0204 (0.0478)	0.0279 (0.0439)
2009-05	-0.0002 (0.0068)	-0.0155 (0.0261)	0.0783 (0.0429)	-0.0625 (0.0383)
2009-06	0.0113 (0.0086)	0.0153 (0.0212)	-0.0851 (0.0393)	0.0585 (0.0417)
2009-07	-0.0059 (0.0073)	0.0159 (0.0210)	-0.0006 (0.0436)	-0.0094 (0.0441)
2009-08	-0.0059 (0.0061)	0.0025 (0.0169)	0.0320 (0.0340)	-0.0286 (0.0357)
2009-09	0.0146 (0.0061)	-0.0243 (0.0195)	0.0422 (0.0478)	-0.0325 (0.0511)
2009-10	-0.0163 (0.0056)	0.0168 (0.0183)	0.0193 (0.0487)	-0.0199 (0.0409)
2009-11	0.0056 (0.0066)	-0.0063 (0.0148)	-0.0421 (0.0482)	0.0428 (0.0424)
2009-12	0.0010 (0.0070)	0.0068 (0.0196)	-0.0021 (0.0632)	-0.0056 (0.0605)
2010-01	-0.0060 (0.0097)	-0.0059 (0.0194)	-0.0072 (0.0343)	0.0191 (0.0359)
2010-02	0.0129 (0.0088)	-0.0052 (0.0218)	-0.0437 (0.0419)	0.0360 (0.0466)
2010-03	0.0006 (0.0066)	0.0251 (0.0177)	0.0088 (0.0605)	-0.0345 (0.0640)
2010-04	-0.0050 (0.0064)	-0.0343 (0.0194)	0.0582 (0.0644)	-0.0190 (0.0675)
2010-05	-0.0085 (0.0069)	0.0212 (0.0167)	0.0240 (0.0337)	-0.0367 (0.0403)
2010-06	0.0035 (0.0075)	-0.0148 (0.0141)	-0.0551 (0.0512)	0.0663 (0.0507)
2010-07	-0.0090 (0.0076)	0.0193 (0.0191)	0.0644 (0.0410)	-0.0747 (0.0369)
2010-07	-0.0090 (0.0076)	0.0193 (0.0191)	0.0644 (0.0410)	-0.0747 (0.0369)
2010-08	0.0061 (0.0081)	0.0153 (0.0186)	-0.0037 (0.0629)	-0.0178 (0.0629)

Continued on next page

Month	0 visits	1–3 visits	4–6 visits	7+ visits
2010-09	0.0035 (0.0144)	0.0091 (0.0222)	0.0558 (0.0508)	-0.0684 (0.0532)
2010-10	0.0157 (0.0179)	0.0276 (0.0183)	0.0255 (0.0367)	-0.0688 (0.0446)
2010-11	0.0426 (0.0210)	0.0191 (0.0179)	0.0849 (0.0512)	-0.1466 (0.0531)
2010-12	0.0284 (0.0254)	0.0273 (0.0192)	0.1015 (0.0423)	-0.1572 (0.0458)
2011-01	0.0277 (0.0272)	0.0008 (0.0211)	0.0704 (0.0472)	-0.0989 (0.0489)
2011-02	0.0648 (0.0362)	0.0513 (0.0391)	0.0270 (0.0687)	-0.1431 (0.0794)
2011-03	0.0031 (0.0268)	0.0417 (0.0232)	0.0638 (0.0400)	-0.1087 (0.0480)
2011-04	-0.0230 (0.0180)	-0.0191 (0.0304)	0.1152 (0.0550)	-0.0731 (0.0609)
2011-05	0.0170 (0.0273)	-0.0181 (0.0283)	0.0709 (0.0567)	-0.0697 (0.0536)
2011-06	0.0004 (0.0191)	0.0420 (0.0259)	0.1025 (0.0510)	-0.1448 (0.0526)
2011-07	-0.0088 (0.0185)	-0.0093 (0.0374)	0.1077 (0.0587)	-0.0895 (0.0600)
2011-08	-0.0096 (0.0192)	0.0196 (0.0256)	0.1327 (0.0460)	-0.1426 (0.0594)
2011-09	0.0094 (0.0130)	0.0086 (0.0248)	0.0937 (0.0515)	-0.1118 (0.0538)
2011-10	0.0061 (0.0198)	-0.0055 (0.0331)	0.0906 (0.0876)	-0.0912 (0.0870)
2011-11	-0.0194 (0.0144)	0.0314 (0.0265)	0.1531 (0.0450)	-0.1650 (0.0563)
2011-12	-0.0051 (0.0140)	0.0092 (0.0532)	0.0704 (0.0468)	-0.0744 (0.0489)
2012-01	-0.0106 (0.0231)	0.0029 (0.0323)	0.1087 (0.0456)	-0.1010 (0.0518)
2012-02	-0.0022 (0.0150)	0.0053 (0.0229)	0.1295 (0.0575)	-0.1326 (0.0624)
2012-03	-0.0018 (0.0203)	0.0166 (0.0224)	0.0918 (0.0570)	-0.1066 (0.0583)
2012-04	-0.0057 (0.0174)	0.0563 (0.0301)	0.1096 (0.0668)	-0.1603 (0.0686)
2012-05	-0.0324 (0.0170)	-0.0314 (0.0307)	0.0872 (0.0544)	-0.0233 (0.0587)
2012-06	-0.0003 (0.0155)	-0.0104 (0.0296)	0.0289 (0.0533)	-0.0182 (0.0645)
2012-07	-0.0197 (0.0191)	-0.0005 (0.0245)	0.1706 (0.0653)	-0.1503 (0.0827)
2012-08	-0.0200 (0.0279)	0.0209 (0.0210)	0.0881 (0.0529)	-0.0890 (0.0579)
2012-09	-0.0141 (0.0167)	-0.0088 (0.0312)	0.1021 (0.0535)	-0.0792 (0.0729)
2012-10	0.0362 (0.0318)	-0.0031 (0.0233)	0.1281 (0.0745)	-0.1611 (0.0797)
2012-11	0.0321 (0.0340)	-0.0028 (0.0321)	0.0912 (0.0583)	-0.1204 (0.0692)
2012-12	0.0764 (0.0232)	-0.0414 (0.0284)	0.1177 (0.0629)	-0.1527 (0.0717)

5 CONCLUSION

This thesis investigated how extreme weather events (with emphasis on floods and heatwaves) affects the utilization of maternal and child health (MCH) services. From a health economics perspective and grounded in an understanding of health systems as complex adaptive systems, the overarching objective was to generate integrated evidence on a) whether these events alter the utilization of essential services (antenatal, delivery, postnatal, and child health care), b) through which mechanisms such changes occur, and c) in which contexts these effects tend to be amplified, with direct implications for public policies on adaptation, resilience, and equity in access.

The common thread connecting the three essays is the proposition that extreme climate events should not be treated as isolated shocks, but rather as systemic disruptions that propagate simultaneously through both demand- and supply-side channels of health services. In economic terms, floods and heatwaves alter access costs (time, transportation, risk), household budget constraints (income loss, displacement), and the production conditions of services (infrastructure, workforce, logistics, and service capacity). This unifying perspective helps explain why, in certain contexts, the system's response is not a simple "collapse" in utilization, but rather a reorganization in the form and intensity of care use, often reproducing or deepening pre-existing vulnerabilities.

The original contributions of the thesis stem from the methodological complementarity across the essays. **Essay 1**, a realist-informed systematic review, organized a fragmented literature and showed that floods tend to more consistently compromise routine services (antenatal care, institutional delivery, and postnatal care), whereas heatwaves are more frequently associated with increased acute and hospital-based utilization, although important gaps remain regarding their effects on the continuity of routine care. By explicitly identifying Context–Mechanism–Outcome configurations, the essay demonstrated that impacts are not uniform; rather, they depend on socioeconomic vulnerability, rurality, infrastructure, response capacity, and the institutional design of the health system.

Essay 2 translated these findings into an integrated explanatory framework through a Causal Loop Diagram (CLD), representing MCH service utilization as the result of dynamic interactions among mobility, displacement, income, health risks, service provision, and disruptions to infrastructure and supply chains. The CLD highlights that a significant share of vulnerability in maternal and child care originates outside the health sector, in domains such as transportation, energy, labor markets, and social protection. At the same time, by mapping reinforcing and balancing feedback loops, the essay shows how climate shocks can generate persistent effects (for example, through income loss and deterioration of local capacity), or be dampened by adaptive mechanisms (service reorganization, outreach, mobile care, inter-municipal coordination), thereby identifying leverage points for resilience-oriented policies.

Essay 3 provided novel causal evidence directly applicable to the Brazilian context by

estimating the effect of the 2010 floods in Pernambuco and Alagoas on antenatal care utilization, using administrative data and a robust inference strategy (controlled interrupted time series and event-study difference-in-differences). The main finding was that the shock did not, on average, produce a substantial increase in exclusion from antenatal care (zero visits), but rather generated a reallocation in the intensity of care: a reduction in the proportion of women with seven or more visits and an increase in the intermediate category (4–6 visits). In terms of system functioning, this pattern suggests a partial disruption in continuity and in the capacity to complete recommended care trajectories, rather than a total breakdown of access. It is a particularly relevant finding in high-coverage systems such as Brazil’s Unified Health System (SUS), where maintaining some contact may be more likely than preserving the ideal intensity of follow-up.

Taken together, the results of the thesis support a central implication: climate adaptation policies in health should not focus solely on the minimal goal of “avoiding complete interruption”, but also on protecting less visible and equally critical dimensions of care performance: continuity, adherence to recommended schedules, recovery capacity, and equity. In practical terms, this points to actions aligned with the agenda of climate-resilient health systems, including: a) strengthening infrastructure and connectivity (safe routes, bridges, patient transport, contingency planning); b) enhancing response capacity in primary care and maternal–child networks (reserve teams, flexible scheduling, rapid rebooking of missed appointments, network coordination); c) developing information systems capable of tracking missed visits and recontacting pregnant women after events; d) ensuring resilient logistics and supplies (energy, essential inputs, communication); and e) integrating health policy with social protection and territorial management strategies, given the role of income, displacement, and mobility in shaping utilization. From an equity perspective, the thesis reinforces that climate shocks tend to exacerbate pre-existing disadvantages; therefore, adaptation strategies must be explicitly designed to prioritize the most vulnerable groups and localities with lower absorptive and adaptive capacity.

Despite its contributions, important research challenges and opportunities remain. Key avenues for future work include: a) advancing the measurement of event intensity and severity (beyond binary classifications), incorporating flood depth, duration, economic losses, and infrastructure failures; b) exploring heterogeneity by socioeconomic vulnerability, rurality, system capacity, and baseline coverage to identify where “downshifting” in care intensity is most likely and most harmful; c) linking utilization changes to maternal and neonatal outcomes in order to quantify welfare consequences and avoidable costs associated with partial disruptions in continuity; d) expanding evidence on heatwaves and routine maternal–child care (not only acute use), particularly in LMICs and underrepresented regions; and e) transforming the CLD into a quantitative system dynamics model (stocks and flows) validated with stakeholders, enabling simulation of scenarios, delays, cumulative effects, and policies under compound and recurrent climate risks.

In sum, this thesis demonstrates that extreme climate events affect maternal and child health not only through direct clinical risks, but through disruptions that cut across local

economies and service organization, altering utilization patterns and threatening continuity of care. By combining realist evidence synthesis, mechanism-based systems modeling, and causal inference applied to Brazil, the research offers an integrated contribution to health economics and to the climate adaptation agenda: strengthening resilience requires not only maintaining access, but also preserving care trajectories, especially for those already living at the margins of the system. In a context of intensifying climate extremes, safeguarding the continuity and equity of maternal and child health care is both a public health priority and a strategic investment in long-term well-being and human capital.

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